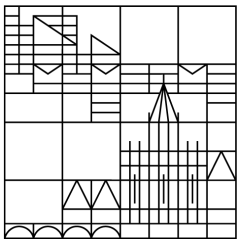
Universität Konstanz



Audit-Aware Topology Selection for LLM Agents

Master's Thesis

Author: **Emrecan Ulu**
Matriculation number: 01/1476626

University of Konstanz
Faculty of Politics, Law and Economics
Department of Politics and Public Administration
MSc Social and Economic Data Science

Industrial context:
Siemens AG, Environmental Protection Data Analysis (EHS EP DA)

1st Assessor: **JunProf. Dr. Carina I. Hausladen**
2nd Assessor: **Prof. Dr. Peter Selb**

September 2026
Period of completion: April 20, 2026 – September 17, 2026

Abstract

Not every compliance answer inside a regulated industrial process carries the same burden of proof. A routine internal check and an audited regulatory filing draw on the same underlying data, yet they do not carry the same burden of proof. Stabilizing an LLM-based system's output for auditability therefore differs from one case to the next: a single fixed structure either over-engineers the routine check or leaves the audited filing insufficiently verifiable. Established practice in LLM systems answers this by dividing the task among subagents and presenting the result through orderly communication. Exactly how that communication should be structured, and how it can be audited, remains open. That communication pattern is *topology*. Topology selectors in the literature condition that choice on task difficulty, cost, or latency. This thesis instead presents EPOCH (Environmental Protection Orchestration Chain Hierarchy), which grounds topology choice in a case's output profile and assurance profile. A three-shape library resolves that choice — Direct, Debate, and Coalition — each tied to a distinct evidence artifact, behind a two-gate, human-in-the-loop pipeline that locks an isolated memory before any topology runs. The architecture held its guarantees across repeated runs and two model families. A genuine in-house business application further validated EPOCH for matching the deliverable's depth to the requester's own output and assurance profile.

Code and evaluation materials: https://github.com/vulonviing/epoch

Acknowledgments

I thank my advisor, Carina Ines Hausladen, for her mentorship, her illuminating courses, and every contribution she made throughout this thesis.

I thank my second advisor, Peter Selb, for agreeing to evaluate this thesis and for his valuable feedback.

I thank our program advisors, Kathrin Krautheimer and Rümeysa Bozer, for all their support throughout the program.

At Siemens, I thank my team lead, Sven Kristen, for shaping this thesis's vision. I thank Christian Gilabert Alarcón and Lisset Salinas Pinacho for supporting me through our weekly thesis meetings, Hansi Senaratne for his substantial help structuring the thesis, and Kaiyuan Xin and Alice Freitas for feedback that meaningfully improved it.

Contents

Abstract 1

Acknowledgments i

List of Figures vi

List of Tables vii

1 Introduction 1

1.1 Motivation and Problem Statement 1

1.2 Research Questions 2

1.3 Contributions 2

1.4 Thesis Outline 3

2 Theory 4

2.1 LLM Agent Pipelines and Multi-Agent Topologies 4

2.1.1 From Many Structures to Three Shapes 5

2.1.2 Direct: One Analytical Line 7

2.1.3 Debate: Independent Readings of One Claim 7

2.1.4 Coalition: Distinct Domain Ownership 8

2.1.5 Topology Selection as a Research Problem 8

2.2 Industrial Compliance Changes the Question 9

2.2.1 Stable Scale-Up and Down 10

2.2.2 A Stable, Expected Output 11

2.2.3 Assurance as an Evidence Requirement 12

2.2.4 Human Commitment 13

2.2.5 Regulatory Reference: The European Union AI Act . . . 14
2.3 Positioning and Contribution . . . 15

3 Data and Methods **17**
3.1 Design Science Research Approach . . . 17
3.2 System Foundations . . . 18
3.2.1 Agent Roles and Deterministic Boundaries . . . 18
3.2.2 Isolated Memory and Human Commitment . . . 19
3.2.3 Artifact Provenance . . . 20
3.2.4 External Perspective at Stage-Final Gates . . . 21
3.2.5 Blind Passes and Anchoring . . . 22
3.3 System Overview . . . 22
3.4 Stage 1 — Auditable Case Construction . . . 23
3.4.1 Demand Orientation and Modality . . . 25
3.4.2 Tabular Scope and Evidence Construction . . . 25
3.4.3 Document Scope and Evidence Construction . . . 26
3.4.4 The Human-Locked Assurance Boundary . . . 27
3.4.5 Case Profiling and Topology Selection . . . 27
3.5 Stage 2 — Topology-Bound Execution . . . 27
3.5.1 Tabular Topology Bindings . . . 28
3.5.2 Real-World Document Evaluation Execution . . . 29
3.5.3 Final Review and External Commentary . . . 29
3.6 Evaluation Design . . . 30
3.6.1 Part I: Gold-Anchored Pipeline Evaluation . . . 30
3.6.2 Part II: Real-World Use-Case Expert Evaluation . . . 31

4 Evaluation and Results **36**
4.1 Evaluation Setup . . . 36
4.2 Gold-Anchored Results . . . 37
4.3 Expert Evaluation . . . 39

5 Conclusion **42**
5.1 Answers to the Research Questions . . . 42

5.2 Implications . . . 43
5.3 Limitations . . . 45
5.4 Future Work . . . 46

A Evaluation Protocol **51**
A.1 Evaluation Taxonomy . . . 51
A.2 Gold-Scoring Judge Prompt . . . 59
A.3 Frozen Gold Reference Register . . . 65

B Configuration **66**
B.1 Backend Presets . . . 66
B.2 Reasoning Switch and Provider Field Mapping . . . 67
B.3 Token Budgets and Batching . . . 67
B.4 Per-Agent Model Selection . . . 68
B.5 Recorded, Not Assumed . . . 68

C Extended Theory and Regulatory Context **69**
C.1 Multi-Agent Literature Families . . . 69
C.2 Selection, Measurement, and Oversight Evidence . . . 70
C.3 EU AI Act Design Mapping . . . 71

D System Architecture and Agent Contracts **73**
D.1 Agent Types and Shelf Records . . . 73
D.2 Role-to-Identifier Map . . . 74
D.3 External Perspective Source Allowlist . . . 74
D.4 Stage 1 Pipeline Contracts . . . 75
D.5 Selection and Stage 2 Bindings . . . 77
D.6 Post-Hoc Architecture Audit (/system-check) . . . 78

E UC2.1 Tabular Replication Artifacts **79**
E.1 UC2.1 Registry Entry . . . 80
E.2 UC2.1 R1 Artifact . . . 81
E.3 UC2.1 DA1 Mapping Artifact . . . 82
E.4 UC2.1 R2 Scope Artifact . . . 83

E.5 UC2.1 D1–D2 Artifacts and UC1 Derived-Facts Example 84
E.6 UC2.1 CP1–TS1 Selection Artifacts 85
E.7 UC2.1 D3 Computation Artifact 87
E.8 UC2.1 P2–P3 Independent Reads 88
E.9 UC2.1 S1 Reconciliation Artifact 89
E.10 UC2.1 F1 Final Deliverable 90
E.11 UC2.1 External Commentary Artifacts 92

F UC4 Document Evaluation Materials **93**
F.1 Source and Version Hierarchy 93
F.2 Frozen Artifact Chain 93
F.3 Expert Review Materials 94

G Peer Review Packets **95**
G.1 Page 1: Common Review Form (All Use Cases) 95
G.2 Returned Review Responses 97
G.3 UC1: German Energy Threshold Priority Screen 101
G.4 UC2: Quarterly ETS 1 Scope 2 Assurance Memo 112
G.5 UC2.1: Annual ETS 1 Scope 2 Assurance Memo 125
G.6 UC3: EMEA Divisional Scope 2 Consolidation 134
G.7 UC4: Siemens 2026 ESRS Reporting Impact 140

List of Figures

2.1 Accuracy loss in long context 4
2.2 Literature families collapse to three topologies 6
2.3 Direct topology 7
2.4 Debate topology 7
2.5 Coalition topology 8
2.6 Each topology scales by agent count 9
2.7 Selector π: case profile to topology 16

3.1 Agent library: reuse, upgrade, add 19
3.2 Isolated memory shelf system 20
3.3 CP1 reads five upstream shelves 21
3.4 Stage 1 case construction (UC2.1) 25
3.5 UC2.1 Stage 2 Debate execution 28
3.6 UC4 complete document pipeline 33
3.7 UC4 registry triangle 34

E.1 UC2.1 F1 record in the published viewer 91

List of Tables

1.1 The four research questions 2

2.1 Topology-selection signals 10
2.2 Signals vs. Siemens EHS requirements 11
2.3 Stable output profiles 12

3.1 Tabular vs. document pipeline families 23
3.2 Use cases introduced in this chapter 24
3.3 The five gold-anchored evaluation tests 30
3.4 Peer-review questions 32
3.5 Five peer-review packets 32

4.1 Evaluation run set per use case 36
4.2 T1–T5 results per use case and backend 37
4.3 Mean peer-review score per criterion 39

A.1 T3's four field comparison classes 59
A.2 T3's five glaring defects 59
A.3 Frozen gold reference register 65

B.1 Configured LLM backend presets 66
B.2 Provider field mapping for reasoning and tokens 67
B.3 Per-agent token budget tiers 67

C.1 Extended multi-agent structure comparison 69
C.2 EU AI Act design-oriented mapping 72

D.1 Agent types and validation boundaries 73

D.2 Shelf record identifiers 74
D.3 Functional roles and identifiers 74
D.4 EX1/EX2 official-source allowlist 75
D.5 Stage 1 input/output contracts 76
D.6 Stage 2 topology bindings 77

E.1 UC2.1 registry entry summary 80
E.2 UC2.1 R1 artifact excerpt 81
E.3 UC2.1 DA1 mapping excerpt 82
E.4 UC2.1 R2 scope excerpt 83
E.5 UC2.1 D1–D2 execution excerpt 84
E.6 UC2.1 CP1–TS1 selection excerpt 85
E.7 UC2.1 D3 artifact excerpt 87
E.8 UC2.1 P2/P3 artifacts excerpt 88
E.9 UC2.1 S1 reconciliation excerpt 89
E.10 UC2.1 F1 structured memo excerpt 90
E.11 UC2.1 external commentary excerpt 92

F.1 UC4 source hierarchy 93
F.2 UC4 artifact chain 94

G.1 Peer-review score matrix 97
G.2 Peer-review criterion-level notes 98
G.3 Peer-review free-text responses (UC1, UC2) 99
G.4 Peer-review free-text responses (UC2.1, UC3, UC4) 100

Chapter 1

Introduction

Section 1.1 states the problem EPOCH (Environmental Protection Orchestration Chain Hierarchy) answers. Section 1.2 states the four research questions that follow from it. Section 1.3 previews what this thesis contributes toward each. Section 1.4 closes with how the remaining chapters are organized. An anonymized, read-only companion viewer provides direct access to the implemented registry, stage decisions, and agent outputs at https://emrecanulu.com/epoch/.

1.1 Motivation and Problem Statement

Integrating AI systems into business processes is now on the industry's agenda. The EU is answering that agenda with new regulatory updates. AI's current limitations are well documented, and techniques exist that overcome many of them. Whether those techniques fit a real industrial setting, and the regulation that governs it, is a separate question. This thesis positions itself there: building a structure that fits a real industrial environment and its regulations.

A quarterly emissions check and an annual sustainability disclosure draw on the same underlying data, but they do not carry the same burden of proof. One is read internally and revised next quarter if it is wrong. The other is filed, audited, and cited against a regulatory deadline. A compliance system built around one fixed process structure, no matter how capable, either over-engineers the quarterly check or under-verifies the annual filing. The question this thesis asks is not whether a multi-agent system can answer a compliance question, but whether it can choose the right structure for the answer it is required to prove.

This thesis is grounded in Siemens AG's Environmental Protection function, part of the broader Environmental Protection, Health Management, and Safety (EHS) organization. That function turns regulation and operational data into decisions that must survive external review: a threshold screen, a dual-method reconciliation, a divisional consolidation, a disclosure impact assessment. The EU AI Act adds a second, formal version of the same demand, requiring risk management, logging, and human oversight for automated systems used in regulated settings [1]. Neither Siemens EHS nor the Act asks for the same rigor on every question, and both ask that the rigor match the question.

Multi-agent systems already decompose a task into specialized agents for exactly this kind of complexity. Chapter 2 shows that the selectors reviewed there choose a communication structure by task difficulty, cost, or latency, never by what the answer must prove. EPOCH's own topology system, set out here, classifies each agent's communication motivation into three shapes instead: single ownership (Direct), independent verification of one claim (Debate), and separately mandated results that must be combined (Coalition). This gap is what the research questions below ask EPOCH to close.

1.2 Research Questions

EPOCH is guided by four research questions, one each for design, build, conformance, and acceptance.

Question	Asks
RQ1 — Design	How can multi-agent topology selection be conditioned on the output profile and assurance profile a compliance answer must carry, rather than on task difficulty, cost, or token budget?
RQ2 — Build	How can a human-in-the-loop pipeline turn an expert-approved isolated memory into an auditable case profile that drives topology selection while satisfying EU AI Act oversight duties?
RQ3 — Conformance	Does an assurance-conditioned pipeline hold its own architectural guarantees — the selected topology, the isolated-memory boundary, and the provenance chain — across repeated runs of the same approved case?
RQ4 — Acceptance	Would domain experts who did not build EPOCH trust and use its finished deliverable in real work, what kinds of shortcomings would they report, and how much expert effort would it displace?

Table 1.1: The four research questions EPOCH is built to answer.

Chapter 5 answers each of these in turn, against the evidence Chapters 2–4 establish.

1.3 Contributions

EPOCH contributes four things, one per research question above.

- A selector, $\pi : (\text{case}, \text{output_profile}, \text{assurance_profile}) \rightarrow \text{topology} \in \Lambda$, that treats required evidence as an input to topology choice rather than a property checked after execution (Section 2.3).
- A two-gate, human-in-the-loop pipeline that locks an isolated memory before any topology runs and keeps every downstream artifact traceable back to that approval through `input_refs` (Section 3.2.2).
- Evidence that the architecture holds across repeated runs and two model families: topology selection correct in every scored run, and a post-hoc audit, `/system-check`, that now catches the one defect class found (Table 4.2, Appendix D.6).

- An expert-reviewed evaluation of five finished deliverables by domain experts who did not build the pipeline, returning a grand mean of 3.88 out of 5 across six reviews (Section 4.3).

1.4 Thesis Outline

The remaining chapters build toward, execute, and evaluate this answer. Chapter 2 develops the theoretical basis: the three-shape topology library and the assurance-conditioned selector π. Chapter 3 builds the two-stage pipeline, from registry demand through human-approved isolated memory to topology-bound execution. Chapter 4 reports the gold-anchored and expert-review evaluation results. Chapter 5 answers the four research questions above and closes with this thesis's limitations and future work.

Chapter 2

Theory

This chapter develops the theoretical basis for EPOCH. Section 2.1 reduces the wider multi-agent design space to three assurance-distinct topology shapes. Section 2.2 connects topology selection to the operational and regulatory requirements of Siemens environment, health, and safety (EHS). Section 2.3 closes with the research gap and the contribution addressed by this thesis.

2.1 LLM Agent Pipelines and Multi-Agent Topologies

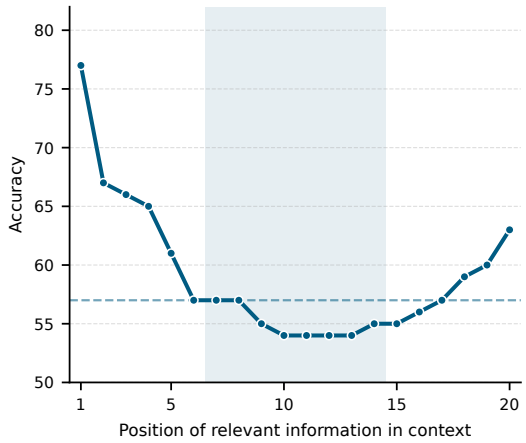


Figure 2.1: Accuracy is highest for information at the start or end of the context and drops in the middle. Adapted from Du and colleagues [2].

Large language models (LLMs) perform well on focused tasks, but longer context does not guarantee more reliable reasoning. Relevant information can disappear inside extended prompts even when retrieval succeeds [3, 2].

Wang and colleagues show that this pressure can appear as a threshold effect: performance falls sharply after the context reaches a certain length rather than declining smoothly [3]. Du and colleagues isolate a second problem. They test whether input length alone can hurt performance even when the model faces no real retrieval difficulty [2].

To remove this difficulty entirely, they keep only the single fact a task needs and replace everything else with content that carries no information. In some trials they mask that filler outright, so the model can attend only to the fact it needs. In both cases, accuracy still falls by 13.9 to 85 percent as the input grows longer, measured across math, question-answering, and coding tasks [2]. This shows that longer input can hurt performance even when the model does find the right information.

One response is to divide a task among specialized agents with narrower inputs. Dividing work gives each agent a smaller epistemic responsibility and creates explicit handoffs between them. Those handoffs introduce new risks: one agent may anchor another, a synthesis may hide disagreement, or a domain contribution may lose its owner. The real challenge is combining what each small agent produces into one final output: the order, structure, and context in which that information is gathered shapes the result [4]. Agents that have split a task this way must therefore communicate with each other in the right order and form to reach a correct outcome.

The pattern of communication among those agents is the *communication topology*: who exchanges information, in what order, and through which dependencies [4]. Topology is not neutral. It affects quality, cost, failure behavior, and the evidence left behind. Recent systems therefore design this connection structure automatically for each task, or route agents to steps dynamically at run time [5, 6, 7, 8, 9].

2.1.1 From Many Structures to Three Shapes

The literature uses many labels for related structures, but they all answer one question: keep the task with a single agent, or split it across several separate agents? ReAct is the standard single-agent answer: one agent interleaves its reasoning steps, such as planning the next move, with actions in the outside world, such as running a search [10].

The remaining three families all split work across agents but differ in how those agents come back together: Debate systems give the same question to independent agents and reconcile their answers [11, 12, 13]; role-based systems such as MetaGPT and ChatDev split a task into distinct owned deliverables [14, 15]; and a newer family either learns or searches for task-specific connection structures [5, 7, 8, 9, 16] or routes among existing agents under a cost budget [6, 17, 18].

The single-agent family establishes a demanding baseline: ReAct combines reasoning traces with external actions in one context and outperforms pure reasoning or pure action strategies on several tasks [10]. A related but separate question asks not whether ReAct specifically works, but whether a task should go to one agent or several at all [19]. Multi-agent systems often look stronger only because each agent thinks on its own, and those separate reasoning steps quietly add up to more computation than a single agent uses. Once the token budget is equalized across both sides, one strong agent matches or beats coordinating multi-agent systems even on multi-hop reasoning, tasks that need several intermediate steps to answer. Reported multi-agent advantages may therefore come from uncounted extra computation rather than from the architecture itself, so building a multi-agent structure needs a task-specific justification.

Debate, role decomposition, and the newer routing families each supply such a reason by shifting the extra cost into a declared structure, whether independent readings of one claim, separately owned deliverables, or a learned connection pattern, rather than adding uncounted reasoning depth. These systems differ in implementation, but three questions capture their assurance-relevant structure. Do agents perform the same task or different subtasks? Does one party or several parties own the result? Does convergence require reconciliation or synthe-

sis? Figure 2.2 maps those choices to the three shapes used in this thesis.

Topology	Task relationship	Ownership	Convergence
Direct	Dependent steps	One analytical line	Sequential completion
Debate	Redundant readings	One claim, several readers	Reconciliation
Coalition	Distinct subtasks	Several domain owners	Synthesis

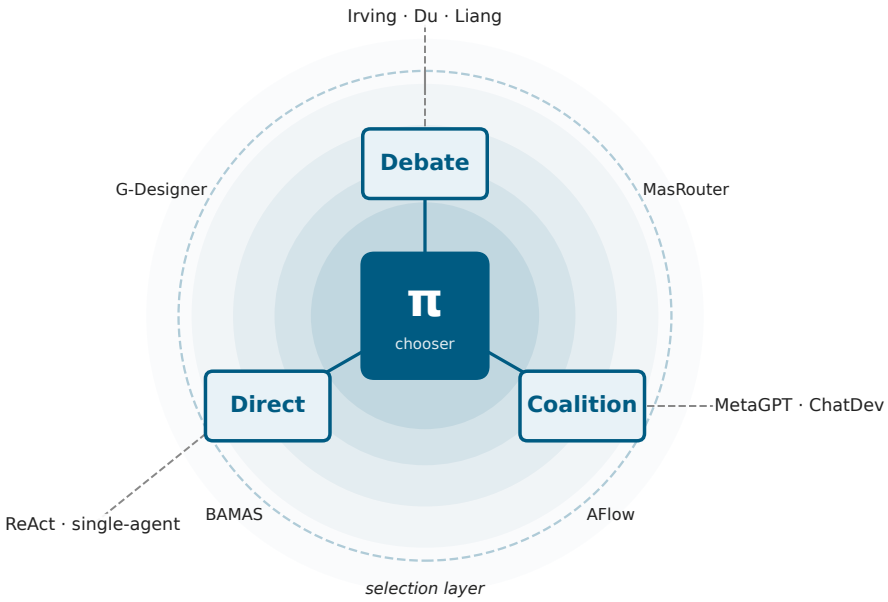


Figure 2.2: Five multi-agent literature families map to three assurance-distinct corners. **ReAct and single-agent baselines** collapse to *Direct*: one owner follows a reason-act loop and produces a single attestation. **Irving, Du, and Liang** (debate and Multi-Agent Debate) collapse to *Debate*: independent readers or independent experts verify the same claim and reconcile into one result. **MetaGPT and ChatDev** (role decomposition) collapse to *Coalition*: independent domain groups each own a sub-result, assembled into a distributed sign-off. The dashed ring marks the *selection layer*: G-Designer, MasRouter, BAMAS, and AFlow generate or route topology choices, making them prior art for the selector π itself.

No single source reviewed here proposes this exact reduction. It is a synthesis of three structural families whose outputs carry different evidence. Appendix C retains the full source-by-source comparison (Table C.1) together with the detailed per-family prose this section compresses.

2.1.2 Direct: One Analytical Line

Direct assigns the result to one accountable analytical line. The execution may contain several dependent steps, but each step extends the result immediately before it. A sequential pipeline is therefore Direct when it preserves one ownership chain; agent count alone does not change the shape. Figure 2.6 shows this directly across all three families.



Figure 2.3: Direct topology: a single-owner sequential execution chain producing one attestation.

Its assurance artifact is a *single attestation*: one conclusion with one traceable chain of responsibility. Direct is appropriate when the approved case does not require an independent second reading or separately mandated domain owners. It is the least structurally complex shape, not a lower-quality default. The chain may separate calculation, interpretation, and composition while later steps receive both the prior result and approved memory. This separation makes intermediate evidence inspectable, but it does not claim independent verification.

Direct therefore has a clear negative boundary. If two independent readers, or independent experts, inspect the same claim without seeing one another, the shape becomes Debate. If independent domain groups must contribute different results, it becomes Coalition. The single attestation describes responsibility across the chain, not the number of software components inside it.

2.1.3 Debate: Independent Readings of One Claim

Debate gives the same approved claim to two or more readers before any reader can see another's conclusion. Irving, Christiano, and Amodei frame debate as a judged argument, while Du and colleagues use cooperative rounds to improve factuality [11, 12]. Liang and colleagues identify a related failure mode: when a model critiques and revises only its own answer, with no outside view, it rarely escapes that answer once it commits to it. Even when the answer is wrong, self-critique tends to just confirm it again. Their fix is the same one debate provides: bringing in an opposing view from outside forces the model to genuinely consider a different answer [13]. Appendix C carries the fuller comparison of these sources.

The defining property is independence before convergence. The final artifact is a *reconciliation record* that preserves the readings, their disagreement, and the reason for the resolved result. Debate verifies one claim twice; it does not divide responsibility for different outputs. Role names alone do not create independence;

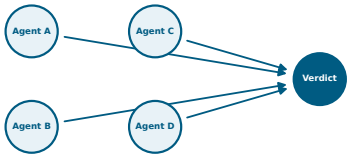


Figure 2.4: Debate topology: four agents in two independent pairs converging on a single reconciliation record.

each reader or expert must form its judgment without seeing the other.

The reconciler has a restricted role. It may compare evidence and state a resolution, but it should not erase one branch or invent a third evidential basis. Its output must point back to both readers or experts. This makes the reconciliation record an assurance artifact. Disagreement remains observable instead of disappearing inside one reasoning trajectory.

2.1.4 Coalition: Distinct Domain Ownership

Coalition applies when the result contains independent, separately mandated contributions. Role-based systems illustrate this structure. A product manager, architect, and engineer do not offer competing answers to one question. Each owns a different artifact that the others cannot authorize [14, 15].

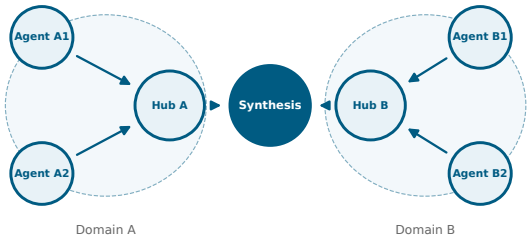


Figure 2.5: Coalition topology: domain-specialist islands converge internally, then contribute to a shared synthesis node.

Its assurance artifact is a *distributed sign-off*: attributable sub-results assembled into one output without erasing their ownership boundaries. This is different from Debate. Independent readings of one claim require reconciliation; independent responsibility for different claims requires synthesis.

Coalition is therefore justified by ownership rather than topic count. Two figures from one team do not require two domain agents. Conversely, one report may require a coalition when several accountable units must each sign off on a separate part. A coordinator may align their contributions, but it cannot absorb their authority. The required evidence, not superficial complexity, determines the shape.

2.1.5 Topology Selection as a Research Problem

Recent selectors make topology a dynamic system property. G-Designer uses task features, MasRouter learns a cost-performance routing policy, AgentConductor adapts graph density, and BAMAS selects under a budget [5, 6, 8, 17]. Their shared move is important: a topology can be selected rather than fixed. The unresolved question is which signals should control that selection in a regulated industrial setting. Appendix C carries the fuller comparison of these systems.

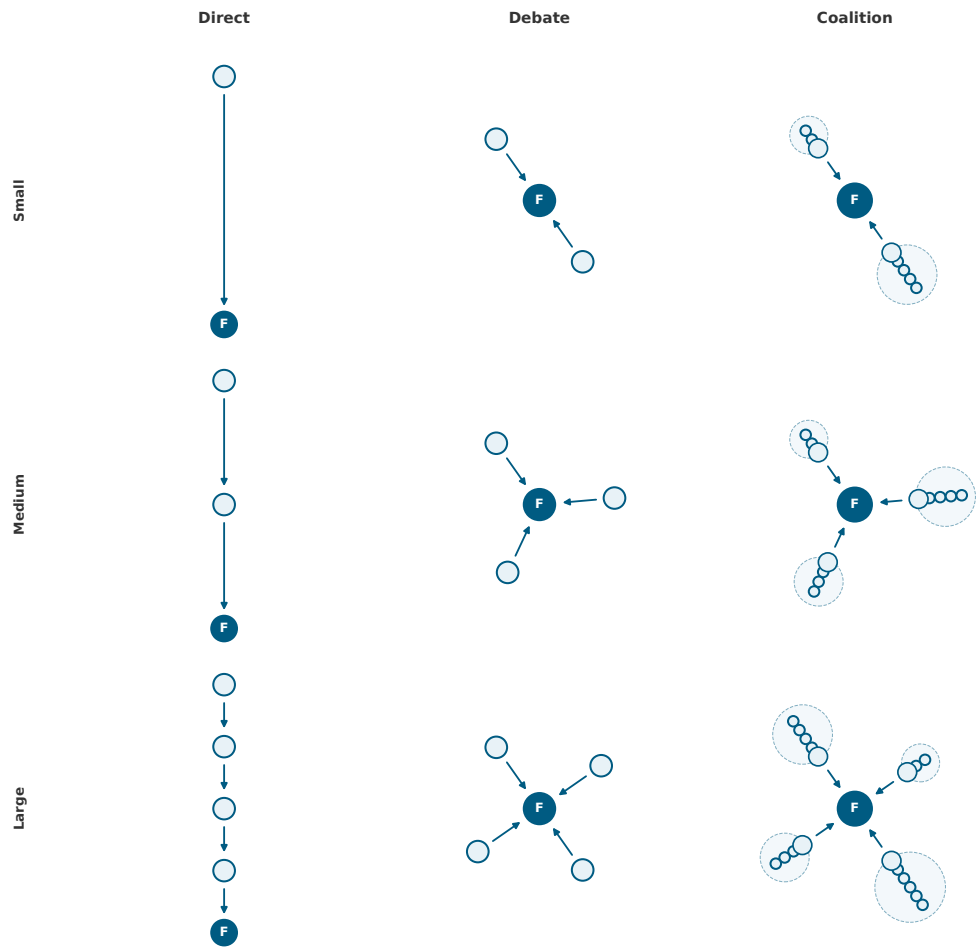


Figure 2.6: Each topology family admits several agent-count configurations without changing shape, and every configuration feeds the same centred finalizer role (**F**). **Direct** remains one ownership chain whether it runs two, three, or five dependent steps. **Debate** still reconciles independent readings of one claim whether two, three, or four readers take part. **Coalition** is the deliberate exception: domain islands are never the same size, so each example combines differently sized domains, all converging through their own local hub. The configurations shown are illustrative examples, not recorded pipeline runs.

2.2 Industrial Compliance Changes the Question

Existing selectors mainly measure task difficulty, benchmark success, token use, coordination cost, or budget. These signals answer how well and how cheaply a configuration may run. They do not answer what evidence the requested output must carry. **Selection signal is not the same as delivery requirement.**

Every row in Table 2.1 measures either expected task success or the resources used to obtain it. ARG-Designer adds per-task accuracy and token efficiency, while RGAO retrieves a topology from code-complexity signals [7, 20]. These additions broaden the features but preserve the objective. They still ask which arrangement is likely to perform well within an acceptable cost.

That question is necessary but incomplete in compliance work. Two topologies can produce similar benchmark scores while leaving different records of responsibility. A selector that ob-

System	Primary selection signal
AgentConductor [8]	task difficulty and pass@1
G-Designer [5]	task type and token cost
GoAgent [9]	query and coordination complexity
MasRouter [6]	collaboration mode, role, and LLM routing
BAMAS [17]	task specification and budget

Table 2.1: Representative signals used by topology-selection systems.

serves only task and budget features cannot distinguish those outputs unless evidence structure becomes a separate input.

Siemens AG's Environmental Protection Data Analysis team works across sites, time windows, datasets, and regulatory frameworks. Its results may enter an internal or external review. A reviewer must be able to ask how a figure was produced, which evidence supported it, and who approved the relevant boundary. Task accuracy and cost remain useful, but they cannot answer those questions.

The operating scope spans thousands of regulatory obligations; a few examples are emissions trading, energy-efficiency obligations, climate reporting, greenhouse-gas accounting standards, and sustainability disclosure [21, 22, 23, 24, 25, 26]. These sources combine numerical thresholds with interpretation and attribution. A cheap result without a reconstructable basis can therefore fail before its numerical quality becomes the central issue.

The same demand recurs at different scales. Take the Scope 2 case as an example: the GHG Protocol requires two separate ways of reporting the indirect emissions from purchased energy. One method uses the average emissions factor of the local grid; the other uses the company's own energy contracts, such as renewable energy certificates. For the same consumption, these two methods can produce two different figures, both accepted as valid, so this case needs an independent comparison rather than one calculation.

A single-site threshold question, by contrast, may need only one attributable calculation, while a portfolio report may require separate divisional owners. The subject matter remains environmental compliance, yet the required evidence shape changes. A static topology would treat these as variations of one workload and miss that difference.

This is why the selector needs three inputs. `case` describes the structure of the work, `output_profile` describes the expected form, and `assurance_profile` describes the evidence floor. None can safely stand in for the others. A binary output can still require distributed ownership, and a complex calculation can still remain one accountable analytical line.

2.2.1 Stable Scale-Up and Down

Siemens EHS does not ask one fixed question. A new site, reporting period, or regulation can change the case without changing the underlying need for a stable process. Some systems already do this: they add more agents or connections as a task gets harder, and remove them

Requirement	Current selectors	Siemens EHS
Task accuracy and cost control	Primary	Required
Scale across recurring cases	Partial	Required
Stable output form	Rarely explicit	Required
Auditable evidence and ownership	Not a selection axis	Required
Human commitment points	Not a selection axis	Required

Table 2.2: The difference between common topology-selection signals and the delivery requirements of the Siemens EHS setting.

as it gets easier, growing and shrinking themselves automatically [5, 27, 20]. That flexibility is useful, but recurring compliance work adds a comparability requirement: structurally equivalent requests should not be processed differently without an explanation – otherwise an auditor cannot tell why two similar cases turned out differently.

In EPOCH, `case` therefore describes structural properties: the kind of time span a case covers, whether it is one reporting period or several years. It also captures which domains a case spans and how many separate parties must own its results. It does not mean difficulty alone. The selector uses a reusable agent library and a fixed, closed set of three topology shapes. A changed case can therefore reuse, replace, or extend a known configuration instead of inventing a new structure each time, without hiding the change.

This differs from unconstrained graph adaptation, the way some systems, such as G-Designer, redesign their connection structure from scratch on every run instead of drawing from a fixed library. A recurring request follows the same declared path unless its approved structural properties change: growth stays limited to reusing a topology-agent binding for a case of the same kind, replacing an agent with an upgraded version that keeps its input-output contract, or adding one to cover a missing capability (Section 3.2.1). What growth cannot do is change the assurance structure a case requires, for example collapsing two independent readings into one unverified agent because that looks cheaper in one particular run.

2.2.2 A Stable, Expected Output

Scale settles how the system adapts; it does not settle what the user receives. The `output_profile` fixes that delivery form before execution. A binary alert, qualitative assessment, and quantitative measurement impose different requirements even when they concern the same regulation.

This distinction follows a classification long established in measurement theory: categorical or binary values, ordinal or qualitative assessments, and numerical measurements each carry a different kind of information [28, 29, 30]. It also supports comparison across runs: a request for a threshold alert should not silently become a narrative memo because an agent found more context. Output form only fixes the shape of the deliverable – a binary alert can come from a single agent's guess or from two independent readings reconciled together. Both look like the same yes-or-no answer from the outside. How much evidence and verification stand behind that

answer is settled not by the output form, but by a separate requirement. The same profile also accepts more than one phrasing of question, as Table 2.3 shows: a binary profile still returns yes or no whether the question is about a legal limit or a reporting threshold, and a quantitative profile still returns an attributed figure whether the question asks for a total or a count of outliers.

Profile	Delivery form	Representative question	Example answer
Binary	Threshold judgment	• Did a facility exceed its legal limit?	• Yes / No
		• Is annual energy use above the reporting threshold?	• Yes / No
		• Does emissions intensity stay under the benchmark?	• Yes / No
Qualitative	Structured assessment	• Does the site's process meet the requirement?	• Meets / Partially meets / Does not meet
		• Is the disclosed method consistent with the standard?	• Consistent / Inconsistent
		• Does the data support external assurance?	• Adequate / Needs remediation
Quantitative	Measured value with attribution	• How much was reported, and by whom?	• e.g., a figure in tCO2e, attributed to a named role
		• By how much does the site exceed the threshold?	• e.g., an amount above the limit
		• How many sites fall outside the tolerance band?	• e.g., a count against the total site number

Table 2.3: Stable output profiles used to define the requested deliverable, with representative questions and example answers showing each profile's flexibility.

2.2.3 Assurance as an Evidence Requirement

Assurance is a documentary requirement, not a synonym for accuracy. An accurate answer without recoverable evidence cannot support a later reconstruction. Reportability asks whether the result and its support can be presented in the required form. Accountability identifies responsibility for material decisions. Auditability allows the result to be reconstructed from stored inputs and steps [31, 1]. Appendix C opens up the concrete mechanism behind these two sources.

The three properties address different failure points. A result may be reconstructable but lack a responsible approver, or attributable but stored in a form that cannot support external review. This thesis therefore treats them as jointly necessary. Assurance describes the evidence carried by the process as a whole, not a confidence score attached to the final sentence.

These properties lead to the three artifacts introduced above. Take a single-site threshold question, calculated and signed off by one engineer: a single attestation, supporting one accountable analytical line. Take the Scope 2 case, where the location-based and market-based figures are compared and resolved into one outcome: a reconciliation record, showing how independent readings of one claim were resolved. Take a portfolio report, where finance approves the cost

figures and the environmental team approves the emissions figures separately: a distributed sign-off, preserving separately owned contributions. These three artifacts are not ranks; they are non-interchangeable evidence structures.

A reconciliation record cannot replace distributed sign-off. In the portfolio example, merging finance's and the environmental team's separate approvals into one reconciled outcome would hide which team actually approved which figure. The real need there is to keep two distinct owners visible, not to resolve a disagreement between them.

A distributed sign-off cannot replace reconciliation either. In the Scope 2 example, presenting the two methods' figures side by side, as if they were two separate teams' independent approvals, hides the fact that they are two competing answers to the same question. That conflict still needs resolving, not two separate sign-offs side by side. Choosing the wrong artifact is therefore not a minor change in presentation – it makes it impossible to show who approved what, and why.

More agents do not automatically increase assurance. Running the same task multiple times and combining the results, such as by majority vote, does not reliably improve the answer – it can just add random variation rather than real new information. Multi-agent coordination also brings its own failure modes (Appendix C) [32, 19, 33]. A more complex shape is justified only when the approved evidence requirement calls for a different artifact. `assurance_profile` names that requirement.

Traceability frameworks often evaluate the evidence record after a multi-agent system has already run [34, 35, 36]. That remains useful for audit. EPOCH moves one decision earlier: it uses the required artifact to choose the process before execution. Post-hoc inspection then tests a declared structure instead of discovering the assurance boundary only after the result exists.

2.2.4 Human Commitment

Evidence requirements cannot authorize themselves. A domain expert must decide which regulatory scope and evidence boundary the system may treat as binding. Human-in-the-loop is therefore not a final quality check attached to an otherwise autonomous process. It is the commitment point that turns a candidate scope into the shared reality of later agents.

A gate is a point in the process where a human must explicitly approve an intermediate or final output before the process can continue. It turns a candidate result into an approved one that later steps can rely on. Oversight also has limited capacity: reviewers can rubber-stamp intermediate outputs when gates are too frequent or poorly placed [37, 38]. EPOCH concentrates human decisions at scope, case-profile, topology-selection, and final-output boundaries. Intermediate deterministic or topology-internal steps remain inspectable without demanding a separate approval.

This placement follows a complementarity argument. Human expertise should own the decisions that establish responsibility, while automation prepares the evidence needed to make those decisions well [39, 40]. Research on scalable oversight reaches a related conclusion:

review is useful when attention is concentrated where it can change the outcome, not spread across every internal action [41, 42]. Appendix C opens up the concrete mechanism behind these two protocols.

A concentrated gate still depends on what evidence reaches it. When every internal component draws on the same regulatory reading, the reviewer inherits that reading's blind spots along with its conclusions. Research on regulated-domain agents shows this narrowing directly: legal-agent classification studies report the same hallucination and verifiability failures in contract and compliance tasks [43]. Tool-invoking agents have also been shown to complete a task's technical goal while missing the regulation it was meant to honor [44]. Appendix C records these two findings as well. EPOCH answers this by adding an outside reading, grounded in a fixed set of official institutions, immediately before each stage-final gate – carried out by two agents, EX1 and EX2, described in Section 3.2.4. That reading is not a second authority, however: deployed LLM judges require their own lifecycle of alignment and monitoring to stay trustworthy [45], a maintenance burden the isolated-memory boundary is not built to carry. EPOCH therefore keeps the outside reading advisory.

2.2.5 Regulatory Reference: The European Union AI Act

Five articles of the EU AI Act supply a legal reference for these process requirements, giving precise language to evidence retention and responsible intervention [1].

The articles also address different layers of control. Risk management and technical documentation concern the process definition; logging supports later reconstruction; human oversight requires an intervention point that can affect the result. No single article prescribes EPOCH's architecture. Their combined requirements motivate the properties that the design makes explicit. Appendix Table C.2 maps each article to its design connection in EPOCH in detail.

The articles above mainly address the deployer, the party that operates the system. The EU AI Act also has a separate obligation track for general-purpose AI models, aimed at the provider that builds the underlying model, not at EPOCH's architecture [1]. Siemens acts here as a deployer using a ready-made model, not as a model provider, so this separate track does not directly bind EPOCH's architecture. That distinction is exactly why scope needs to be drawn carefully: the architecture can support the deployer's documentation and oversight duties without claiming that every article – including provider obligations that do not even apply here – applies identically to every component or use case.

EPOCH does not depend on classifying a specific Siemens EHS use case inside the Act's high-risk boundary; that classification depends on the system's intended purpose, a determination this thesis does not make. The Act instead functions as a forward-looking design reference, with its timetable in Appendix C and its article-by-article mapping in Table C.2.

The industrial and regulatory requirements now point to one question: not only which topology performs a task efficiently, but which topology produces the evidence structure the approved case requires.

2.3 Positioning and Contribution

No topology selector reviewed here conditions its choice on the complete evidence structure the output must carry. Existing systems select with task features, accuracy, cost, latency, or budget. Post-hoc frameworks assess whether a record is adequate after execution [34, 35, 36]. Neither family uses a human-approved regulatory and data boundary to choose the evidence shape before execution.

EPOCH instead uses:

$$\pi : (\text{case}, \text{output_profile}, \text{assurance_profile}) \rightarrow \text{topology} \in \Lambda.$$

The difference is structural. **Budget is a cost ceiling**; assurance is an evidence floor. A budget can constrain resources, but it cannot replace a required reconciliation record or distributed sign-off. EPOCH makes that evidence requirement an input to selection rather than a property checked only after the output exists.

The two axes can conflict. A low budget may favor Direct while an approved assurance requirement calls for independent verification. EPOCH does not resolve that conflict by silently lowering the evidence requirement. It exposes the tension for human review. This treatment makes assurance a constraint on the acceptable process rather than another score that can be traded against cost.

Two supporting contributions follow. First, this thesis distinguishes all three shapes through ownership. Direct fits when no independent second reading or separately mandated domain owners are required. Debate fits when the same claim needs independent verification, which requires reconciliation. Coalition fits when separately mandated results need combining, which requires synthesis.

Second, it examines that distinction specifically in regulatory compliance work – settings where the evidence behind a result, and who is responsible for it, must be checkable. That check must be available not only to an internal reviewer, but to an outside party such as an auditor or regulator. The contribution is not a claim that one topology is generally best. It is a method for selecting the shape whose artifact matches the approved case.

The supporting architecture follows from that claim. Selection occurs only after a reviewer approves the regulatory and evidence boundary. The chosen shape then determines which intermediate artifacts must exist and how they converge. Auditability is therefore present before, during, and after topology execution: the case explains the choice, the topology structures the evidence, and the stored lineage supports reconstruction.

This framing also makes the contribution testable. A case profile can be checked against the evidence approved by the reviewer. The topology decision can then be compared with the required artifact, and the run can be inspected for the promised dependency structure. Failure can therefore occur at three visible points: profiling, selection, or execution. One final score does not explain all three.

The thesis evaluates this logic under a bounded library rather than claiming a complete taxonomy of multi-agent organization. Direct, Debate, and Coalition are retained because their evidence artifacts differ. A future topology belongs in the library only if it adds a distinct assurance structure or implements one of these structures more effectively.

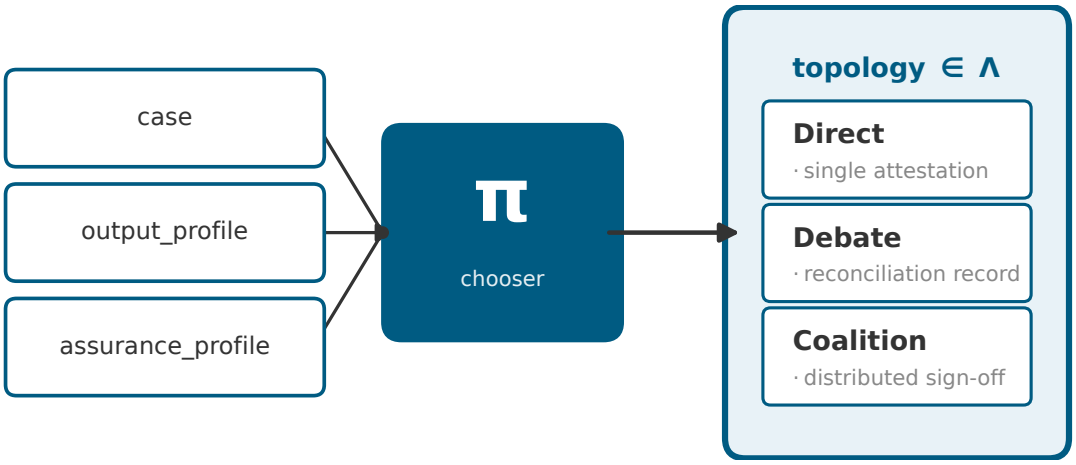


Figure 2.7: EPOCH's selector π maps the approved case profile, namely `case`, `output_profile`, and `assurance_profile`, to a topology in the library Λ. Each topology carries a structurally distinct assurance artifact.

Chapter 3 turns this selection rule into a two-stage system with human-locked evidence boundaries and separate tabular and document pipelines.

Chapter 3

Data and Methods

This chapter explains how the selection rule developed in Chapter 2 becomes an auditable system. Section 3.1 frames the artifact through design science research. Sections 3.2 and 3.3 establish the shared assurance mechanisms and two pipeline families. Sections 3.4 and 3.5 follow the system from case construction to final output. Section 3.6 closes with the evaluation design.

3.1 Design Science Research Approach

This thesis follows the design science research approach developed by Hevner and colleagues [46]. The artifact is not evaluated apart from its setting. Its requirements come from an industrial compliance problem, its design draws on Chapter 2, and its evaluation returns to a real reporting case.

The relevance cycle begins with Siemens environment, health, and safety (EHS). The operating environment requires repeatable analysis across sites, periods, regulations, and data forms. A reviewer must also be able to reconstruct how the system reached a result and who approved its scope. Auditability and human commitment are therefore design requirements, not optional quality features.

The registered examples represent four recurring tasks: a threshold assessment, a comparison between two calculation methods, a consolidation across organizational divisions, and a public-document change assessment. These tasks differ in output and ownership, and the final one also shifts the evidence from operational records to public text, so the artifact must vary its execution without losing a stable approval and provenance structure.

The rigor cycle supplies the three topology shapes and the distinction between task signals and evidence requirements: existing selectors show that architecture choice can be dynamic, but none reviewed here uses an approved assurance artifact as a selection input, a gap EPOCH turns into the policy π.

The knowledge base shapes the internal controls the same way. Research on long context

supports narrow agent inputs, work on human oversight supports concentrated approval points, and regulatory process requirements motivate reconstructable records; the resulting artifact combines these ideas instead of treating the selector as an isolated classification model.

The design cycle builds and evaluates that artifact: Stage 1 creates a human-approved case and selects a topology, and Stage 2 binds the decision to an execution shape and produces the requested output. The evaluation first controls the choice of large language model (LLM), then asks domain experts to assess a public-document case.

Development itself follows repeated build-and-check iterations: structured outputs are inspected through their declared schemas and shelves (Section 3.2.2), deterministic stages are reproduced from fixed inputs, and complete runs are reviewed at human gates. This cycle does not establish empirical effectiveness by itself; it produces a stable artifact that the evaluation design can examine.

3.2 System Foundations

The tabular and document pipelines process different evidence, but they share one assurance foundation. Both separate language-model judgment from deterministic operations – whether an output is an LLM's interpretation or a value computed exactly by code is never blurred, a distinction Section 3.2.1 covers in detail. Every agent is also restricted to only the inputs explicitly given to it, with no direct access to another agent's output. Both pipelines record which inputs and which agent produced each output, and place human decisions where an approval changes what later agents may treat as true – both described in detail in Section 3.2.2.

3.2.1 Agent Roles and Deterministic Boundaries

The system uses three agent types. LLM agents interpret text or structured evidence and return schema-constrained judgments. Deterministic agents parse, join, calculate, or summarize through reproducible code. Hybrid agents combine both forms of work while keeping numerical operations and linguistic judgment distinct.

This classification controls what the implementation may validate. A schema checks the structure and types of an LLM response, but code does not compare its prose with a hidden answer key. Deterministic outputs can be checked by repeating the same operation. A hybrid output preserves that boundary by showing which result came from code and which judgment came from the model. Table D.1 states this validation boundary for each type.

Roles are grouped by purpose rather than by one use case, covering regulation review, data mapping, case profiling, topology selection, and downstream production. Table D.3 maps each group to its implementation identifiers.

The library makes these roles reusable. A later case can reuse an agent, upgrade it for a specialized task, or add a missing capability. Reusability supports scale, but it does not allow

an agent to broaden its own authority. The pipeline still decides which inputs reach which role, and exact interfaces and family-specific bindings stay in Appendix D.

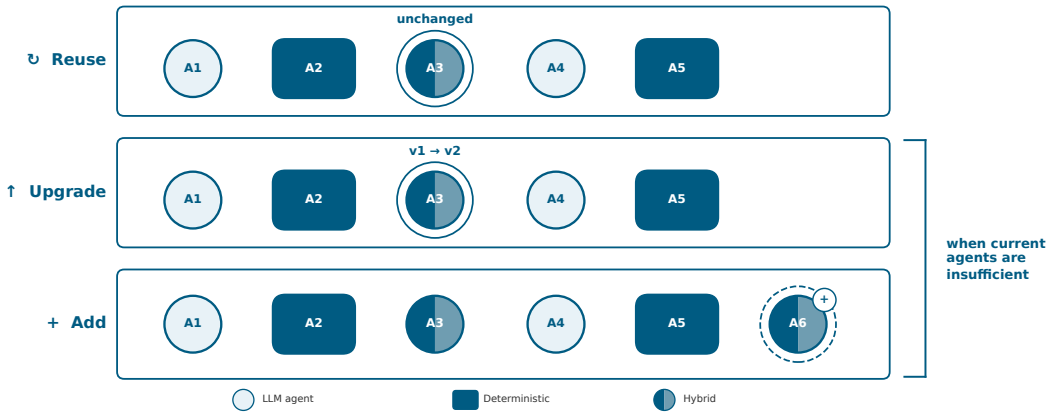


Figure 3.1: The agent library supports three operations: reuse an existing agent unchanged, upgrade it as a task-specific version, or add an agent for a missing capability.

3.2.2 Isolated Memory and Human Commitment

EPOCH organizes persistent outputs as agent-local shelves. An agent can read only the material passed to it through the pipeline; it cannot inspect another shelf on its own. This creates an *isolated memory*: a deliberately bounded view whose contents can be traced to a named upstream artifact.

The boundary becomes authoritative through human approval. This approval is the pipeline's first gate (Section 2.2.4): the reviewer fixes the exact scope before execution proceeds at all, not afterward. In the tabular family, the reviewer approves the mapping between regulatory needs and available data, then approves the supported legal scope. In the document family, the reviewer approves the regulation comparison, report mapping, and their joined action record. Later stages use those approved records instead of reopening the raw request.

The tabular handoff illustrates the principle: a regulation agent may identify a need broader than the available data. In UC2.1, a compliance memo comparing two CO_2e methods for large manufacturing sites, the regulation agent proposes checking which industrial activity category a site falls under, a classification drawn from Annex I of the EU Emissions Trading Directive. The site database has no matching field for that classification, so a mapping agent can only propose what the operational catalog exposes, marking the Annex I field as unmapped, and the reviewer approves that narrower set. The later scope review must preserve that decision rather than restore the broader candidate.

The document branch applies the same rule to text. Independent readers may disagree with deterministic matching candidates, and a later comparison may revise both proposals after reading the authoritative paragraphs. Only the joined, human-approved result becomes downstream evidence. Earlier proposals remain visible without competing with the active boundary.

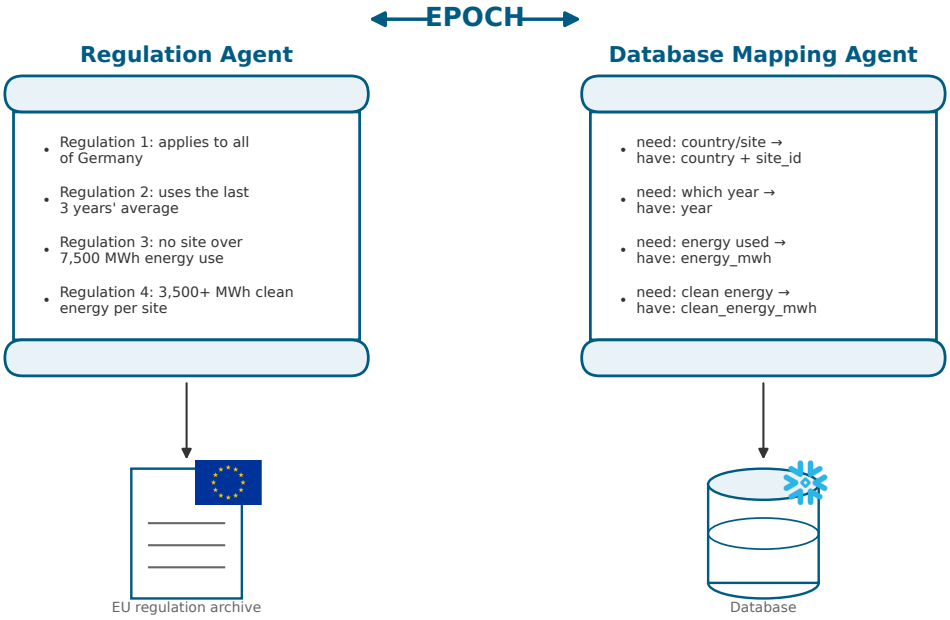


Figure 3.2: The isolated memory shelf system. A regulation-analysis role reads the European Union regulation archive, while a database-mapping role reads the operational catalog and receives only the approved regulatory handoff. Neither role can reach the other's source. The implementation codes visible in the figure are mapped to their full contracts in Appendix D. The rules and data answers are illustrative, not the recorded output of an actual pipeline run.

The registry is the first input that starts the entire agent chain – a demand snapshot whose fields Section 3.4.1 defines (a worked registry entry appears in Appendix E.1). It still describes what the user wants. It does not become a second source of truth after the evidence boundary is approved. If the original demand and approved evidence diverge, later stages carry the difference as a limitation. They do not correct the approved scope toward the request. **Assurance is the approved choice.**

3.2.3 Artifact Provenance

Audit-awareness concerns how an output was produced, not whether its conclusion is automatically correct. Each agent stores its result in a shelf tied to the case. The active record represents the approved or current artifact, while earlier and rejected candidates remain archived.

Each record names four nested things: the request it belongs to, the run that produced it, the agent that wrote it, and the record itself. It also carries a SHA-256 digest of its payload, so altering one character changes the digest and binds the record to its content rather than to a filename. Every upstream artifact the agent consumed appears in that same record, named by identifier, digest, and run, so following those references rebuilds the evidence chain without rerunning a single LLM call. Table D.2 (Appendix D) lists one complete record. Figure 3.3 shows CP1's five-shelf read in full.

Promotion and rejection both leave evidence: an earlier active record moves to archive rather

than disappearing, and a rejected candidate is archived with its reason instead of being deleted. Appendix D describes how the shelf stores that history and how the orchestration layer alone reads across shelves.

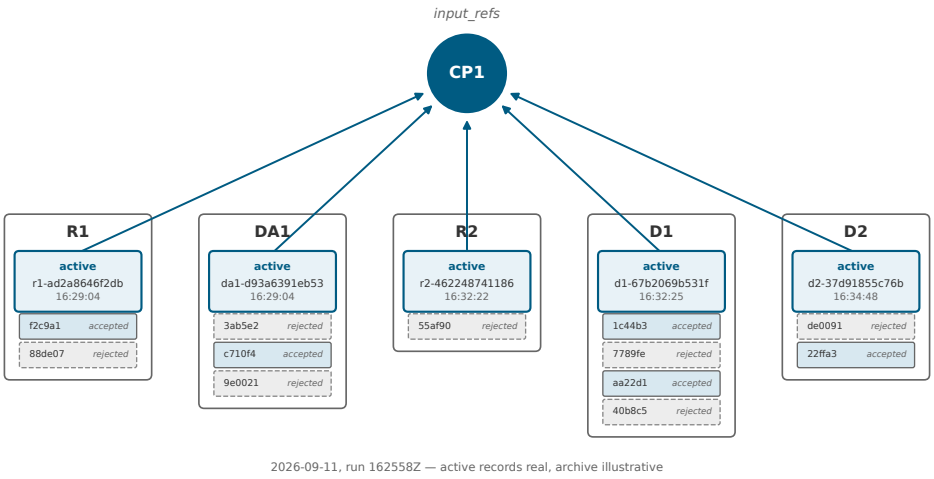


Figure 3.3: CP1 is the orchestration layer's real five-shelf reader: it names one active record from each of R1, DA1, R2, D1, and D2 in `input_refs`, all from the same run. Each shelf also keeps an illustrative archive of earlier candidates, accepted (once active, later superseded) or rejected (never promoted), and no two shelves carry the same history depth. CP1 always reads the current, active record at the top of a shelf, never the archive beneath it, however deep that archive runs.

3.2.4 External Perspective at Stage-Final Gates

A human reviewer can inherit a narrow evidence frame when every internal component uses related sources. An external perspective agent therefore adds an outside reading immediately before the final human gate of each stage. It searches only a fixed set of official institutions (Appendix D.3 lists all of them), rather than the open web whose retrieved citations often cannot be verified against their source [47], and marks whether a source supports, challenges, extends, or exposes a gap in the candidate.

This channel is advisory by construction. The reviewer sees the commentary beside the candidate, but no downstream agent consumes it. The external record stays on its own shelf, and the candidate itself is never changed. Outside commentary can therefore inform approval without entering isolated memory.

The two stage-level instances answer different questions, one reviewing the evidence boundary and the other the completed deliverable (Appendix D.3 states the exact pairing rule); neither selects a topology, changes an action, or writes into another agent's artifact. The nearest comparable system uses a similar retrieval step, but wires it differently: several agents retrieve regulatory evidence and use it directly to detect, and even revise, a requirement's compliance status, with no human approval before that verdict takes effect [48]. EPOCH keeps that same kind of retrieved evidence outside isolated memory instead: it can inform the human's approval,

but it never becomes the verdict itself, and it never substitutes for that approval.

3.2.5 Blind Passes and Anchoring

An LLM role can inherit the same narrow frame a human reviewer risks: reading a deterministic candidate before forming its own view lets that candidate anchor its judgment rather than merely inform it. RD2, for example, might propose that a provision of the European Sustainability Reporting Standards (ESRS) is unchanged in the 2026 revision (Section 3.4.3 gives UC4's full context). An agent that reads that candidate first, without forming its own independent reading, could simply confirm that label even when the source texts actually show a substantive amendment. EPOCH answers this by splitting an at-risk role into two passes instead of one agent, a distinct agent formation called a blind-pass pair, named X.1 and X.2.

The first pass reads only the authoritative source text and never sees the deterministic candidate; the second pass reads that independent proposal beside the candidate and the same source text, and only this second pass assigns the binding status. Neither reading is ground truth, and the source text can override both. Section 3.4.3 applies this split to two document-family role pairs; Appendix D records the exact payload contracts each pass receives and why the structure extends to any future role that carries the same anchoring risk.

3.3 System Overview

EPOCH separates construction from execution. Stage 1 turns a registry demand into an approved case profile and topology decision. Stage 2 binds that decision to an execution structure and produces the requested output. The split keeps the reason for choosing a process distinct from the result produced by that process.

The system supports two pipeline families. Tabular cases select and calculate evidence from operational records. Document cases compare public regulation and report text. Evidence form selects the construction path; a numbered use-case label is not a hidden routing rule.

Both paths converge on the same two roles. The case profiler describes what the approved case requires through its subject, output form, and assurance need. The topology selector maps that profile to a shape in Λ. Their shared position supports selection across data forms without pretending that tables and legal paragraphs require the same preparation.

The common stage names describe epistemic functions rather than identical agent chains. Both families discover evidence, form a boundary, profile a case, and produce a reviewed result. Their internal roles differ because the evidence supports different deterministic operations and interpretive risks.

The two-stage boundary also limits feedback. Stage 2 receives a topology and an approved evidence package; it does not send an interpretation backward to change the scope that justified its own selection. A later run may begin from a revised demand, but the current run retains one

Stage function	Tabular family	Document family
Evidence discovery	Regulation needs and runtime catalog	Regulation versions and report sections
Boundary formation	Approved mapping and supported findings	Approved change, exposure, and joined actions
Preselection evidence	Quality-qualified data and derived facts	Approved disclosure-requirement join
Stage 2 output	Selected topology and requested deliverable	Three risk postures and reconciled report

Table 3.1: The shared stage functions and modality-specific implementations of the two pipeline families.

visible decision history.

The two current families are a product of the everyday automations requested so far, not a ceiling on the architecture itself: agents built for one family stay recombinable into a hybrid pipeline for the other, a case examined further in Appendix C.

3.4 Stage 1 — Auditable Case Construction

Stage 1 establishes the evidence on which topology selection may act. It begins with the user's demand, tests that demand against the available source form, and ends with an approved topology decision. The tabular and document branches use different construction methods, but both expose uncertainty before it can become an execution instruction.

Table 3.2 introduces the five use cases named throughout this chapter and Chapter 4, each spelling out the regulation it names on first use.

Use case	Request	Family / topology
UC1	Screens every German Siemens site against the German Energy Efficiency Act (EnEfG). Sections 8 and 16 of that act require companies above a yearly energy-consumption threshold to run an energy-management system and report waste heat; this screen lists which sites approach or exceed those thresholds.	Tabular / Direct
UC2	A quarterly compliance memo for large manufacturing sites covered by Phase 1 of the EU Emissions Trading System (ETS, Directive 2003/87/EC): it calculates each site's CO_2-equivalent emissions by two independent methods, one derived from energy use and one directly reported, and compares them.	Tabular / Debate
UC2.1	The same request as UC2, but on an annual grain over three years (FY2022–FY2024) instead of quarterly – used to show the dual-method comparison recurring across several years.	Tabular / Debate
UC3	Under the EU Corporate Sustainability Reporting Directive (CSRD)'s European Sustainability Reporting Standards (ESRS), Climate Change disclosure (E1): each Siemens EMEA business unit calculates and signs off its own Scope 2 emissions figure independently, and the group then consolidates those separately owned figures into one total.	Tabular / Coalition
UC4	Assesses how the ESRS environmental requirements, amended in 2025 and further revised in 2026, change what Siemens' FY2025 sustainability statement needs to disclose.	Document / Debate

Table 3.2: The five use cases used as worked examples in this chapter and evaluated in Chapter 4.

Figure 3.4 follows the expanded dual-method tabular case, called Use Case 2.1 (UC2.1) in the implementation (Table E.1 gives its full registry entry). An external commentator reaches the scope-approval gate as advisory evidence only; it never enters the isolated memory that scope review (R2) and data mapping (DA1) lock. Appendix D provides the complete identifier map.

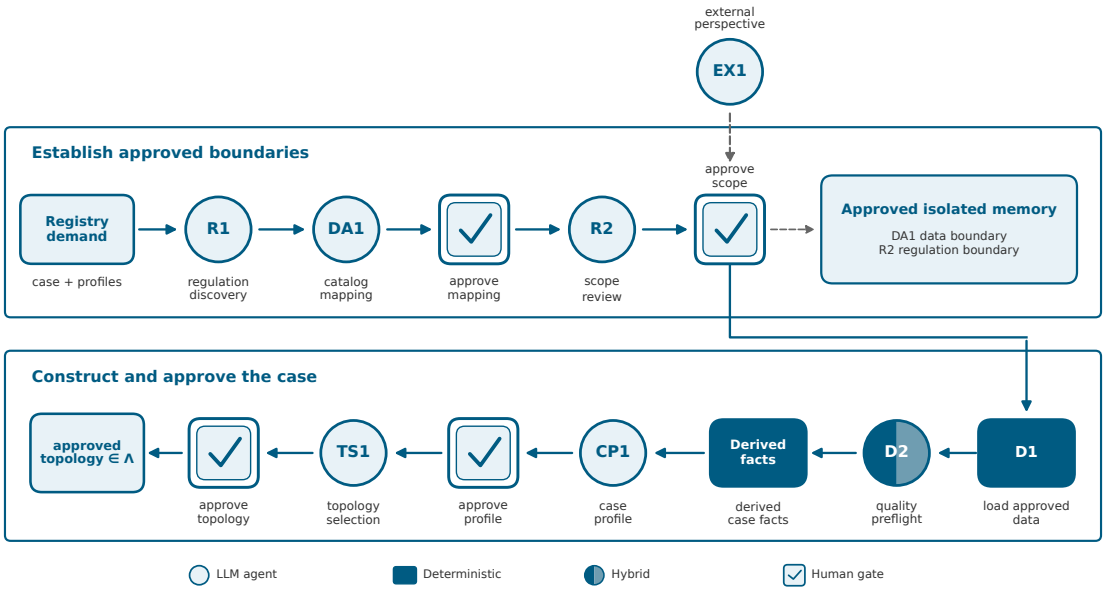


Figure 3.4: Agent and human-gate sequence that turns a registry demand into an approved topology in Λ, using UC2.1 as the working example. External commentary reaches the scope-approval gate as advisory evidence and does not enter the isolated memory. The anonymized execution behind Figure 3.4 can be inspected from its Stage 1 overview at `https://emrecanulu.com/epoch/uc2-1/overview/`.

3.4.1 Demand Orientation and Modality

Every run begins with a registry entry naming the natural-language request, expected output, preferred output form, assurance need, and known scope. These fields are a demand snapshot, not an answer key: Stage 1 may still show that part of the request is unsupported or that the available evidence permits only a narrower claim.

The registry also declares the pipeline family. Tabular cases use protected operational records and write their run artifacts outside version control; document cases use tracked public texts and write public experiment artifacts. The configuration layer chooses the location from evidence form, so agents never hard-code a numbered case or an absolute path.

A new tabular case therefore inherits catalog mapping and data-quality assessment, while a document case inherits text parsing and matching. Both remain subject to the same provenance, approval, and final-agent gate rules.

3.4.2 Tabular Scope and Evidence Construction

The tabular branch first asks what the regulation may require. Regulation agents read the demand and relevant legal sources, identify candidate findings, and express their operational data needs (Table E.2). This pass is deliberately broad, so a limited database catalog cannot decide which obligations exist before the regulation has been read.

A data-mapping agent then compares those needs with the runtime catalog, proposing available tables, fields, joins, filters, and measures while leaving unsupported needs visible. A human reviewer approves, removes, or remaps the proposal (Table E.3); that decision fixes the data boundary, and deterministic execution later follows the approved handoff exactly.

The next regulation review asks whether the approved operational evidence supports each legal finding. Supported findings move forward, while blocked and excluded findings remain in the audit record (Table E.4); a broad finding may be divided so an assessable part survives without hiding the unsupported remainder.

Human approval locks that regulatory boundary. Deterministic data roles then load and transform the approved evidence, and a quality role measures completeness, duplicates, validity, consistency, timeliness, coverage, and reconciliation (Table E.5). A language-model judgment relates observed quality conditions to the approved claims without imputing values or changing the law.

This order keeps three decisions separate. Regulation discovery is not limited by database availability, data execution cannot revise the approved mapping, and quality is judged only against claims that survived review. The result is a case-evidence package with explicit legal and operational limits.

3.4.3 Document Scope and Evidence Construction

The real-world document evaluation case, called Use Case 4 (UC4) in the implementation, requires a parallel construction path: it compares the 2025-amended European Sustainability Reporting Standards (ESRS) environmental requirements with the 2026 Revised ESRS, then assesses their effect on Siemens' fiscal year 2025 Sustainability Statement, and no stage in this branch reads confidential operational data. Deterministic preparation first extracts disclosure requirements and report sections from the tracked corpus, and also proposes possible links between old and new requirements and between requirements and report sections; these candidates make the corpus easier to inspect, but they do not decide regulatory change or reported status.

Two blind-reading pairs (Section 3.2.5) then address different questions: `RC1.1/RC1.2` compares the old and new regulation, while `RM1.1/RM1.2` checks how the earlier requirement appears in the Siemens statement. The two questions remain separate because a change in regulation does not show whether the company reported the earlier requirement. A deterministic join (`RD3`) combines their final classifications, retains unmatched requirements on either side, and assigns the reporting action implied by the two statuses; interpretive judgment determines the statuses, and reproducible code determines the join and action category. Figure 3.7 in Section 3.6.2 draws this shape as a triangle: the 2025 baseline anchors both real comparisons, and the join reconstructs the one relationship no reader ever inspects directly.

One human gate reviews the binding comparisons and joined action table as a package; first-pass readings remain stored for reconstruction but do not crowd the decision surface. Ap-

pendix D records the blind-pass, join, and status contracts, and Appendix F records the source hierarchy and frozen run. The complete public-text comparison path can be inspected in the UC4 Stage 1 overview at https://emrecanulu.com/epoch/uc4/overview/.

3.4.4 The Human-Locked Assurance Boundary

Both branches lock this boundary as Section 3.2.2 describes, differing only in what the approved package contains: the approved mapping, supported findings, and quality-qualified data for tabular cases, the approved regulation-to-report comparison and joined actions for the document case. Appendix D details how the gate display narrows that package for review.

3.4.5 Case Profiling and Topology Selection

The case profiler converts the approved package into a selection profile: a tabular profile reflects supported findings, approved data, quality, and derived facts, while a document profile reflects the approved comparison and action summary. The original demand still defines the requested deliverable, but approved evidence remains the assurance authority.

The profile separates three kinds of signal. Routing signals describe the required assurance artifact and whether one claim needs independent methods or several organizational owners; escalation signals describe urgency and consequence; context signals record task type, remaining uncertainty, and the confirmed output form. The assurance artifact is the central routing signal, since residual uncertainty does not by itself require Debate and several fields do not by themselves require Coalition, so the profiler asks whether the case requires one accountable line, independent verification of one claim, or separately mandated contributions.

A human reviewer approves this profile before the topology selector uses it. The selector scores Direct, Debate, and Coalition, explains which approved signals drove the choice, and opens a separate human gate before Stage 2 begins; the score distribution keeps close alternatives and conflicting signals visible instead of hiding them behind one label.

The separation preserves two questions: the profiler states what the case requires, while the selector chooses the execution shape. Combining them would hide whether an error arose during profiling or during the mapping from that profile to Λ. Table D.5 in Appendix D records the generic contract; Appendix E.6 reproduces the worked UC2.1 profile and topology decision, and the selection record itself is published at https://emrecanulu.com/epoch/uc2-1/ts1/.

3.5 Stage 2 — Topology-Bound Execution

Stage 2 converts an approved selection into a reader-facing result. It no longer decides the regulatory or data boundary; its task is to apply an execution shape to approved evidence, preserve that evidence through interpretation, and place the completed output before a human reviewer.

Figure 3.5 continues the dual-method tabular case from Figure 3.4 into its Debate binding. Two independent method readings (Table E.8) meet at reconciliation (Table E.9), and the finalizer converts that record into the requested memo (Table E.10). External commentary reaches only the final human gate (Appendix E.11), never the deliverable itself.

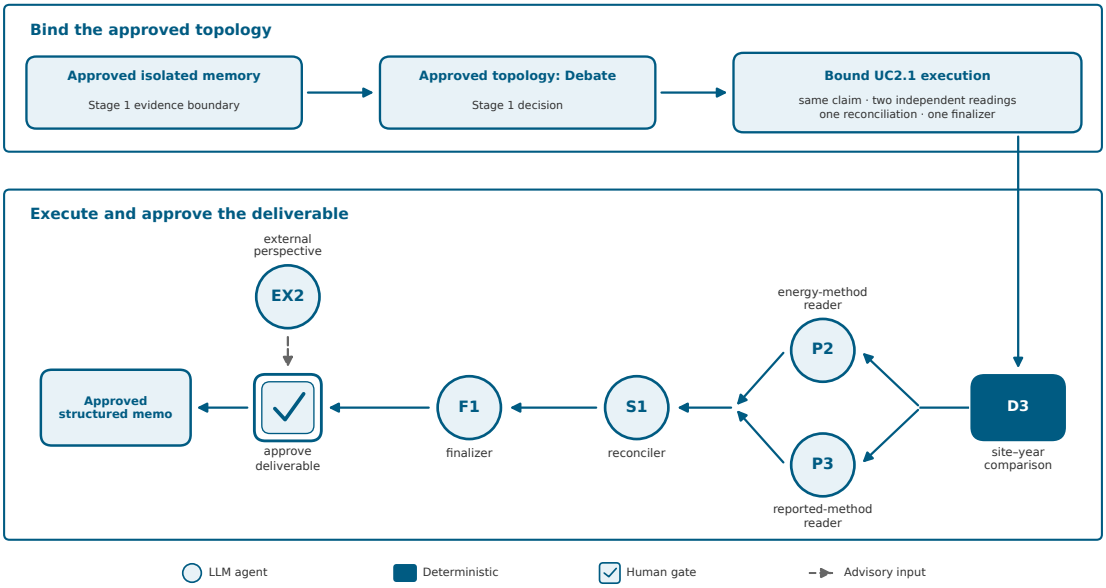


Figure 3.5: UC2.1 Stage 2 execution after the approved Debate selection. The deterministic comparison branches into two independent method readings, which first meet at reconciliation. The finalizer converts that record into the requested structured memo. External commentary reaches the human gate as advisory evidence and does not enter the approved deliverable. Its topology-bound continuation is available in the corresponding Stage 2 overview at https://emrecanulu.com/epoch/uc2-1/stage2/.

3.5.1 Tabular Topology Bindings

Every tabular path begins with a deterministic result builder (Table E.7), which applies the registered computation to approved data and preserves the relevant grain, such as a site-rule, site-year, division, or portfolio. The selected topology does not change that formula; it changes how the resulting figures are interpreted and assembled.

In the dual-method case, Debate's independent readings (Section 2.1.3) become two readers who each begin from a different method but see the same approved boundary and paired values; a reconciler, the first role allowed to compare them, preserves both readings, explains material differences, and passes one reconciliation record to the finalizer. Direct (Section 2.1.2, UC1) and Coalition (Section 2.1.4, UC3) bind the same way, trading the reconciler for a single interpreter or a coalition synthesizer whose contributions follow the approved data rather than a fixed agent count.

These bindings differ through dependency structure alone: only the finalizer opens a human

gate in any of the three, while intermediate interpretations and syntheses stay inspectable on their own shelves. Exact batching, inputs, outputs, and implementation identifiers for all three are in Table D.6 (Appendix D).

3.5.2 Real-World Document Evaluation Execution

Document evaluation does not run identically for every request, since a different demand may call for different agents or reuse agents from other use cases, and the same flexible agent library and topology system already covers this (Section 3.3). UC4's demand asks how the ESRS environmental requirements, amended in 2025 and revised again in 2026, change what Siemens' FY2025 sustainability statement needs to disclose (Table 3.2).

That assessment produces a comparison table read from conservative, balanced, and maximum-assurance postures, fixing this use case's three-reading structure: the three assessors may assign different urgency to the same approved action table, but none can change the underlying reporting action, so risk posture stays observable without a second regulation comparison. Debate was the topology decision here, and the three mandated readings converge through a document finalizer that reconciles the three priorities, exposes disagreement, and produces a requirement-level comparison with a business summary: a deterministic rollup supports the standard-level view, while language-model reasoning explains the reconciled result, keeping the required action separate from the agreed urgency.

The chain stays interpretable because every assessor reads the same action classification, so divergence reflects risk posture rather than a different reconstruction of the regulation-to-report join. Exact priority values, lineage fields, and finalizer contracts are in Table D.6 (Appendix D). The three risk-posture readings and their reconciled output are visible together in the UC4 Stage 2 overview at https://emrecanulu.com/epoch/uc4/stage2/. Figure 3.6 shows this chain end to end.

3.5.3 Final Review and External Commentary

When a pipeline family ends, it reaches the human approval gate immediately after the stage-level external perspective agent (Section 3.2.4) adds its advisory outside view on the exact candidate. Approval promotes the final candidate and completes the run; rejection archives it while leaving approved upstream evidence intact (Appendix D details the shelf mechanics behind both). Placing the gate after the finalizer ensures that the reviewer sees the requested deliverable rather than an unfinished topology-internal result. Appendix D lists the approved final deliverable each use case actually produced at this gate.

Stage 2 consequently leaves two connected trails. The execution trail explains how an approved topology became a dependency structure. The artifact trail links the final interpretation to each stored input. The evaluation uses both trails to judge the result and whether the system made its basis visible.

3.6 Evaluation Design

The evaluation has two complementary parts: the first repeats each use case several times against a frozen gold reference and asks whether the pipeline's own architecture held during the run, and the second places the selected system in a public real-world setting and asks domain experts to judge the result. Neither part substitutes for the other: the first checks structural discipline against a known-correct case, and the second checks whether that discipline produces something a human would actually trust.

3.6.1 Part I: Gold-Anchored Pipeline Evaluation

Gold is not the answer this system is required to reproduce. It is one frozen example of a structurally correct run, approved by a human reviewer at the same gates that govern every other run. For each use case, gold's input is the approved DA1 mapping or R2 finding set a reviewer actually signed off inside the pipeline, and gold's output is the checkpoint chain that approved input produced.

A run that surfaces a different, well-grounded regulation set, a different LLM-assigned identifier, or a broader field list than gold's is not thereby wrong: it is judged against its own cited source, never against gold's wording, counts, or list membership. Table 3.3 summarizes the five tests this principle produces, applied at least five times per use case against the same frozen gold, together answering whether the pipeline produces a substantive result, whether topology and agent selection stay consistent, and whether the run holds together end to end.

Test	What it measures	Unit of report
`T1`	Selected topology matches the expected shape and the exact agent set specified ran, no fewer and no more	run → use case
`T2`	Five architectural invariants held during the run, among them checkpoint order and `EX1`/`EX2` adjacency	run → use case
`T3`	Checkpoints are complete and their content is grounded	per checkpoint
`T4`	Seven-item, gold-independent checklist confirming the pipeline completed cleanly, among them a resolvable provenance chain and a recomputable payload hash	run → use case
`T5`	No agent outside the selected chain, and none from the other pipeline family, ever fired	run → use case

Table 3.3: The five tests EPOCH's evaluation applies to each use case's run set, adapted from `evaluation/TAXONOMY.md`.

`T3` (Output Substantiveness) first checks that every checkpoint exists and that its required fields are populated, then produces two complementary assessments: a field score against the comparison class fixed in the use case's contract, and a defect record from a closed list checked regardless of that class. The two may flag the same problem, but they serve different reporting purposes; Appendix A.1 gives both the four comparison classes (Table A.1) and the five glaring defects (Table A.2) in full.

`T3`'s scoring surfaces in a three-part cell: the ratio of structurally complete checkpoints to the

total, the content-grounding check's result, and the count of confirmed glaring defects. Table 4.2 in Chapter 4 reports this cell format for every use case.

Claude Opus 5 served as the single LLM judge and executed all five tests. Given only a use-case name, the shared prompt resolves the registry identifier, frozen gold package, expected topology, checkpoints, and run set from disk. Appendix A.1 gives the taxonomy, while Appendix A.3 identifies the five frozen packages.

Appendix A.2 reproduces the executing prompt in full. Before opening a run's `TS1` record, the judge derives and records the expected `T1` topology independently. It also excludes the gold source run and uses a comparison class fixed before the run is inspected.

This part selects no model. The scored, five-repetition run set reported here ran on `claude-opus-5` across all five use cases. After the default backend moved to `claude-opus-4-8`, one additional run per use case was captured on the new backend and scored against the same frozen gold as a confirmation pass, not a comparison. This test examines whether the same architecture keeps making the same structural decisions when it runs repeatedly against the same approved input. Appendix B traces the backend history that led to this choice.

3.6.2 Part II: Real-World Use-Case Expert Evaluation

The expert evaluation asks whether Siemens colleagues who did not build the pipeline would trust and use its completed output. The unit of evaluation is the report presented at the final human gate: `F1`, a topology-agnostic pure-LLM finalizer that composes the reader-facing deliverable from whichever chain fed it, for the four tabular use cases, and `F2`, a hybrid agent that turns the three persona reads into a final verdict and adds a deterministic summary rollup, for the document use case. Reviewers assess that finished deliverable, not gold or the intermediate agent records, and five use cases produced six reviews because two colleagues assessed UC4 independently.

Ten fixed questions govern every review. Table 3.4 states them in the order shown to the reviewer. Each answer uses a five-point scale with fixed descriptions for scores `1`, `3`, and `5`; scores `2` and `4` mark positions between those anchors.

Criterion	Question shown to the reviewer
Correctness and grounding	Are the claims accurate and supported by the cited sources or figures?
Scope fit	Does the deliverable answer the requested question at the right boundary?
Clarity and usability	Could the reviewer act on it without asking the author what it means?
Honesty about limitations	Does it state what could not be verified, where data were thin, and where uncertainty remains?
Overall confidence	Would the reviewer rely on it in an actual work decision?
Siemens implementation fit	Could Siemens integrate the output into its existing systems and workflows?
Flexibility	Could the output adapt to different scopes, data, or regulatory changes?
Evidence structure fit	Is the evidence organized at the right depth for the output and its assurance needs?
Auditability and human control	Can key conclusions and the effect of human review be traced?
Approach improvement	Does the approach improve traceability and confidence without unnecessary complexity?

Table 3.4: The ten questions used for every expert review.

Beyond the ten scores, the form asks how long an expert would need to produce an equivalent report manually, with required written fields for what the work did best and worst and optional fields for explanations and further comments, separating a numerical judgment from the reason behind it. Every review packet follows the same fixed sequence: a scoring and written-assessment form on page 1, the case-specific system design on page 2, then the registry request, scope, and selected run's final `F1` or `F2` output, closing with a References and Assurance page naming sources, provenance, model, run, and gate mode. Table 3.5 shows how the result view changes with the use case; Section G.1 reproduces the shared first page, and Sections G.3–G.7 reproduce the remaining case-specific pages.

Case	Report	Result view
UC1	`F1`	Site-by-rule priority screen with status, threshold gap, signals, and finding.
UC2	`F1`	Quarterly site reconciliation with two methods, their difference, scope decision, and finding.
UC2.1	`F1`	Annual version of the same two-method reconciliation view.
UC3	`F1`	Division-level consolidation with subtotal, period, data status, provenance, and finding.
UC4	`F2`	Requirement-level ESRS change, Siemens reporting status, impact, decision, and confidence, followed by a standard summary.

Table 3.5: The case-specific result view inside each otherwise fixed peer-review packet.

Each packet draws from one preselected, recorded `claude-opus-5` run and remains fixed after

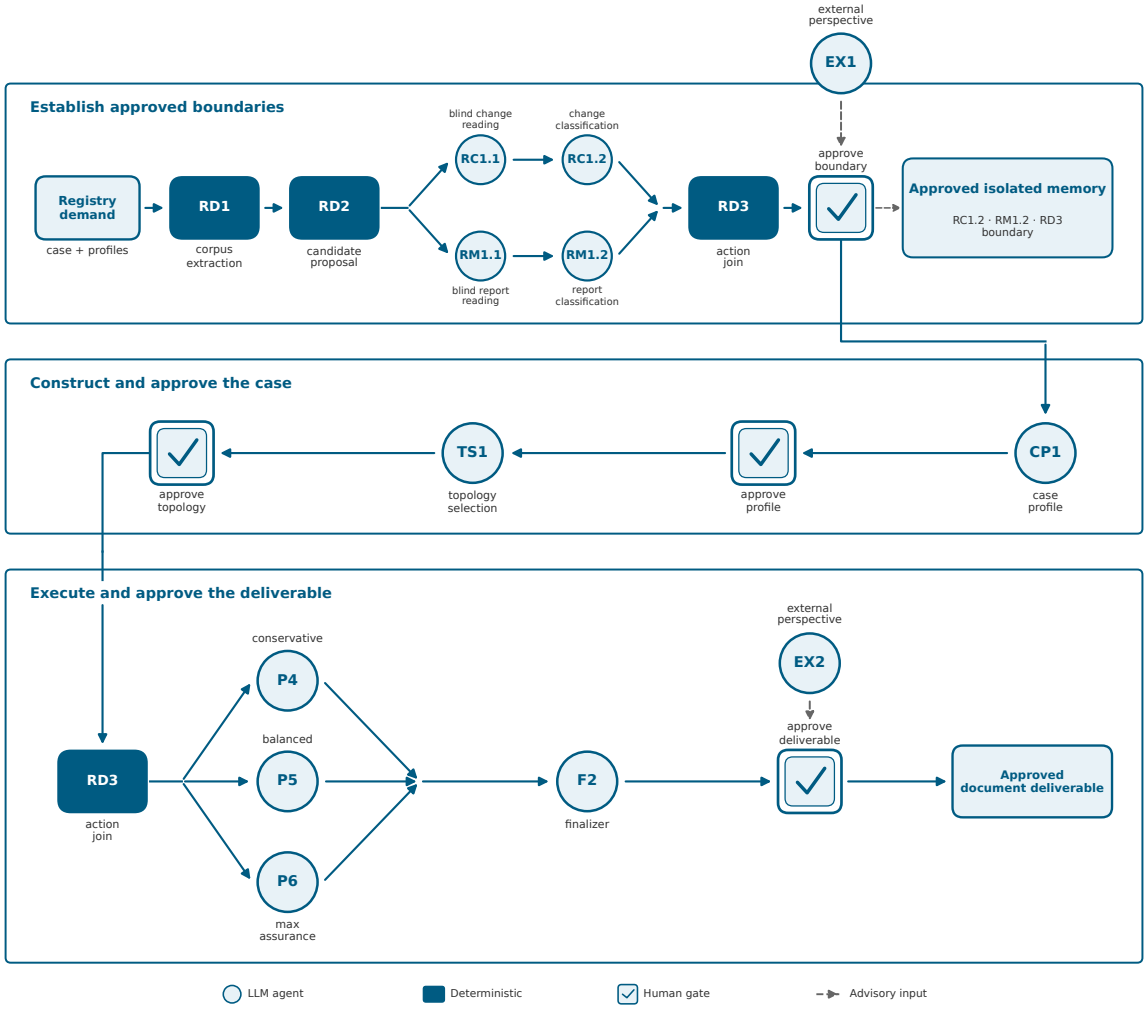


Figure 3.6: UC4's complete chain: Stage 1 boundary construction (top), case construction (middle), and Stage 2 topology-bound execution (bottom), from registry demand to the approved document deliverable.

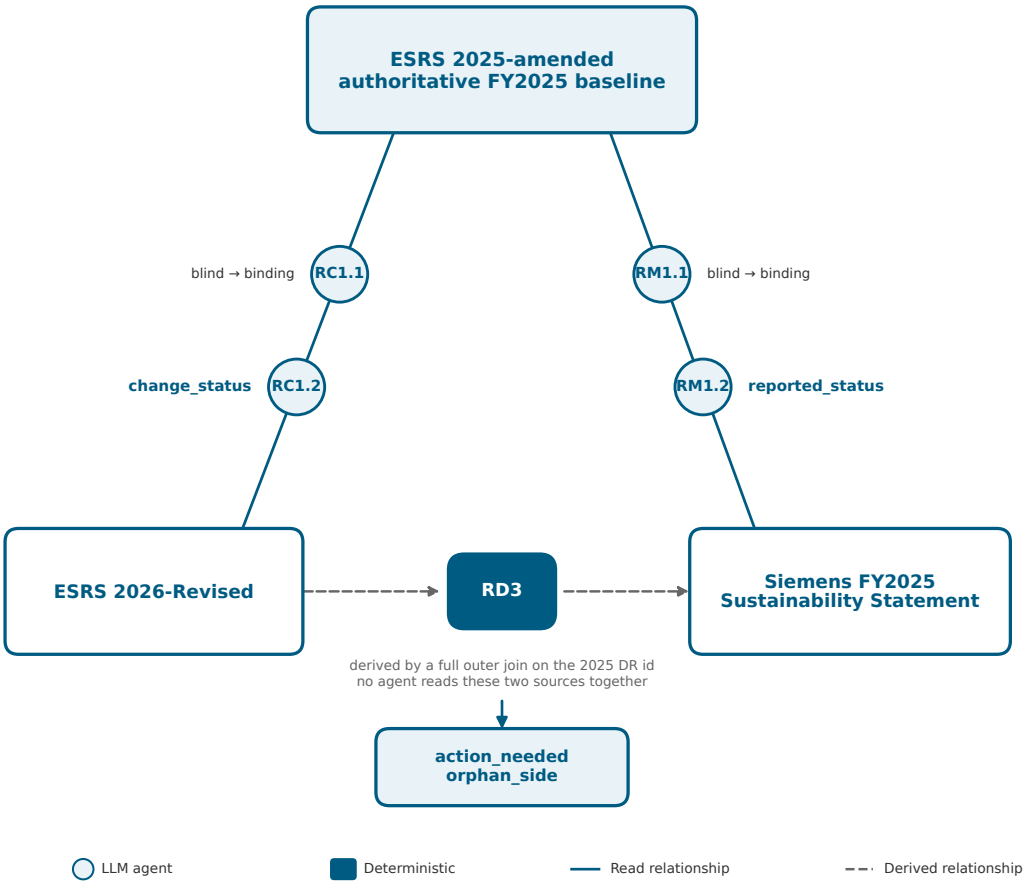


Figure 3.7: UC4's registry compares three sources: the **2025-amended ESRS** baseline, the **2026-Revised ESRS**, and **Siemens' FY2025 Sustainability Statement**. The 2025 baseline sits at the apex because it is the shared key both real comparisons join on: `RC1.1`→`RC1.2` produces `change_status` against the 2026 text, and `RM1.1`→`RM1.2` produces `reported_status` against the Siemens statement. The dashed base edge marks the relationship no agent reads directly. `RD3` derives it by joining the two verdicts on the 2025 disclosure-requirement id, producing `action_needed` and `orphan_side`.

review begins, so no score triggers another run or a revised report, and one reviewer assesses one use case in one pass without comparing it with another case. Appendix G reproduces the shared first page once and the remaining masked pages for all five use cases.

Returned forms are transcribed before publication: reviewer names are reduced to initials, site names, division names, and numerical values are masked in the four tabular packets, and UC4, which uses public material, needs no masking. The scores, manual-time estimates, and written assessments produced by this procedure are reported in Section 4.3.

Chapter 4

Evaluation and Results

This chapter reports what Section 3.6's two-part evaluation found. Section 4.1 states the run sets each result below is drawn from. Section 4.2 reports the gold-anchored tests against a frozen reference. Section 4.3 reports the returned expert reviews.

4.1 Evaluation Setup

Two run sets exist for every use case, on two different backends. The first set ran `claude-opus-5` through the Foundry Anthropic backend, five times per use case where possible, following Section 3.6.1's method: `uc1_enefg_threshold_check` (Direct), `uc2_ets1_scope_memo` (Debate), `uc2_1_ets1_scope_memo_annual` (Debate), `uc3_csrd_scope2_measure` (Coalition), and `uc4_esrs_2026_impact` (Debate, document family). Both human gates were auto-approved in every run; `EX1`/`EX2` still ran wherever the feature was present. The second set is a single confirmation run per use case on `claude-opus-4-8`, captured after a snapshot-writing defect discovered in the first set was fixed in code. Table 4.1 states, per use case, how many runs were captured on each backend.

Use case	Backend	Topology	Runs capt.
UC1	`claude-opus-5`	Direct	5
UC1	`claude-opus-4-8`	Direct	1
UC2	`claude-opus-5`	Debate	5
UC2	`claude-opus-4-8`	Debate	1
UC2.1	`claude-opus-5`	Debate	5
UC2.1	`claude-opus-4-8`	Debate	1
UC3	`claude-opus-5`	Coalition	5
UC3	`claude-opus-4-8`	Coalition	1
UC4	`claude-opus-5`	Debate	5
UC4	`claude-opus-4-8`	Debate	1

Table 4.1: Runs captured for each use case and backend (Table 3.2 states what each use case requests).

Five repetitions is not a power calculation; it is where repeating a case stopped changing the output in any way that mattered, not a target fixed in advance. That justification is a real limitation, not a formality: a sample of four or five scored runs cannot establish a failure rate, only surface whether one occurred. The `claude-opus-4-8` set is a single confirmation run per use case, locked to that scope by design. It checks whether the fix held on a different backend, not whether that backend is more or less stable than `claude-opus-5`.

4.2 Gold-Anchored Results

Table 3.3 named five tests answering three questions: whether topology and agent selection are correct, whether the run holds together end to end, and whether the output is substantive. Gold is judged only as a structural template, never as a content answer key. A run that surfaces a different, well-grounded regulation set or a broader field list than gold's is not thereby wrong. It is judged against its own cited source. Table 4.2 reports T1–T5 for both run sets side by side.

Use case	T1	T2	T3. Complete	T3. Grounded	T3. Defects	T4	T5
claude-opus-5, $n = 4$–5 per use case							
UC1	5/5	5/5	10/10	1.9–2.0/2	2	n/a	5/5
UC2	0/4	4/4	12/12	2.0/2	1	0/4	0/4
UC2.1	5/5	4/5	12/12	9/10	0	4/5	5/5
UC3	0/5	5/5	11/11	2.0/2	0	0/5	0/5
UC4	4/4	1/4	13/13	2.0/2	0	4/4	4/4
claude-opus-4-8, $n = 1$ confirmation, captured after the snapshot-writing fix							
UC1	1/1	5/5	10/10	2.0/2	0	7/7	1/1
UC2	1/1	4/4	12/12	2.0/2	0	7/7	1/1
UC2.1	1/1	4/4	12/12	2.0/2	0	7/7	1/1
UC3	1/1	5/5	11/11	2.0/2	0	7/7	1/1
UC4	1/1	5/5	13/13	2.0/2	0	7/7	1/1

Table 4.2: T1, T2, T4, and T5 report runs (or, where the source report enumerates items instead, items) passing over the total scored.

T3's three columns mean:

- **Complete** — checkpoints that exist and have every required field populated, over the total.
- **Grounded** — how well grounded the free-text and LLM-composed fields are: a rubric mean out of 2 where fields were sampled and scored, or a count of substantive sub-checks passing where the report enumerates them instead (UC2.1's `9/10` reflects one numeric-tolerance miss among ten such sub-checks, not a rubric score).
- **Defects** — confirmed defects from the closed list in Table A.2 (fabricated citation, empty required field, contradiction with the record's own data, and similar).

Why UC2/UC2.1/UC3 fail T1/T4/T5 in Block 1:

- A snapshot writer filtered a run's active agent set by registry id, not run directory, so an earlier run's record could leak into a later one.
- The leaked record never appears in any downstream agent's own `input_refs`, a null join to the run's actual computation: it never reached a downstream LLM call, the pipeline's output, or the final deliverable.
- Fixed before the `claude-opus-4-8` set was captured; see prose below.

The topology decision was correct in every scored run of every use case, in both sets. UC1 selected Direct, UC2 and UC2.1 selected Debate, UC3 selected Coalition, and UC4 selected Debate, each independently re-derived from the topology library's own specifications before the judge opened any run's `TS1` record. Block 1's T1 failures for UC2 and UC3 are not selection failures at all: the topology chosen in every run was the correct one. What failed was a snapshot-writing defect that filtered a run's active agent set by registry id instead of run directory, so an earlier run's stale record could leak into a later run's set file. UC3's observed post-topology set in that block was `{C1,C2,F1,P3,S1}` against an expected `{C1,C2,F1}`, with the extra `P3`/`S1` records byte-identical to a different run's shelf entries.

The leaked record is inert, not corrupting. UC2's real `F1.input_refs` is `{r2,d3,p2,p3,s1}` for the correct Debate reads, and the stale `P1` record it also carries never appears there. UC3's real `F1` input is `{C1,C2}`, and the stale `P3`/`S1` records appear in no agent's `input_refs` at all. The leaked record is a *null join* to the run's actual computation. It sits in the shelf snapshot but nothing downstream ever reads it. Set that shelf-bookkeeping failure aside, and almost every one of these runs succeeded on content: the correct topology ran, the correct agents produced the correct output, and T3's grounding checks passed. What was wrong was the record of which files sat in the shelf, not what any agent actually read or wrote. The fix landed before the `claude-opus-4-8` set was captured: `_scoped_snapshot` and `_write_run_snapshot` now filter every write by the record's own `run_id`. Block 2's clean T1/T4/T5 column for UC2/UC2.1/UC3 is the measured evidence that it held. This class of shelf-provenance defect is exactly what the pipeline's own post-hoc architecture audit, `/system-check`, was built to catch on every future run; Appendix D.6 describes it.

UC4's Block 1 T2 failure is a different defect, already root-caused and closed. `EX1` has no human gate of its own and promotes to its shelf immediately, once it holds the mapping agents' com-

puted payload. RC1.1, RC1.2, RM1.1, RM1.2, and RD3, by contrast, wait for the atomic mapping gate's human approval before their own records are written. created_at therefore measures shelf-write order, not computation order: EX1's recorded timestamp sat at or before RC1.1's in three of four runs simply because it wrote first, not because it ran first. The call path genuinely held the mapping agents' already-computed payload at that point, and EX1 never leaked into any other agent's input. Block 2's clean T2 result for UC4 confirms the fix, not merely a lucky single run.

4.3 Expert Evaluation

Section 3.6.2 describes the design: a ten-criterion, one-to-five rubric scored cold by a Siemens colleague who did not build the pipeline, against one frozen final deliverable per use case. All six reviews are returned. Table 4.3 reports the mean score per criterion across all six; Appendix G.2 reproduces every score and every written response in full.

Criterion	Mean (1–5, six reviews)
Correctness and grounding	4.00
Scope fit	4.00
Clarity and usability	3.50
Honesty about limitations	4.33
Overall confidence	3.50
Siemens implementation fit	4.17
Flexibility	4.17
Evidence structure fit	3.67
Auditability and human control	3.33
Approach improvement	4.17
Grand mean	**3.88**

Table 4.3: Mean score per criterion across all six returned reviews (UC1, UC2, UC2.1, UC3, and two independent reviews of UC4). Per-review means: 3.8, 3.5, 4.0, 3.9, 3.7, 4.4.

Reviewers trusted the system's account of its own limits more than any other property. Honesty about limitations scored highest at 4.33. Every "what this work did best" answer names a specific, concrete gap the deliverable itself surfaced, such as a misspelled city, an empty zip code, or a data-completeness issue, rather than a generic disclaimer.

Auditability and human control scored lowest at 3.33, but the two written notes attached to that criterion do not fault the architecture. Both come from the same reviewer, and both attribute the low score to how the reviewed runs were produced. Every human gate in every scored run was auto-approved, so no reviewer could see what an actual human decision would have caught or changed. That is a property of the packets shown to reviewers, not of the gates themselves. A separate reviewer independently confirmed the trace exists: whether a human review occurred, and whether it changed the output, is visible in the record.

Auditability and human control is not a criterion incidental to this thesis's argument: it is the criterion π's entire justification rests on, the human-approved gate that this thesis treats as the

evidence floor. A low score here therefore carries more weight than a low score on any other criterion would, and the auto-approval explanation above does not fully discharge it. That explanation accounts for why no reviewer observed a human decision in the reviewed runs; it does not establish that the architecture would satisfy this criterion if one had occurred.

Two findings limit how far this concern generalizes. T1, the automated structural test, confirms that the correct topology ran in every scored run of every use case (Section 4.2). This criterion asks a different question: whether a human decision was visible, not whether the correct decision was structurally reached. The same reviewer who assigned this score also described the selected topology as "a good architectural choice" for the case it served. **The gap is in visible participation, not in correctness.**

This observation does not fault the architecture on its own terms. The isolated-memory design (Section 3.2.2) already establishes that a human-approved correction on the next run is the intended remedy; that is not what this review tested. The packets were deliberately built to evaluate what the system produces on its own, auto-approved. This criterion's reading is an observation about no human being in the loop, not a finding about the content the system produced.

The most telling signal here comes from UC4's second review, by Reviewer C. UC4 was that reviewer's own case: the person who designed its registry entry, defined its request, and was the intended recipient of its output. Reviewer C's remark that the deliverable was "really great and fast overview with high accuracy" carries particular weight here, because an overview was what that request asked for, not a deep multi-domain investigation. Had it asked for the latter, `TS1` would have selected the more complex, multi-team Coalition topology instead. The Debate topology it selected matched the overview scale actually requested. The one substantive gap Reviewer C raised was itself flagged "not related to the model," and this review carried the highest mean of all six, 4.4. Appendix G.2 reproduces every score, including this contrast, and Reviewer A's two auditability notes verbatim.

Manual production-time estimates use mixed units, but a single conversion rule makes them comparable. Once a typical break (about an hour) and the attention it takes to refocus after clocking in (about another hour) are subtracted from a nominal eight-hour day, an effective working day comes to six hours. By that rule, UC4's "8h" review converts to roughly 1.3 days. Read this way, the five cases average: UC1 2.5 days, UC2 1.5 days, UC2.1 1.5 days, UC3 2.5 days, and UC4 roughly 2.9 days across its two reviewers (4.5 days and roughly 1.3 days).

The free-text answers cluster on two axes. What the work did best clusters on grounded, specific limitations and site-level explanation, echoed independently by reviewers who never saw each other's packets. What it did worst is a presentation problem, not a content one: length, and the absence of an executive summary. Four of the six reviews ask for a shorter overview or a prioritized top-line view before the full detail, not for a correction to any finding.

A later companion interface renders the same anonymized UC2.1 result through a portfolio summary, filters, and navigable tables at `https://emrecanulu.com/epoch/uc2-1/stage2/f1/`. Reviewers did not see this interface, so it does not alter the scores reported here. This is a

scope-of-time limit, not a defect in the system itself. Within the thesis's seven-month window, the system first built its shared infrastructure. Only after that did it have time to build one deliverable tailored to the reviewer who actually defined its request, UC4's Reviewer C, not a customized case for every reviewer.

That reviewer raised no length or missing-summary complaint at all. More time spent on output customization would plausibly earn the same kind of favorable read from the other reviewers. The registry already captures each case's own natural-language request and preferred output form, so an industrial deployment that builds each case around its actual requester should not face this complaint at all. UC4 is itself the evidence: the same deliverable read as "really great and fast overview" to the reviewer who asked for it, and "extensive and not concise enough" to the one who did not.

Chapter 5

Conclusion

Section 5.1 answers each of EPOCH's four research questions in turn. Section 5.2 works through what those answers mean beyond the pipeline itself. Section 5.3 states what this thesis cannot claim, and Section 5.4 closes with what remains open.

5.1 Answers to the Research Questions

RQ1 asked how multi-agent topology selection can be conditioned on the output profile and assurance profile a compliance answer must carry, rather than on task difficulty, cost, or token budget. EPOCH's answer is the selector $\pi : (\text{case}, \text{output_profile}, \text{assurance_profile}) \rightarrow \text{topology} \in \Lambda$, resolved by a bounded three-shape library: Direct, Debate, and Coalition, each tied to a distinct evidence artifact rather than a distinct performance profile (Section 2.1). Ownership is what separates the shapes: Direct fits single ownership, Debate fits independent verification of one claim, and Coalition fits separately mandated results that must be combined (Section 2.3). Assurance functions as an evidence floor a run must clear, not a score traded against cost the way budget is.

RQ2 asked how a human-in-the-loop pipeline can turn an expert-approved isolated memory into an auditable case profile that drives topology selection while satisfying EU AI Act oversight duties. The R1→DA1→human→R2→human→D1→D2→CP1 chain is that answer: two human gates lock the regulatory and data boundary before any topology runs, and every later agent reads only that approved boundary, never the raw request (Section 3.2.2). Each shelf record carries `input_refs` back to the exact upstream artifact it consumed, so the chain from approval to deliverable stays reconstructable end to end. Table C.2 maps this same structure article by article against the Act's oversight duties, without claiming to settle the classification question those duties depend on.

RQ3 asked whether an assurance-conditioned pipeline holds its own architectural guarantees, namely the selected topology, the isolated-memory boundary, and the provenance chain, across repeated runs of the same approved case. On the property that matters most, yes: topology selection was correct in every scored run of every use case, on both backends (Table 4.2). Two

real defects surfaced along the way, and both were shelf-provenance or timestamp artifacts, not selection or reasoning failures. One was a snapshot writer that let a stale record leak across runs; the other was a gate-less agent whose write order did not match its read order. Both are root-caused, fixed in code, and confirmed clean in the `claude-opus-4-8` confirmation pass.

RQ4 asked whether domain experts who did not build EPOCH would trust and use its finished output, what shortcomings they would report, and how much expert effort it would displace. Six returned reviews across five deliverables gave a grand mean of 3.88 out of 5, with the highest trust placed in the system's own honesty about what it could not verify. Most of what reviewers criticized was how the report looked, not what it found. Section 5.2 works through what that split, and the effort it displaces, actually means.

5.2 Implications

Section 4.3 reported what six domain experts said about five finished deliverables. Read together, those reviews argue for EPOCH on a basis the ten-criterion scores alone do not show: what the system costs to adopt, and what its critics actually asked for.

A reviewer's manual-time estimate, for example UC2's "1–2 full working days" or UC3's "2–3 working days," is not a one-time number. It is what that same report costs again next quarter, and the quarter after that, for as long as a human produces it by hand. EPOCH's cost runs the other way. UC1 and UC2 carried the expensive part, because building the pipeline itself happened there. Once that backbone existed, UC2.1 and UC3 were each onboarded in roughly three to four hours of author effort. UC4 was the most demanding case, an entire document-family pipeline of nine new agents, and took four to six hours. That figure is corroborated, not only recalled: git history shows the ESRS corpus landing at 16:54, the full document pipeline at 20:12, and an end-to-end verified run at 20:17 the same day, with audit fixes complete by 14:40 the next morning. The report itself then costs minutes: the UC4 run both reviewers assessed ran from 16:31 to 16:49, about eighteen minutes wall clock, against their own estimates for the same deliverable by hand, which average to roughly 22 hours (about 2.75 working days). **A one-time onboarding cost measured in hours, set against a manual cost that recurs on every reporting cycle, is where the pipeline's return sits.**

Reviewer criticism clustered on how the report looked far more often than on what it found. Four of the six "what did this work do worst" answers ask for the same missing thing: a shorter summary or top-line view before the full detail (Appendix G.2 quotes all four in full). None of these four disputes a finding; each asks for the same finding presented shorter, with a summary in front of it.

A second cluster asks for evidence the system already computes to be made visible, not for new computation. UC1's reviewer wanted a worked calculation for one site, step by step; the pipeline already runs that arithmetic, simply without surfacing it in the deliverable. Its threshold configuration is visible in Table E.5, and UC1's own record is visible at https://emrecanulu.com/epoch/uc1/stage2/f1/. UC2's reviewer wanted the emission factor's source traceable,

not only its output. The same record for UC2 is visible at https://emrecanulu.com/epoch/uc2/stage2/f1/.

The same dual-method computation is visible at site level in UC2.1's own D3 table (Table E.7) and at https://emrecanulu.com/epoch/uc2-1/stage2/f1/, the record a normal user of either use case would already have open. Neither record is the exact walkthrough or traceable field either reviewer pictured, but both show the underlying arithmetic already exists inside the system. Both are choices about which fields an agent's prompt tells it to print, changeable without touching the topology or the gate structure that produced the number in the first place. The same presentation grammar supports UC3's division-owned consolidation, where portfolio totals and division-level results remain separately inspectable at https://emrecanulu.com/epoch/uc3/stage2/f1/. The review itself was necessarily brief, packets designed to be read cold in roughly ten to fifteen minutes inside a reviewer's ordinary working day, a limit this level of gap is consistent with. Had the review protocol asked for full access to the live viewer instead of a single output PDF, most of this presentation-level criticism would likely never have surfaced.

Not every criticism was cosmetic, and the pattern is worth more for surviving the exception than for having none. UC2's "year" field was empty in every row of the reviewed run, a real content gap. The same reviewer noted that labeling the deliverable an "assurance memo" promises more verification than the report actually performs. That naming concern shares the same root as the packet-versus-viewer gap named above: the review saw only the final ten-to-fifteen-minute artifact, not the underlying provenance record that the live viewer already exposes. Naming both here, rather than smoothing them over, is what keeps the larger pattern claim credible: a claim survives naming its own exception, or it does not survive at all.

Few of the six reviews criticized the model's reasoning itself; most of what they found fault with was missing context, not faulty logic. Two examples show this directly. UC4's second reviewer prefixed their one substantive gap with "not related to the model" before naming it. UC1's reviewer raised a similar point: the system cannot know that a site installed a new energy management system last month, because that information never reached it. Both point at the same thing: the problem is not how the system processes the information it has, but which information reaches it at all. That gap is not a defect in the system either: a system cannot be held responsible for an output it was never asked to provide. Had the registry entry itself carried a note such as "notify the system if a new change occurs," Stage 1's R1 agent would have surfaced that exact gap. A human reviewer could then have closed it before the run ever started. Had that information been available, the system would already know and use it, because every agent keeps a traceable record of every input it saw and consumed in its own shelf record (Section 3.2.2). This observation holds only for the six reviews collected here, not as a general claim.

RQ1's answer points at a gap wider than this pipeline: none of the selectors reviewed in Section 2.3 takes the concrete evidence a result must carry, a reconciliation record or a distributed sign-off, as an input to selection itself, which is why EPOCH adds assurance as a third, independent axis instead of a cost traded down. In practice, a routine case runs through a cheap, single-agent chain, while a case needing independent verification is routed automatically to a

heavier shape, Debate or Coalition, so the system is never forced to impose the same rigor on every case.

RQ2's and RQ3's answers explain why that routing stays trustworthy after the fact. Because the isolated-memory boundary and its `input_refs` chain are checkable inside the running pipeline in real time (Section 3.2.2), an auditor can trace any deliverable field back to the human-approved record that authorized it, exactly what the EU AI Act's record-keeping duties (Table C.2) ask for. The same rule, that a record counts only once another agent's `input_refs` points to it, is what kept both of RQ3's defects out of production output; `/system-check` (Appendix D.6) now carries that protection forward automatically on every future run.

5.3 Limitations

Five limitations trace back to one cause: the seven months available for this thesis. EPOCH is not a narrow script but a full platform, with its own pipeline, agent library, and human-review interface. Building that infrastructure, testing it broadly enough to trust its results, and adapting research findings into a working design took most of that time. Doing all of this inside a large company's ordinary pace left little room for anything beyond it.

- **Infrastructure over scale.** Most of that time went into building the pipeline's shared infrastructure. UC1 and UC2 carried that cost, as Section 5.2 states, leaving little time to run the system beyond first-onboarding scale.
- **A planned model comparison was dropped before it started.** The evaluation was originally planned around `gpt-5.6-sol`, but that model had not become available on Azure AI Foundry in time for this thesis's submission deadline. The plan shifted to `claude-opus-5` once it became available on Azure, because the Foundry Anthropic adapter was already in production use and adopting Opus 5 needed only a new backend preset. This was not a performance downgrade: Opus 5 remained competitive on independent leaderboards such as Arena's [49].
- **No cross-model comparison was run.** Every scored result in this thesis comes from one model family, confirmed only on `claude-opus-4-8` (Table 4.2); running the dropped `gpt-5.6-sol` comparison was further ruled out for budget reasons (`evaluation/opus-5/README.md`, "Limitations"). Whether the architecture's guarantees hold on a structurally different model family is untested.
- **Only one genuinely external registry.** Four of the five use cases were shaped jointly between the author and Siemens EHS as infrastructure-building cases. Only UC4's registry was authored end to end by the person who actually wanted its output, the reviewer named `C` in Section 4.3.
- **Reviewers saw a static packet, not the live system.** The same seven-month schedule limited each review to a ten-to-fifteen-minute packet showing the final output alone (Section 5.2). The live viewer at `https://emrecanulu.com/epoch/` already exposes the underlying records behind that output; most of what reviewers asked to see is a page that

interface already provides.

5.4 Future Work

Each limitation above names a direct next step.

- With the infrastructure now built, run the pipeline across many more cases at production scale instead of further foundation work.
- Test cheaper, smaller models against individual agents for cost optimization. Strong models already produced good results here, as expected; the open question is how far that quality still holds.
- Run a genuine cross-model comparison across multiple model families, testing whether topology-selection correctness and the provenance guarantees named in Section 5.2 hold independently of `claude-opus-5` specifically.
- Onboard additional use cases genuinely requested end to end by their own stakeholder, the way UC4 was, to test whether the registry-driven request pattern generalizes beyond one case and one requester.
- Give reviewers direct access to the live viewer instead of a static packet, so the same underlying evidence reaches them without a length limit.

Bibliography

[1] European Parliament and Council of the European Union, "Regulation (EU) 2024/1689 of the european parliament and of the council laying down harmonised rules on artificial intelligence (AI act)," https://eur-lex.europa.eu/eli/reg/2024/1689/oj, 2024, art. 6, 9, 11, 12, 14, 15, 53, 55. Accessed 2026-08-18.

[2] Y. Du, M. Tian, S. Ronanki, S. Rongali, S. Bodapati, A. Galstyan, A. Wells, R. Schwartz, E. A. Huerta, and H. Peng, "Context length alone hurts LLM performance despite perfect retrieval," 2025.

[3] W. Wang, J. Min, and W. Zou, "Intelligence degradation in long-context LLMs: Critical threshold determination via natural length distribution analysis," 2026.

[4] K.-T. Tran, D. Dao, M.-D. Nguyen, Q.-V. Pham, B. O'Sullivan, and H. D. Nguyen, "Multi-agent collaboration mechanisms: A survey of LLMs," 2025.

[5] G. Zhang, Y. Yue, X. Sun, G. Wan, M. Yu, J. Fang, K. Wang, T. Chen, and D. Cheng, "G-Designer: Architecting multi-agent communication topologies via graph neural networks," 2024.

[6] Y. Yue, G. Zhang, B. Liu, G. Wan, K. Wang, D. Cheng, and Y. Qi, "MasRouter: Learning to route LLMs for multi-agent systems," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025, pp. 15 549–15 572. [Online]. Available: https://aclanthology.org/2025.acl-long.757/

[7] S. Li, Y. Liu, Q. Wen, C. Zhang, and S. Pan, "Assemble your crew: Automatic multi-agent communication topology design via autoregressive graph generation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026, arXiv:2507.18224.

[8] S. Wang, R. Lu, Z. Yang, Y. Wang, Y. Zhang, L. Xu, Q. Xu, G. Yin, C. Chen, and X. Guan, "AgentConductor: Topology evolution for multi-agent competition-level code generation," 2026.

[9] H. Chen, X. Zheng, Y. Liu, P. Jiao, S. Li, H. Liu, Z. Zhao, Z. Xu, I. Khalil, and S. Pan, "GoA-gent: Group-of-agents communication topology generation for LLM-based multi-agent systems," 2026.

[10] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," 2022.

[11] G. Irving, P. Christiano, and D. Amodei, “AI safety via debate,” 2018.

[12] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, “Improving factuality and reasoning in language models through multiagent debate,” 2023.

[13] T. Liang, Z. He, W. Jiao, X. Wang, Y. Wang, R. Wang, Y. Yang, S. Shi, and Z. Tu, “Encouraging divergent thinking in large language models through multi-agent debate,” 2023.

[14] S. Hong, M. Zhuge, J. Chen, X. Zheng, Y. Cheng, C. Zhang, J. Wang, Z. Wang, S. K. S. Yau, Z. Lin, L. Zhou, C. Ran, L. Xiao, C. Wu, and J. Schmidhuber, “MetaGPT: Meta programming for a multi-agent collaborative framework,” 2023.

[15] C. Qian, W. Liu, H. Liu, N. Chen, Y. Dang, J. Li, C. Yang, W. Chen, Y. Su, X. Cong, J. Xu, D. Li, Z. Liu, and M. Sun, “Communicative agents for software development,” 2023.

[16] J. Zhang, J. Xiang, Z. Yu, F. Teng, X. Chen, J. Chen, M. Zhuge, X. Cheng, S. Hong, J. Wang, B. Zheng, B. Liu, Y. Luo, and C. Wu, “AFlow: Automating agentic workflow generation,” 2024.

[17] L. Yang, J. Luo, X. Liu, Y. Lou, and Z. Chen, “BAMAS: Structuring budget-aware multi-agent systems,” 2025.

[18] X. Zhao, S. Liu, Y. Zhang, Q. Ma, Z. Lin, N. Wang, G. Cheng, C. Liu, and Y. Xu, “ATOM: Instantiating budget-controllable multi-agent collaboration via nucleus-electron hierarchy,” 2026.

[19] D. Tran and D. Kiela, “Single-agent LLMs outperform multi-agent systems on multi-hop reasoning under equal thinking token budgets,” 2026.

[20] A. Talluri, P. Anne, B. C. Pendiyala, and R. Chilukuri, “Retrieval-conditioned topology selection with provable budget conservation for multi-agent code generation,” 2026.

[21] European Parliament and Council, “Directive 2003/87/ec establishing a system for greenhouse gas emission allowance trading within the union,” https://eur-lex.europa.eu/eli/dir/2003/87/oj, 2003, consolidated framework accessed 2026-06-23.

[22] ——, “Directive (EU) 2023/959 amending directive 2003/87/ec and establishing ETS2,” https://eur-lex.europa.eu/eli/dir/2023/959/oj, 2023, accessed 2026-06-23.

[23] Federal Republic of Germany, “Gesetz zur steigerung der energieeffizienz in deutschland (EnEfG),” https://www.gesetze-im-internet.de/enefg/, 2023, accessed 2026-06-23.

[24] ——, “Bundes-klimaschutzgesetz (KSG),” https://www.gesetze-im-internet.de/ksg/, 2019, accessed 2026-06-23.

[25] GHG Protocol, “Scope 2 guidance,” https://ghgprotocol.org/scope-2-guidance, 2015, accessed 2026-06-23.

[26] European Commission, “Commission delegated regulation (EU) 2023/2772: European sustainability reporting standards,” https://eur-lex.europa.eu/eli/reg_del/2023/2772/oj, 2023, accessed 2026-06-23.

[27] G. Yu, “AdaptOrch: Task-adaptive multi-agent orchestration in the era of LLM performance convergence,” 2026.

[28] S. S. Stevens, “On the theory of scales of measurement,” *Science*, vol. 103, no. 2684, pp. 677–680, 1946.

[29] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Waltham, MA: Morgan Kaufmann, 2012.

[30] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning: With Applications in R*, 2nd ed. Springer, 2021.

[31] J. A. Kroll, J. Huey, S. Barocas, E. W. Felten, J. R. Reidenberg, D. G. Robinson, and H. Yu, “Accountable algorithms,” *University of Pennsylvania Law Review*, vol. 165, no. 3, pp. 633–705, 2017.

[32] L. Chen, J. Q. Davis, B. Hanin, P. Bailis, I. Stoica, M. Zaharia, and J. Zou, “Are more LLM calls all you need? towards scaling laws of compound inference systems,” 2024.

[33] M. Cemri, M. Z. Pan, S. Yang, L. A. Agrawal, B. Chopra, R. Tiwari, K. Keutzer, A. Parameswaran, D. Klein, K. Ramchandran, M. Zaharia, J. E. Gonzalez, and I. Stoica, “Why do multi-agent LLM systems fail? (MAST),” 2025.

[34] C. Paduraru, P.-L. Bouruc, and A. Stefanescu, “A trace-based assurance framework for agentic AI orchestration: Contracts, testing, and governance,” 2026.

[35] O. Solozobov, “Decision evidence maturity model for agentic AI: A property-level method specification,” 2026.

[36] S. Zabolotnii, “TRACE: A metrologically-grounded engineering framework for trustworthy agentic AI systems in operationally critical domains,” 2026.

[37] B. Green, “The flaws of policies requiring human oversight of government algorithms,” *Computer Law & Security Review*, vol. 45, 2022, arXiv:2109.05067.

[38] E. Turan, “Oversight has a capacity: Calibrating agent guards to a subjective, fatiguing human,” 2026.

[39] D. Acemoglu and S. Johnson, *Power and Progress: Our Thousand-Year Struggle Over Technology and Prosperity*. PublicAffairs, 2023.

[40] D. Acemoglu, “The simple macroeconomics of AI,” National Bureau of Economic Research, Working Paper 32487, 2024.

[41] Z. Kenton, N. Y. Siegel, J. Kramár, J. Brown-Cohen, S. Albanie, J. Bulian, R. Agarwal, D. Lindner, Y. Tang, N. D. Goodman, and R. Shah, "On scalable oversight with weak LLMs judging strong LLMs," 2024.

[42] N. Kale, C. B. C. Zhang, K. Zhu, A. Aich, P. Rodriguez, Scale Red Team, C. Q. Knight, and Z. Wang, "Reliable weak-to-strong monitoring of LLM agents," 2025.

[43] S. Liu, R. Zhang, R. Ma, Y. Deng, L. Zhu, J. Li, Z. Li, Z. Shen, and M. Du, "LLM agents in law: Taxonomy, applications, and challenges," 2026.

[44] D. Song, Y. Huang, B. Chen, T. Cong, R. Goebel, L. Ma, and F. Khomh, "Evaluating implicit regulatory compliance in LLM tool invocation via logic-guided synthesis," 2026.

[45] E. Y. Kong, J. Tan, I. Gupta, L. Olds, C. Campbell, D. Fagnan, R. Kavuri, V. Balin, R. Gosain, L. Garcia, and M. Jang, "The lifecycle of LLM-as-a-judge for large-scale recommendation explanations," 2026.

[46] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," *MIS Quarterly*, vol. 28, no. 1, pp. 75–105, 2004.

[47] H. Onweller, E. Lumer, A. Huber, P. Ramchandani, V. K. Subbiah, and C. Feld, "Cited but not verified: Parsing and evaluating source attribution in LLM deep research agents," 2026.

[48] S. Das, N. Deb, N. Chaki, and A. Cortesi, "A multi-agent RAG framework for regulatory compliance checking of software requirements," *ACM Transactions on Software Engineering and Methodology*, vol. 35, 2026, doi:10.1145/3785472.

[49] Arena, "Arena leaderboard," https://arena.ai/leaderboard, 2026, independent crowd-sourced model-comparison leaderboard. Accessed 2026-09-15.

[50] European Parliament and Council of the European Union, "Regulation (EU) 2026/1744 of the european parliament and of the council of 8 july 2026 amending regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as regards the simplification of the implementation of harmonised rules on artificial intelligence (Digital Omnibus on AI)," https://eur-lex.europa.eu/eli/reg/2026/1744/oj, 2026, adopted 8 July 2026; published in OJ 24 July 2026; in force 27 July 2026. Accessed 2026-08-18.

Appendix A

Evaluation Protocol

This appendix preserves the two source artifacts that govern the Part I evaluation in Section 3.6.1. The taxonomy defines what the five tests measure and how their results are scored. The judge prompt turns those definitions into one fixed execution sequence.

A.1 Evaluation Taxonomy

The taxonomy is the specification, not an executable scorer. It defines the tests, source artifacts, scoring units, and reporting limits used by the evaluation. The listing below reads the complete specification directly from `evaluation/TAXONOMY.md`.

```
# Evaluation Taxonomy

This file is the framework that sits above ‘evaluation/model_comparison/‘ and
‘evaluation/uc4_document_audit/‘. It names and defines the tests (T1-T5) that
gold-labeling runs are measured against. It does not run anything, does not
score anything, and does not replace any existing gold or freeze mechanism --
it names the vocabulary those mechanisms already use, and adds the two axes
(agent/topology selection and process conformance) that sit outside content
comparison and therefore have no home in ‘gold/contract.json‘ today.

Run planning, run execution, and gold labeling itself remain outside this
document by design. This file defines what each test measures, where it
reads from, and how it is scored -- not how many runs to do or in what mode.
T1-T5 are executed by ‘evaluation/JUDGE_PROMPT.md‘, a use-case-agnostic
LLM-judge prompt that resolves each use case’s gold and run set from disk.

## 1. Scope and claim

The claim this taxonomy exists to test is: agent and topology selection was
never hardcoded per use case. Each use case’s registry entry states a demand
(‘natural_request‘, ‘expected_output‘, ‘output_profile‘, ‘assurance_profile‘)
and each agent carries its own description; TS1 derives
‘selected_topology_id‘ (‘Literal["Direct", "Debate", "Coalition"]‘,
‘src/epoch_switch/agents/orchestration/topology_selector/topology_selection.py:31‘)
from those descriptions at run time, with no per-use-case branch in code that
names a topology or an agent list in advance.

What this taxonomy does **not** claim to cover: an all-pairs or combinatorial
sweep across the full agent roster. The system has no such call surface to
```

```
begin with -- a run does not invoke agents against each other freely; TS1
selects one topology, and the topology library then runs a fixed, ordered
chain for that topology. There is no N×N interaction to test, so exhaustive
pairwise coverage is not a gap being left out of this taxonomy; it is
undefined for this architecture, the same way asking for the "coverage of all
function pairs calling each other" is undefined for a single deterministic
call graph. T5 (below) is the correctly-scoped substitute: it checks that no
agent outside the selected topology's chain ever activates for a run it does
not belong to.

## 1a. Gold is a structural template, not a content answer key

Gold is one frozen example of a structurally correct run, not the answer this
system is required to reproduce. A run that surfaces a different, well-grounded
set of regulation findings than gold's, assigns different LLM-generated ids, or
approves a broader or narrower field list is not thereby wrong -- it is a
different but potentially equally valid pass over the same source material.
Run content is judged against its **own cited source** (the regulation text,
the data catalog, the registry demand it answers), never against gold's
wording, counts, ids, or list membership. Only a glaring, undeniable defect --
a field contradicting its own record's narrative, a fabricated or
misattributed citation, a required deliverable field left empty, an aggregate
that does not reconcile with its own components, an enum value outside its
declared set -- counts as a problem (T3's 3c list, below).

This reframes what `field_classes`' `exact` label may cover. Marking a field
`exact` in `contract.json` used to be read as "score this by gold-value
equality," which forced id lists (`approved_field_ids`, `finding_refs`,
`citations[]`), materiality flags (`entries[].material`), and similar
LLM-composed content into literal diffs against one frozen sample. From here
on, `exact` covers **only closed enums, booleans, and fixed Literals** --
`verdict`, `risk_level`, `selected_topology_id`, `entity_grain`, and similar
fields whose value space is fixed by a Pydantic `Literal`/`bool`/`Enum`
declaration, independent of any one run's content. Anything an LLM assigns,
composes, or counts -- ids, list membership, list length, free-text flags -- is
never a gold-equality target; it is judged for grounding against its own
source, the same way a `semantic` field is. This generalizes the caveat
`JUDGE_PROMPT.md` already carries for "LLM-derived exact lists" (§T3) into the
taxonomy itself, so grounding-based judgment is the default the judge applies,
not an exception it has to remember to invoke.

Likewise, a mismatch against `collections.expected_count` is a **structural
signal only** -- did the collection get produced at all, roughly at the shape
the contract anticipated -- never a score. A run returning 6 findings where
gold recorded 4, or 8 approved fields where gold recorded 10, is not a T3
finding by count alone; it is a finding only if the extra or missing items
fail their own grounding check.

## 2. Vocabulary map

Three comparison vocabularies already exist in the repo. This taxonomy does
not add a fourth -- it says where each one applies and adds one narrow
sub-scale (§3, T3) inside the one that already covers prose.

| Vocabulary | Defined in | Scope | Used by |
|---|---|---|---|
| Field-comparison classes: `exact` / `numeric` / `data_derived` / `semantic` / `ignored` | `evaluation/model_comparison/<usecase>/gold/contract.json` → `comparison_classes` | Every field of every tabular checkpoint (UC1/UC2/UC2.1/UC3) | T3, T4 |
| Gold-fitness rubric: 5 weighted dimensions /10 (factual & arithmetic correctness 30%, internal consistency 20%, scope & boundary fidelity 20%, evidence & citation quality 15%, fitness as gold 15%) with freeze bands 9-10 / 7-8 / 5-6 / 0-4 | `evaluation/model_comparison/GOLD_HUMAN_REVIEW.md` | Whole-checkpoint content judgment during gold drafting, ahead of the human `review.json` signature | Freeze Readiness (§6), unchanged |
| Audit verdict vocabulary: `supported` / `partially_supported` / `contradicted` / `unlocatable` / `unsupported` / `
```

```
     confirmed‘, fix-list impact grades ‘invalidates‘ / ‘degrades‘ / ‘cosmetic‘ | ‘evaluation/uc4_document_audit/
     AUDIT_PROMPT.md‘ | UC4 packet-level claim auditing -- a separate cycle from UC4’s own ‘gold/contract.json‘ (§4) |
     UC4’s own audit cycle, unchanged; kept alongside, not merged into, T1-T5 |
| **3-level rubric (this document): ‘2‘ / ‘1‘ / ‘0‘** | §3, T3 only | **Only fields the contract already marks ‘semantic
     ‘.** It is a scoring resolution for one existing class, not a new class. | T3 |

‘exact‘, ‘numeric‘, and ‘data_derived‘ fields are never rubric-scored -- they
are checked by code against the frozen gold value (with the stated tolerance
for ‘numeric‘, and the data-drift carve-out for ‘data_derived‘). Only
‘semantic‘ fields receive a ‘2‘/‘1‘/‘0‘ human or LLM-judge rubric score:

- **2** -- fully meets the gold: same claim, adequately grounded.
- **1** -- partially meets it: right general claim, weak or missing
  grounding, or a material omission.
- **0** -- wrong or missing: contradicts the gold or absent where the gold
  requires content.

## 3. Test definitions (T1-T5)

| | T1 | T2 | T3 | T4 | T5 |
|---|---|---|---|---|---|
| Name | Selection Correctness | Process Conformance | Output Substantiveness | Execution Integrity | Negative
     Activation Control |
| Measures | Right topology + right agent set activated | Architectural rules held during the run | Checkpoints are
     structurally complete and their content is grounded/self-consistent; gold used only as a structural template | The
     pipeline actually completed cleanly, run to run | No agent outside the selected chain ever fired |
| Needs frozen gold? | No | No | Structurally (shape only, not content) | No | No |
| Unit of report | run → use case → aggregate | run → use case → aggregate | checkpoint × sub-check table (3a/3b/3c
     ) | run → use case (deterministic checklist) | run → use case → aggregate |

### T1 -- Selection Correctness

What: does ‘selected_topology_id‘ for the run equal the use case’s
‘expected_topology‘, and does the set of agents that actually ran downstream
of TS1 equal the use case’s designed post-topology agent set -- with no
overcorrection needed from the human gate.

Data source:
- ‘<run_dir>/active_profile_set.json‘ (tabular family) or
  ‘<run_dir>/active_document_set.json‘ (document family; filename chosen by
  ‘pipeline_family‘ at ‘src/epoch_switch/regulation_cli.py:543‘) →
  the ‘"ts"‘ record’s ‘payload.selected_topology_id‘, and the ‘agent_id‘
  field of every record in the file (read ‘agent_id‘ from each record, not
  the file’s own shorthand key -- keys are lowercase shorthand like ‘ts‘,
  ‘da1‘, ‘rc1_1‘; ‘agent_id‘ is the canonical uppercase id, e.g. ‘"RC1.1"‘).
- The gold side of the same pair is already assembled at draft time:
  ‘gold/real/draft/checkpoints/ts1.json‘ → ‘_gold.observed_agents[]‘ (the
  realized post-topology agent set, computed in
  ‘evaluation/model_comparison/<usecase>/create_gold.py:272-276‘ as
  ‘sorted(set(records) - PRE_TOPOLOGY_AGENTS)‘, where
  ‘PRE_TOPOLOGY_AGENTS = {"R1","DA1","R2","D1","D2","CP1","TS1","D3"}‘) and
  ‘payload.selected_topology_id‘ in the same file.
- Expected values come from ‘gold/contract.json‘ → ‘expected_topology‘ and
  ‘supporting_agents[]‘ minus ‘PRE_TOPOLOGY_AGENTS‘.

Scoring: pass/fail per run -- pass iff ‘selected_topology_id ==
expected_topology‘ and the observed post-topology agent set equals the
contract’s expected set exactly (no missing, no extra agent). Aggregate as a
ratio per use case and across all runs. On divergence, the contract’s
existing ‘topology_policy.on_divergence‘ rule applies verbatim:
‘TS1.selected_topology_id‘ mismatch counts as a miss, and any
‘topology_scoped‘ checkpoint from an un-selected topology is
‘not_applicable‘ -- **‘not_applicable‘ is not zero**, those agents never ran
so they were neither right nor wrong; do not fold ‘not_applicable‘ into the
```

T1 pass/fail denominator as a failure.

T2 -- Process Conformance

What: the architectural invariants that are not about content, held for the run. Checklist (each item pass/fail per run):

- Chain order matches `pipeline_order` in the contract (checkpoints appear in ascending order in the run's timeline).
- Only the final agent of the selected topology's chain opened a human `[a]pprove/[r]eject` gate (Direct intermediate steps like P1, or Coalition domain agents like C1/C2, are auto-promoted without a gate -- contract `human_gate: null` on those checkpoints matches what actually happened).
- Post-R2 boundary held: CP1 and downstream records' `input_refs` cite only R2 `in_scope_findings` and the DA1 handoff/approved mapping -- never raw registry `site_filter` / `time_window` / `regulation_refs`.
- P1's `input_refs` cite only R2 `in_scope_findings` + D3 payload (no CP1, no DA1 handoff, no registry demand). F1's `input_refs` cite P1's output, R2 `in_scope_findings`, and the registry demand snapshot only (no CP1, no DA1 handoff, no raw D3).
- EX1 ran immediately before the Stage-1 gate and EX2 immediately before the Stage-2 gate (or both were skipped together with a logged reason, e.g. no websearch-capable backend); neither EX1 nor EX2 content appears inside any other agent's `payload` or `input_refs` -- advisory content stayed out of the pipeline's own shelves.

Data source: the same run-set file as T1, read for `input_refs`, `review_round`, `approved_at`/`created_at` ordering, and gate presence/absence. There is no CLI transcript on disk anywhere in this codebase (the CLI's Rich console output is never recorded) -- do not look for one. Whether a gate opened for an agent is read from what is actually recorded: `human_decisions_round_NN.json` (per-field DA1 decisions), the record's own `review_round` and the `*_output_round_NN.json` suffix count for that agent (more than one round implies at least one prior rejection/revision cycle), and `human_rejected` records in the owning agent's shelf archive (`registry_profiles/<registry_id>/archive/`).

T3 -- Output Substantiveness

What: is each checkpoint structurally real and is its content grounded and self-consistent -- gold used only as a structural template (§1a), never as a content answer key. Three sub-checks per checkpoint:

3a -- Structural conformance. The checkpoint exists; every field the contract declares for it is present and non-empty; declared `collections` exist and are joinable on the contract's `join_key`. Missing or empty is a failure. A different row count than gold's `expected_count`, a different list length, or different id values are **not** a 3a failure -- §1a already settles that those are structural signals at most, never a defect by themselves.

3b -- Substantive quality. For fields the contract still scores by class: `exact` (now restricted to closed enums/booleans/Literals, §1a) is pass/fail against the fixed value space, not against gold's chosen value; `numeric` is checked within the contract's `tolerances` entry; `data_derived` is comparable only if the run used the same data catalog snapshot as gold, else reported as data drift, never a miss; `semantic` and everything else (LLM-composed ids, lists, counts) get the `2`/`1`/`0` rubric from §2, scored against the field's **own cited source** -- is the claim true against the regulation text or data catalog it cites, internally consistent with the agent's own other fields and with upstream agents, and responsive to the registry demand. Reuse the contract's per-checkpoint `comparison_policy` sentence and the 5-item `review_checklist` for this -- both already ask exactly this question. `ignored` fields (timestamps, hashes, snapshot ids) are skipped entirely.

3c -- Glaring defects. A closed list; nothing outside it is a T3 defect no matter how much it differs from gold:
- A field contradicts its own record's narrative (e.g. a materiality flag that disagrees with that same entry's stated scope determination).
- A citation whose source text does not say what the field claims it says (fabricated or misattributed reference).
- A contract-required deliverable field is empty where the run reached that checkpoint.
- An aggregate/rollup does not reconcile with its own stated components.
- An enum/Literal value falls outside its declared set.

Scoring/reporting unit: one table per checkpoint (3a pass/fail per field presence check, 3b rubric/pass-ratio per scored field, 3c a flat list of any defects found) rolled up to a per-checkpoint summary and a per-use-case summary across checkpoints.

T4 -- Execution Integrity

What: did the pipeline actually run, end to end, cleanly -- independent of gold entirely. A concrete, fully deterministic per-run checklist, computable from the run directory alone:

1. **Completion.** The terminal agent's record (`f1` tabular / `f2` document) exists and its `run_id` equals the run directory name -- exactly `_snapshot_is_end_to_end()` (`src/epoch_switch/regulation_cli.py:606`). File existence alone is not completion: the same snapshot is also written on rejection and post-gate exception paths, so the `run_id` match is load bearing, not a formality.
2. **No stale records.** Every non-null record's `run_id` matches the run directory; any record whose `run_id` mismatches, or that carries `carried_over: true` outside the legitimately-resumable pre-topology set, is reported.
3. **Status health.** Every record's `status == "active"`. Stop/failure statuses to check for: `human_rejected`, `pipeline_stopped`, `insufficient`, `unapproved_candidate`, `validation_failed`, `superseded`. A stopped run's *reason* is not in the run directory -- it lives in the owning agent's shelf archive (`registry_profiles/<registry_id>/archive/`), matched on `run_id`.
4. **Artifact set matches topology.** The `*_output_round_NN.json` files present agree with the topology recorded in `evidence_store.json`'s `topology_selection_*` key (a Debate run has `p2/p3/s1`, no `p1/c1/c2`).
5. **Provenance chain resolves.** Each record's `input_refs` cite `artifact_id`s that actually exist in the same snapshot, forming a DAG; `approved_at` is monotonic along it.
6. **Payload integrity.** `payload_sha256` recomputes from `payload` (`json.dumps` with `ensure_ascii=False, sort_keys=True, default=str`).
7. **Round consistency.** Each record's `review_round` agrees with the `_NN` suffix count of that agent's round files.

Reporting unit: one row per run, rolled up to a use-case pass ratio. As a closing note only (never a defect on its own): run-to-run variance in LLM-composed fields (ids, lists, materiality-style flags, counts) is expected and is not reported here -- it is only worth a mention if a field that must be deterministic (a fixed enum, a code-computed total) varies across runs.

T5 -- Negative Activation Control

What: the complement of T1 -- agents that do **not** belong to the selected topology's chain, or to the pipeline family at all, never activated. Concretely: RC1.1/RC1.2/RM1.1/RM1.2/RD3 never appear in a tabular (UC1/UC2/UC2.1/UC3) run's set file; DA1/D1/D2/D3 never appear in a UC4 run's set file (UC4 is `pipeline_family="document"` and never touches those tabular-only agents); a Debate use case's set file never contains `p1`, `c1`, or `c2`; a Direct use case's set file never contains `p2`, `p3`, `s1`, `c1`, or `c2`; a Coalition use case's set file never contains `p1`, `p2`, `p3`, or

`s1`.

This test exists because the failure mode is real, not hypothetical:
`evaluation/model_comparison/GOLD_AUDIT.md` records an actual case of a `p1`
shelf record leaking into a Debate use case's draft from a stale cross-run
shelf. T5 is the standing check against that recurring.

Data source: `set(agent_id for record in run_set.values()) -
set(contract["supporting_agents"])` must be empty, for both families (see §4
for UC4's own `contract.json`, now built).

Scoring: pass/fail per run; on failure, name the intruding agent.

4. Checkpoint tables (two pipeline families, kept separate)

Tabular (UC1, UC2, UC2.1, UC3) -- as authored on disk today

Checkpoint files under `gold/real/draft/checkpoints/` are named by agent id
(the retired `g1_..`-`g5_..` naming was replaced under `schema_version: 2`,
see `GOLD_AUDIT_PASS2.md`). Human gates G1-G5 map to DA1/R2/CP1/TS1/F1
respectively. D1 and D3 are `EVIDENCE_ONLY_AGENTS` -- deterministic, never a
checkpoint, kept in `source_artifacts/` only. EX1/EX2 are advisory and have no
gate of their own, but as of `v002` they are full scored gold checkpoints
(see "EX1/EX2 as gold checkpoints" §4a below); being scored does not make
them binding inside the pipeline -- see AGENTS.md "External commentary agents".

Use case	Topology	Checkpoints (v001)	v002 adds
UC1 (`uc1_enefg_threshold_check`)	Direct	`r1, da1, r2, d2, cp1, ts1, p1, f1`	`ex1, ex2`
UC2 (`uc2_ets1_scope_memo`)	Debate	`r1, da1, r2, d2, cp1, ts1, p2, p3, s1, f1`	`ex1, ex2`
UC2.1 (`uc2_1_ets1_scope_memo_annual`)	Debate	same set as UC2	`ex1, ex2`
UC3 (`uc3_csrd_scope2_measure`)	Coalition	`r1, da1, r2, d2, cp1, ts1, c1, c2, f1`	`ex1, ex2`

Each use case's own `field_classes` map is the source of truth for T3 and is
not repeated here -- see `gold/contract.json` for that use case.

Document (UC4) -- `gold/contract.json` built, draft generated, not yet frozen

UC4 (`uc4_esrs_2026_impact`) now has a `gold/contract.json` and its own
`create_gold.py` under `evaluation/model_comparison/uc4_esrs_2026_impact/`,
following the same contract → draft → human review → freeze flow as
UC1/UC2/UC2.1/UC3. `evaluation/uc4_document_audit/` remains a separate,
still-active packet-audit cycle (§2, third row) -- the two are not merged.

Checkpoint set, following AGENTS.md's document chain
(`RD1 -> RD2 -> RC1.1 -> RC1.2 -> RM1.1 -> RM1.2 -> RD3 -> human -> CP1 ->
TS1 -> P4/P5/P6 -> F2 -> human`), all 11 checkpoints widened to every
LLM/Hybrid agent plus RD3 (the same "widen checkpoints" decision already
applied to the tabular contracts):

Checkpoint	agent_type	human_gate	pipeline_order
`RC1.1`	llm	`null`	3
`RC1.2`	llm	`mapping`	4
`RM1.1`	llm	`null`	5
`RM1.2`	llm	`mapping`	6
`RD3`	deterministic	`mapping`	7
`CP1`	llm	`cp1`	9
`TS1`	llm	`ts`	10
`P4` / `P5` / `P6`	llm	`null`	11 / 12 / 13
`F2`	hybrid	`f2`	15
`EX1` (v002)	llm	`null`	8
`EX2` (v002)	llm	`null`	16

‘RD1‘/‘RD2‘ stay evidence-only (deterministic, no LLM, fully derivable from tracked files -- same status as D1/D3 in the tabular family) and never appear as shelf records, so they are outside ‘PRE_TOPOLOGY_AGENTS‘/checkpoints entirely. ‘RC1.1‘/‘RM1.1‘ are the blind-pass X.1 halves: checkpointed for audit but never gate-displayed on their own (‘human_gate: null‘), matching the tabular pattern for an intermediate step. ‘P4‘/‘P5‘/‘P6‘ are ‘topology_scoped: false‘ for UC4 -- unlike UC1’s P1 or UC3’s C1/C2, they always run regardless of TS1’s choice (AGENTS.md), so T1 for UC4 checks their presence unconditionally rather than as a topology-dependent set. ‘expected_topology‘ is ‘Debate‘ (observed ‘topology_scores: {Direct: 20, Debate: 72, Coalition: 8}‘ on the reference run). No ‘DA1‘, no ‘R2‘ -- UC4’s ‘pipeline_family="document"‘ never enters the tabular DA1/D1/D2/D3 chain (AGENTS.md "UC4 -- document pipeline").

Field-class assignment follows the same five-class vocabulary from §2, fully enumerated in ‘evaluation/model_comparison/uc4_esrs_2026_impact/gold/contract.json‘ -- not repeated here, same convention as the tabular use cases in the table above. Two UC4-specific notes worth carrying into any T3 read: RD3’s row fields are ‘data_derived‘ even where they look like prose, because RD3 only copies RC1.2/RM1.2’s content through a deterministic join -- a mismatch there means the upstream agent drifted, not RD3; and P4/P5/P6 disagreeing with each other on ‘effort_estimate‘/‘action_priority‘ is the intended Debate signal, not a T4 instability finding.

4a. EX1/EX2 as gold checkpoints (v002, both pipeline families)

All five packages (UC1, UC2, UC2.1, UC3, UC4) were amended to ‘v002‘ to add EX1 and EX2 as scored checkpoints -- ‘v001‘ is untouched. EX1 sits immediately after the Stage-1 human gate and EX2 immediately after the Stage-2 human gate in ‘pipeline_order‘, in both pipeline families. ‘human_gate‘ stays ‘null‘ for both -- EX has no gate of its own, and the checkpoint sits outside T1’s topology-membership test (‘topology_scoped: false‘).

‘bullets[]‘ carries no stable id (‘CommentaryBullet‘ has no ‘bullet_id‘), so it is matched as an **unordered set**, not by ‘join_key‘: a model is judged on whether it surfaced comparable concerns from allowlisted institutions, not on reproducing the gold’s bullet text, order, or count. Field classes: ‘stance‘/‘relevance‘/‘grounding‘/‘sources[].institution‘ are ‘exact‘ (closed Literals / fixed allowlist); ‘point‘/‘sources[].title‘/ ‘institutions_consulted[]‘/‘search_notes[]‘ are ‘semantic‘; ‘sources[].url‘/‘sources[].published‘ are ‘ignored‘.

The one script-level deviation from the tabular ‘create_gold.py‘: UC4 has no ‘R1‘, so the registry-drift-anchor read (‘records["R1"]["input_refs"]["registry_sha256"]‘ in the tabular copies) instead reads ‘RC1.1‘ (falling back to ‘RM1.1‘) -- both are the first document-chain agents to consume the registry seed. ‘PRE_TOPOLOGY_AGENTS‘ and the ‘active_document_set.json‘ filename are the other two family-specific constants; see the UC4 ‘create_gold.py‘’s inline comments for the exact diff against the tabular copies.

5. Data source map

Everything T1/T2/T5 need is under one evaluation run directory:

‘‘‘text
evaluation/opus-5/{real,public}/<registry_id>/<TIMESTAMP>Z_regulation_cli/run_NNN/
+-- active_profile_set.json   # tabular family
+-- active_document_set.json   # document family (UC4) -- filename chosen
| # by pipeline_family, regulation_cli.py:543
+-- <agent>_output_round_NN.json # e.g. ts1_output_round_01.json
+-- evidence_store.json
‘‘‘

(‘experiments/{real,public}/‘ is where a run is originally produced by the
CLI; the run sets this taxonomy scores against are the copies recorded under
‘evaluation/opus-5/‘, per use case and registry id.)

Reading a run’s agent set and topology, minimal form:

‘‘‘python
p = run_dir / "active_profile_set.json"
p = p if p.exists() else run_dir / "active_document_set.json"
s = json.load(open(p))
records = [v for v in s.values() if isinstance(v, dict)]
registry_id = {r["registry_id"] for r in records}   # not in the path -- read it
agents = sorted(r["agent_id"] for r in records)   # canonical ids, not shorthand keys
topology = (s.get("ts") or {}).get("payload", {}).get("selected_topology_id")
‘‘‘

Gold-side equivalents, per use case under
‘evaluation/model_comparison/<usecase>/gold/real/‘:
- ‘draft/manifest.json‘ → ‘available_agents[]‘, ‘missing_checkpoints[]‘,
  ‘missing_supporting_agents[]‘, ‘registry_drift‘, ‘lineage_consistent‘.
- ‘draft/checkpoints/<cp>.json‘ → ‘_gold.*‘ (includes ‘observed_agents[]‘ on
  ‘ts1.json‘ only) and ‘payload.*‘.
- ‘draft/source_artifacts/<agent>.json‘ → raw shelf/run record, including
  ‘input_refs‘ (needed for T2).
- ‘frozen/v<NNN>/‘ → same shape, plus ‘checkpoint_file_sha256{}‘ once frozen.

Note for this checkout: tabular run set files are recorded under gitignored
‘evaluation/opus-5/real/<registry_id>/‘ and are not present on every machine --
verify T1/T2/T5 extraction against them on a machine holding the real-data
run sets. UC4’s document-family run sets are public and committed under
‘evaluation/opus-5/public/uc4_esrs_2026_impact/‘.

6. Freeze Readiness

A single rollup derived from T1-T5, reported alongside -- never instead of --
the existing gates:

- **Hard gate (must all pass):** T1, T2, T4, T5. A run that fails any of
  these did not exercise the pipeline as designed, or did not actually
  complete -- its content is not a fair T3 input regardless of how good that
  content looks. T4 joins the hard gate under the new definition (§3) because
  it is now a deterministic completion/integrity checklist, not a variance
  signal -- a run that never finished, carries a stale record, or has a broken
  ‘input_refs‘ chain should not be scored for substantive quality at all.
- **Quality band:** T3’s 3a/3b rollup (§3) -- structural conformance plus the
  substantive rubric mean, read against the same bands ‘GOLD_HUMAN_REVIEW.md‘
  already uses (9-10 freeze as-is · 7-8 freeze with a note · 5-6 needs fix ·
  0-4 reject) -- this taxonomy does not invent new bands, it feeds the same
  ones from a field-level T3 table instead of a single per-checkpoint
  impression. Any 3c glaring defect caps the band at 5-6 regardless of the
  rubric mean.

This rollup does not touch ‘create_gold.py freeze‘’s own eight validation
gates (version format, missing checkpoints/supporting agents,
‘overall_decision == "approved"‘, every checkpoint’s ‘review.json‘ decision
‘approved‘, registry drift resolved + live re-check, lineage reviewed,
refusal to overwrite an existing version) -- those remain the sole mechanism
that actually writes a frozen package.

7. Validity limits

TS1 is an LLM agent, not a deterministic function (see AGENTS.md "Agent Type
Classification"). A T1 result of "N/N correct" across a batch of runs is an
observed empirical reliability under this use case set and this

```
registry/agent-description design -- not a proof that selection is
deterministic or that it generalizes to a differently worded or more
ambiguous use case. State results this way: "TS1 selected the expected
topology and agent set in N of N observed runs," not "TS1 always selects
correctly."

Related evaluation-methodology references (precision/recall, pass^k, schema
matching, execution accuracy) are indexed in
‘evaluation/archive/references.md‘.
```

`T3` (Output Substantiveness) reads this specification into two tables the main text reports as one two-layer score. Every checkpoint field carries one of four comparison classes fixed at authoring time (Table A.1), and every checkpoint is separately checked against a closed list of five glaring defects (Table A.2), regardless of which comparison class its fields carry.

Comparison class	How it is scored
`exact`	A closed enum, boolean, or Literal, checked against its fixed value space.
`numeric`	Checked within its declared tolerance.
`data_derived`	Comparable only against the same data-catalog snapshot gold used; a mismatch is reported as drift, not a miss.
`semantic`	Free text and LLM-composed lists, scored `2/1/0` against their own cited source: full grounding, partial grounding, or a claim contradicting its source or its own record.

Table A.1: How `T3` scores a checkpoint field, by the comparison class fixed for it at authoring time.

Glaring defect	When it is recorded
Record contradiction	A field contradicts the narrative in its own record.
Unsupported citation	The cited source does not support the field’s claim.
Empty required field	A deliverable field required by the contract is empty.
Unreconciled aggregate	An aggregate or rollup does not reconcile with its stated components.
Invalid fixed value	An enum or Literal value falls outside its declared set.

Table A.2: The closed list of defects `T3` records separately from comparison-class scores. A run’s difference from gold does not create another defect outside these five.

A.2 Gold-Scoring Judge Prompt

The judge prompt operationalizes the taxonomy for one named use case. Claude Opus 5 used the same prompt for every scored case. The listing below reads the complete instructions directly from `evaluation/JUDGE_PROMPT.md`.

```
# Gold-Scoring Judge Prompt (T1-T5)

Score a use case’s frozen gold pipeline against its own real/public run set. This
prompt is use-case-agnostic: it works for UC1, UC2, UC2.1, UC3, or UC4 unchanged.
```

```
Trigger it with the use case named directly ("audit UC4 with the prompt" / "prompt ile
uc4'ü auditle" / "score UC1 against gold"). Everything else -- registry id, expected
topology, checkpoints, run set -- is resolved from disk in Step 0, never assumed or
hardcoded here.

This prompt implements the five tests `evaluation/TAXONOMY.md` *defines* but does not
execute (TAXONOMY.md §0: "does not run anything, does not score anything"). Read
TAXONOMY.md first if you have not already -- this file assumes its vocabulary
(field-comparison classes, the semantic 2/1/0 rubric, T1-T5 names) without repeating
the definitions.

Work inside this repository. Follow every step in order. Do not skip ahead, do not
backfill a later step into an earlier one.

## Step 0 -- Resolve inputs from disk

1. Map the spoken use case name to a `registry_id` by listing
   `evaluation/model_comparison/*/`: UC1 = `uc1_enefg_threshold_check`, UC2 =
   `uc2_ets1_scope_memo`, UC2.1 = `uc2_1_ets1_scope_memo_annual`, UC3 =
   `uc3_csrd_scope2_measure`, UC4 = `uc4_esrs_2026_impact`.
2. Read `evaluation/model_comparison/<registry_id>/gold/real/frozen/<latest vNNN>/manifest.json`.
   **Require `status == "frozen"`. If no frozen gold exists for this use case, stop and
   tell the user -- T3 has no anchor without one; do not improvise a comparison against
   the draft.** Take `expected_topology`, `source_run_ids`, `source_registry_sha256`,
   `version`.
3. Read the same directory's `gold/contract.json` for `supporting_agents`,
   `checkpoints{}` (each with `agent_id`, `agent_type`, `human_gate`, `topology_scoped`,
   `pipeline_order`, `comparison_policy`, `collections{join_key, expected_count}`,
   `field_classes`, `tolerances`), `topology_policy`, `review_checklist`.
4. Read `evaluation/model_comparison/<registry_id>/create_gold.py`'s own
   `PRE_TOPOLOGY_AGENTS` constant -- it differs by pipeline family (tabular:
   `{R1,DA1,R2,D1,D2,CP1,TS1,D3}`; UC4 document: `{RC1.1,RC1.2,RM1.1,RM1.2,RD3,CP1,TS1}`)
   -- do not assume one family's set for the other.
5. Locate the run set: `evaluation/opus-5/{public,real}/<registry_id>/<TS>Z_regulation_cli/run_001/`.
   The set file is `active_document_set.json` when `pipeline_family == "document"`,
   otherwise `active_profile_set.json` (`src/epoch_switch/regulation_cli.py:543`).
   **UC1-UC3 real-data runs are gitignored and may not exist on this machine. If the
   use case's run directory is absent or empty, stop and tell the user plainly -- do not
   half-score from whatever happens to be present.**
6. **Exclude every run listed in the gold's `source_run_ids` from the scored set** --
   scoring a gold against the run it was derived from is circular. Name the excluded
   run(s) in the report header and do not report a per-run score for them.
7. Determine the next output directory:
   `evaluation/model_comparison/<registry_id>/scoring/scoring_N/findings/` (N = 1 if
   none exist yet). This directory and `ai_usage/steps_logs.md` are the only paths you
   write to -- each use case's evaluation stays self-contained under its own folder
   (AGENTS.md, "Evaluation isolation").

## Execution discipline -- apply this to every step below

- Strict order: T1/T2/T5 → T3 → T4 → English report → Turkish translation. Never write
  the report before every findings file exists.
- Each test is an isolated unit of work. Finish it, write its findings JSON, and only
  then move to the next. To reuse a fact from an earlier test, re-read that test's
  findings file -- do not rely on your own recollection of it.
- **Independent derivation before anchoring.** For T1, derive the expected topology
  yourself from `src/epoch_switch/selection/topology_library.py`'s specs and the
  registry's own demand fields, and write that derivation down -- before opening the
  run's `ts`/`TS1` record. Do not let TS1's stated answer anchor your judgment.
- Deterministic checks (T1, T2, T5, and T3's `exact`/`numeric`/`data_derived` fields)
  may use throwaway analysis code (Python, in your scratchpad). **Do not commit a
  scorer script** -- this is a one-off judged analysis that produces a Markdown report,
  not new pipeline tooling.
- Semantic scoring is LLM-as-judge: read the gold text's *meaning*, never its surface
```

```
  form or wording. Every score below ‘2‘ needs a one-line reason and the exact field
  path that decided it. No unsourced claims.
- A field’s comparison class is fixed at authoring time in ‘contract.json‘ and is
  **never re-decided per run** -- you are applying the class, not choosing it.
- **Gold is a structural template, not a content answer key (TAXONOMY.md §1a).** A
  different but well-grounded regulation set, a different LLM-assigned id, a broader or
  narrower field list, a different collection row count -- none of these are findings by
  themselves. Ground a claim against the run’s **own cited source** (the regulation text,
  the data catalog, the registry demand) before calling anything a defect. Do not report a
  difference from gold as a finding unless it lands on T3’s 3c glaring-defect list
  (field contradicts its own narrative, fabricated/misattributed citation, a required
  field left empty, an aggregate that does not reconcile with its own components, an enum
  outside its declared set). When in doubt, it is not a finding.

## T1 / T2 / T5 → ‘findings/t1_t2_t5.json‘

One entry per run per test, ‘{test, run_dir, verdict, reason}‘; pass/fail unless noted.

**T1 -- Selection Correctness.** Pass iff the run’s ‘ts‘/‘TS1‘ record’s
‘payload.selected_topology_id == expected_topology‘, AND the observed post-topology
agent set equals the contract’s expected set exactly (no missing, no extra). Compute
the observed set as ‘{agent_id of every record in the run’s set file} -
PRE_TOPOLOGY_AGENTS‘ (read the canonical uppercase ‘agent_id‘ field inside each record,
never the lowercase shorthand key the file uses, e.g. ‘rc1_1‘). Compute the expected
set as ‘supporting_agents - PRE_TOPOLOGY_AGENTS‘ from the contract. On divergence, the
contract’s ‘topology_policy.on_divergence‘ rule applies verbatim:
‘TS1.selected_topology_id‘ mismatch counts as a miss; any ‘topology_scoped‘ checkpoint
from an un-selected topology is ‘not_applicable‘.

**T2 -- Process Conformance.** Five-item checklist per run, each pass/fail:
1. Checkpoints appear in ascending ‘pipeline_order‘ in the run’s own timeline
   (‘created_at‘/‘approved_at‘ ordering in the set file).
2. Only the selected topology’s final agent opened a human ‘[a]pprove/[r]eject‘ gate --
   intermediate steps with ‘human_gate: null‘ in the contract were auto-promoted, not
   gated, in the actual run.
3. Post-R2/RD3 boundary held: CP1 and downstream records’ ‘input_refs‘ cite only the
   approved isolated memory (R2 ‘in_scope_findings‘ + DA1 handoff for tabular; RD3’s
   join summary for UC4) -- never raw registry ‘site_filter‘/‘time_window‘/
   ‘regulation_refs‘.
4. Direct-chain input boundaries held where applicable: P1’s ‘input_refs‘ cite only R2
   ‘in_scope_findings‘ + D3 payload; F1’s cite P1’s output, R2 ‘in_scope_findings‘, and
   the registry demand snapshot only.
5. EX1 ran immediately before the Stage-1 gate and EX2 immediately before the Stage-2
   gate (or both were skipped together with a logged reason), and neither EX1 nor EX2
   content appears inside any other agent’s ‘payload‘ or ‘input_refs‘.

**T5 -- Negative Activation Control.** Pass iff
‘{agent_id of every record} - supporting_agents‘ is empty **after whitelisting ‘EX1‘
and ‘EX2‘** -- they always run and are advisory (AGENTS.md, "External commentary
agents"), but are deliberately absent from ‘supporting_agents‘, so a naive subtraction
reports them as false intruders. On failure, name the intruding agent. Also confirm the
family exclusion holds: no ‘DA1‘/‘D1‘/‘D2‘/‘D3‘ in a document-family run’s set file; no
‘RC1.1‘/‘RC1.2‘/‘RM1.1‘/‘RM1.2‘/‘RD3‘ in a tabular run’s set file.

Apply these carve-outs across all three tests:
- ‘not_applicable‘ **is not zero** -- those agents never ran, so they were neither right
  nor wrong. Never fold ‘not_applicable‘ into a pass/fail denominator as a failure.
- A run predating a feature legitimately lacks it (for example, a run captured before
  EX1/EX2 existed has no ‘ex1‘/‘ex2‘ records) -- that is ‘not_applicable‘, not a T2
  failure. Check the run’s date against when the feature landed (git log / AGENTS.md)
  before scoring it as missing.
- Record each run’s backend (model string, or its absence) from
  ‘evaluation/opus-5/README.md‘’s run manifest if this use case has one. A mixed-backend
  set is a reporting caveat, never a score adjustment.
```

```
## T3 -- Output Substantiveness → `findings/t3.json`

For every checkpoint in the contract, for every scored run, run three sub-checks
(TAXONOMY.md §3). Gold is used only as a structural template here -- never as the
content this run is required to reproduce.

**3a -- Structural conformance.** Does the checkpoint exist; are all fields the
contract declares for it present and non-empty; do declared `collections` exist and
join cleanly on the contract's `join_key`. Missing or empty is a failure. A row count
that differs from `expected_count`, a different list length, or different id values
are **not** a 3a failure by themselves -- report the count difference once as a
structural note, not as a per-row miss.

**3b -- Substantive quality.** Score by class:
- `exact` (now restricted to closed enums/booleans/Literals -- `verdict`,
  `risk_level`, `selected_topology_id`, `entity_grain`, and similar fixed value
  spaces) -- pass/fail against the fixed value space itself, not against the value
  gold happened to record.
- `numeric` -- within the checkpoint's `tolerances` entry (for example
  `topology_scores.*: 15`). Pass/fail.
- `data_derived` -- comparable only if the run used the same data catalog snapshot as
  gold; on divergence, report as **data drift, not a T3 miss**. UC4 carve-out: RD3's
  row fields are `data_derived` even where they read as prose, because RD3 only
  copies RC1.2/RM1.2 output through a deterministic join -- a mismatch there means the
  upstream LLM agent drifted, not that RD3 computed the join wrong.
- Everything else (`semantic`, and any LLM-composed id/list/count field previously
  marked `exact`) -- score **every** such field, on every scored run (full coverage,
  no sampling), `2`/`1`/`0` against the field's **own cited source**, never gold's
  wording or list membership: `2` -- the claim is true against its cited source,
  internally consistent with the agent's own other fields and with upstream agents,
  and responsive to the registry demand; `1` -- right general claim, weak or missing
  grounding, or a material omission; `0` -- contradicts its own source, contradicts
  itself, or is absent where the contract requires content. Reuse the checkpoint's
  `comparison_policy` sentence and the contract's `review_checklist` items as the
  grounding questions to ask. For LLM-composed lists/ids/counts specifically: spot-check
  the run's actual values against the underlying source text (the regulation markdown,
  the report section, whatever the field cites) -- a court reporter transcribing the
  same testimony twice in different words is not wrong the second time. If the content
  is real and correctly identified but differs from gold's list in which sub-points it
  includes or how it groups them, or a persona agent emits extra substantive on-topic
  rows under its documented posture, that is the agent behaving as designed -- record it
  as **grounded, not a T3 finding**, with the source line(s) checked. Only score it a
  genuine `0` if the cited content does not actually say what the run claims.
- `ignored` -- skip entirely (provenance/volatile fields: timestamps, hashes, snapshot
  ids).

**3c -- Glaring defects.** A closed list -- nothing outside it is a T3 defect: a field
contradicting its own record's narrative; a citation whose source does not say what is
claimed; a contract-required deliverable field left empty; an aggregate/rollup that
does not reconcile with its own stated components; an enum/Literal value outside its
declared set. Scan every checkpoint's payload for these independent of any gold
comparison, and list any found explicitly.

Join list-valued fields (`collections`) on the contract's `join_key`, never by list
index.

Roll up: one table per checkpoint (3a pass/fail per required field, 3b rubric/pass-ratio
per scored field, 3c a flat defect list) → per-checkpoint summary → per-use-case summary
across checkpoints. Separately, walk the contract's `review_checklist` and mark each
item PASS/fail against what you found, in `GOLD_HUMAN_REVIEW.md`'s convention.

## T4 -- Execution Integrity → `findings/t4.json`
```

```
Gold-independent: did the pipeline actually run, end to end, cleanly. One row per
**run** (not per use case), each item pass/fail, computable entirely from the run
directory:

1. **Completion** -- the terminal agent's record (`f1`/`f2`) exists and its `run_id`
   equals the run directory name (`_snapshot_is_end_to_end()`,
   `src/epoch_switch/regulation_cli.py:606`). File existence alone is not completion --
   the same snapshot is written on rejection and post-gate exception paths too.
2. **No stale records** -- every non-null record's `run_id` matches the run directory;
   flag any mismatch or any `carried_over: true` outside the legitimately-resumable
   pre-topology set.
3. **Status health** -- every record's `status == "active"`. Watch for
   `human_rejected`, `pipeline_stopped`, `insufficient`, `unapproved_candidate`,
   `validation_failed`, `superseded`. A stopped run's reason lives in the owning
   agent's shelf archive (`registry_profiles/<registry_id>/archive/`), matched on
   `run_id` -- not in the run directory.
4. **Artifact set matches topology** -- the `*_output_round_NN.json` files present
   agree with `evidence_store.json`'s `topology_selection_*` key.
5. **Provenance chain resolves** -- every `input_refs` entry cites an `artifact_id`
   that actually exists in the same snapshot (a DAG, not dangling); `approved_at` is
   monotonic along it.
6. **Payload integrity** -- `payload_sha256` recomputes from `payload`
   (`json.dumps(payload, ensure_ascii=False, sort_keys=True, default=str)`).
7. **Round consistency** -- each record's `review_round` agrees with the `_NN` suffix
   count of that agent's round files.

State plainly in the findings and the report: T4 is now a deterministic
did-it-actually-complete-cleanly checklist, not a content-variance signal. Report
run-to-run variance in LLM-composed fields (ids, lists, materiality-style flags,
counts) only as a closing note, never as a T4 finding -- it is expected. The one
exception: if a field that must be deterministic (a fixed enum, a code-computed
total) varies across runs, that is worth naming. If scoring UC4, P4/P5/P6
disagreeing with each other on `effort_estimate`/`action_priority` is the intended
persona-diversity signal, not a T4 or T3 finding.

Carry forward known, already-fixed defects rather than re-deriving them: if a run
predates the stale-shelf-leak fix in `regulation_cli.py` (`_scoped_snapshot`/
`_write_run_snapshot`), a stale `P1`/`P3`/`S1` (or similar) record legitimately fails
checks 2/3 here -- report it as a defect already fixed in code, name the fix, and do
not treat it as an open issue requiring further action.

## Report -- write English first, in full

Write `evaluation/model_comparison/<registry_id>/scoring/scoring_N/SCORING_REPORT.md`
only after all findings files (`t1_t2_t5.json`, `t3.json`, `t4.json`) exist. Use this
exact structure:

1. **Title + metadata** -- registry id, gold version and freeze date/reviewer, the runs
   scored, the excluded gold-source run(s), each run's backend, judge.
2. **Does the pipeline work? -- the four questions, answered directly, before any score
   table.** One short paragraph each, results-first, no hedging:
   - Does the pipeline really run end to end? (T4)
   - Are the actually-required agents running -- the right topology, the right chain?
     (T1, T5)
   - Does the flow follow our architecture's working principles? (T2)
   - Are the results real and useful -- grounded, self-consistent, responsive to the
     registry's demand? (T3, gold used only as a structural template -- see
     TAXONOMY.md §1a)
3. **Score table.** Rows: T1, T2, T3, T4, T5, bold **Overall**. Columns: Result,
   one-line reason. Base every row on the findings files, not a fresh impression.
4. **Method** -- coverage numbers only (runs/checkpoints/fields/rows scored vs. total).
   Terse.
5. **T1 / T2 / T3 / T4 / T5**, each in this fixed sub-shape:
   - `### Issues` -- bullets, each starting with its verdict tag in brackets (`[fail]`,
```

```
     `[drift]`, `[not_applicable]`), one line, with the field path, agent id, or run
     directory referenced inline. For T3, only 3a structural failures and 3c glaring
     defects belong here -- a difference from gold that grounds cleanly against its own
     source is not an issue; say so under "What held up" instead.
   - `### What held up` -- bullets tagged `[pass]`, same convention.
   - One closing paragraph (2-3 sentences): the net judgment for this test -- a summary
     verdict, not a hedge.
6. **Per-checkpoint T3 tables** -- rows = fields, columns = sub-check (3a/3b/3c) +
   per-run result.
7. **Freeze Readiness rollup** (TAXONOMY.md §6) -- hard gate: T1 + T2 + T4 + T5 must all
   pass; quality band from T3's 3a/3b rollup, read against `GOLD_HUMAN_REVIEW.md`'s
   bands (9-10 freeze as-is / 7-8 freeze with note / 5-6 needs fix / 0-4 reject); any
   3c glaring defect caps the band at 5-6 regardless of the rubric mean.
8. **Prioritized fix list** -- grouped as: blocks the pipeline / degrades the result /
   cosmetic. Each item names the exact code or prompt location and the smallest fix.
   Do not list a difference from gold as an item unless it is a 3c glaring defect or a
   T4 completion failure -- "gold disagrees" alone is never a fix-list item.
9. **Threats to validity** -- terse bullets. Always include: the gold-source-run
   circularity exclusion, any mixed-backend set, and the small sample size (n runs).
10. **Per-agent scorecard** -- one row per `supporting_agents` entry. Columns: T3 score
    (/checkpoint mean, 3a/3b only), T5 clean (yes/no), Issue (fix-list item number if
    named there, blank otherwise).

Phrase every claim in the register TAXONOMY.md §7 requires: "TS1 selected the expected
topology and agent set in N of N observed runs," never "TS1 always selects correctly."

## Translation -- Turkish, last, full

Only after `SCORING_REPORT.md` is complete and final, write
`evaluation/model_comparison/<registry_id>/scoring/scoring_N/SCORING_REPORT_TR.md`: a
full Turkish translation of the finished English report -- same structure, same section
order, same tables -- not a summary, not written concurrently with the English version.

## Finish

Log the completed scoring pass to `ai_usage/steps_logs.md` in the project's required
format: English only, header `## Step <N> -- YYYY-MM-DD HH:MM`, a short title, at most 4
bullets of at most 15 words each. Write nothing outside
`evaluation/model_comparison/<registry_id>/scoring/scoring_N/` and
`ai_usage/steps_logs.md` during this pass -- propose fixes in the fix list, do not apply
them to pipeline code, agent prompts, or schemas.
```

A.3 Frozen Gold Reference Register

The judge prompt defines how a package is read, but it does not identify which five frozen packages Chapter 4 used. Table A.3 records that missing link from use case to version, source run, topology, and checkpoint set. The masked copies preserve those decisions without publishing confidential site identities or measurements. They document the evaluation; they do not replace the private tabular packages used for scoring.

Case	Gold	Review	Frozen	Source run	Topology	Count
UC1	v002	Author	2026-09-08	2026-09-06_165447Z_regulation_cli	Direct	10
UC2	v002	Author	2026-09-08	2026-09-06_095741Z_regulation_cli_02	Debate	12
UC2.1	v002	Author	2026-09-08	2026-09-06_095742Z_regulation_cli	Debate	12
UC3	v002	Author	2026-09-08	2026-09-06_163313Z_regulation_cli	Coalition	11
UC4	v002	Author	2026-09-08	2026-09-01_104148Z_regulation_cli	Debate	13

Table A.3: The five frozen gold references used for the reported evaluation. Each masked disclosure package is stored by case under public/gold/, followed by v002/; its SHA256SUMS file fixes the public derivative's contents.

Gold record means:

- Version: v002 adds EX1/EX2 to the scored checkpoints.
- Reviewer: the author approved each package; public JSON masks the name.
- Source run: the run from which gold was drafted and later excluded from scoring.

Public package means:

- Tabular packages mask site identity, measurements, people, paths, and private fingerprints.
- UC4 retains public document content while masking people and machine-local paths.
- Public hashes are calculated after masking and verify only the disclosure copy.

Appendix B

Configuration

This appendix records the runtime configuration behind every scored run: the backend presets EPOCH can call, how each provider family's reasoning mode is switched on, the per-agent token budgets, and how model identity, reasoning, and token usage are recorded rather than assumed. All values are read directly from `src/epoch_switch/config.py` and `.agents/llm-backend-reference.md`.

B.1 Backend Presets

All six presets route through Azure AI Foundry under `azure_ad` (Entra ID) authentication, so no static API key travels on the wire. Claude Opus 5 (`opus5`) was the backend for the scored evaluation runs reported in Chapter 4; the other five are configured and independently selectable (Section B.4) but were not themselves scored. Development itself started on `sonnet46` and ran there in full before moving to `gpt55`. Both backends produced consistent architectural decisions, which is why the pipeline design carried over unchanged across them. Once evaluation moved to broader, repeated runs, a stronger model became the practical choice, which is how `opus5` became the scored backend.

Preset	Kind	Model	Reasoning effort
`opus48`	`foundry_anthropic`	`claude-opus-4-8`	high
`opus5`	`foundry_anthropic`	`claude-opus-5`	high
`sonnet46`	`foundry_anthropic`	`claude-sonnet-4-6`	high
`gpt55`	`azure_openai`	`gpt-5.5`	xhigh
`gpt56sol`	`azure_openai`	`gpt-5.6-sol`	xhigh
`gpt54mini`	`azure_openai`	`gpt-5.4-mini`	medium

Table B.1: The six named backend presets in `config.py`'s `BACKENDS` registry.

B.2 Reasoning Switch and Provider Field Mapping

The two backend kinds spell reasoning, output limits, and token usage differently; EPOCH keeps each provider's own field names in its provenance record (Section B.5) rather than inventing a shared vocabulary.

Concern	`foundry_anthropic`	`azure_openai`
Output cap	`max_tokens`	`max_completion_tokens`
Reasoning switch	`thinking={"type": "adaptive"}` — the only valid on-mode; `budget_tokens` is rejected. Omitting `thinking` leaves reasoning **off**.	no separate switch — `reasoning_effort=` is both switch and level.
Effort	`output_config={"effort": ...}`	`reasoning_effort=`
Token usage	`usage.input_tokens`, `usage.output_tokens` (split)	`usage.prompt_tokens`, `usage.completion_tokens`, `usage.total_tokens` (split plus total)

Table B.2: How the two active provider kinds spell the reasoning switch, effort, output cap, and token usage.

Because omitting `thinking` silently leaves an Anthropic-family call's reasoning off rather than raising an error, Rule R23 of the `/system-check` catalog (Appendix D.6) requires the switch to be explicit and recorded on every call, not assumed from the backend preset's configuration alone. A plain `openai`-kind backend sends no reasoning parameter at all and may not back an LLM agent under R23.

B.3 Token Budgets and Batching

The global default is `TEMPERATURE=0.0`, `MAX_TOKENS=32,000`. Four agents run at a heavier 64,000-token cap because Opus 5's adaptive reasoning counts against `max_tokens`. These four emit one whole-portfolio or whole-table JSON payload per call, the same four that actually truncated at 32,000 during development.

Tier	**Tokens**	**Agents**
Medium	32,000	R2, DA1, D1, D2, CP1, TS1, P1, P2, P3, S1, RC1.1/RC1.2, RM1.1/RM1.2, P4–P6, C1, EX1/EX2
Heavy	64,000	R1, F1, F2, C2

Table B.3: The two token-budget tiers and the agents on each.

Batching and concurrency are also per-agent: P1 batches 5 locations per call at concurrency 8 with up to 5 web-search uses per batch; P2, P3, and S1 batch 5 sites per call at concurrency 4 (halved from P1's because P2 and P3 run concurrently and would otherwise double Foundry's web-search load); the UC4 blind-pass pairs (RC1.1/RC1.2, RM1.1/RM1.2) and the persona

assessors (P4–P6) batch one unit per call at concurrency 5; EX1/EX2 allow up to 10 web-search uses per call. Every figure is an `EPOCH_*` environment override with the default shown here.

B.4 Per-Agent Model Selection

Every LLM agent can be repointed at a specific backend preset independently of the process-wide default, through `config.py`'s `set_agent_backend()` and `backend_for_agent()`, for example R1 running on `sonnet46` while DA1 runs on `gpt55` in the same run, because `core/llm_client.py` caches one adapter per preset rather than one per process. This per-agent picker is what the model-family comparison in Section 3.6.1 exercises. Deterministic agents (D1, D3, RD1, RD2, RD3) are absent from the selectable set by design, because they never call an LLM. EX1/EX2 depend on `call_json_websearch()`, implemented only on the `foundry_anthropic` adapter, so they fall back to a `foundry_anthropic` preset even when the process-wide default is an `azure_openai` preset.

B.5 Recorded, Not Assumed

Every LLM or Hybrid agent writes an `llm_provenance` block that records model identity (`provider`, `backend_preset`, `model`), whether reasoning was enabled and at what effort, and token usage read from the provider's own usage object, never estimated. This block sits as a top-level key in its shelf record, sibling to `payload`. It is never nested inside `payload`, which would change `payload_sha256` and let configuration metadata leak into whatever consumes that agent's payload downstream. Deterministic agents record `llm_provenance` as an explicit `null`, so "no LLM was involved" is itself on the record. Rules R22–R24 of the `/system-check` catalog (Appendix D.6) audit exactly this: R22 that identity is recorded, R23 that reasoning is on and effort is never `none`, and R24 that token counts come from the provider's usage object rather than a placeholder.

Appendix C

Extended Theory and Regulatory Context

This appendix retains the source-level comparisons behind the compressed argument in Chapter 2. It separates supporting detail from the main selection claim without treating that detail as optional evidence.

C.1 Multi-Agent Literature Families

Pattern	Representative sources	Structure	Assurance boundary
Single-agent execution	ReAct; equal-budget baseline [10, 19]	One model follows a reason–act loop and owns one result.	No independent verification or split ownership.
Same-claim debate	Irving; Du; Liang [11, 12, 13]	Several readers inspect one claim before convergence.	One resolved verdict, not separately owned outputs.
Role decomposition	MetaGPT; ChatDev [14, 15]	Specialized roles produce different artifacts.	Multi-owner work without same-claim verification.
Adaptive graph generation	G-Designer; ARG-Designer; AgentConductor; GoAgent; AFlow [5, 7, 8, 9, 16]	A generator learns or searches for a task-specific graph.	Task success and efficiency, not a declared evidence floor.
Budget routing	MasRouter; BAMAS; ATOM [6, 17, 18]	A policy selects agents or topology under cost constraints.	Budget is binding; output evidence remains outside the objective.

Table C.1: Extended comparison of the multi-agent structure families used in the three-shape reduction.

Same-claim debate has two variants that share one defining property. Irving, Christiano, and Amodei use adversarial argument judged by a human, while Du and colleagues use cooperative exchanges in which agents see each other's answers and converge on one [11, 12]. The interaction style differs, adversarial or cooperative, but both preserve multiple readings of one claim before a final verdict. Liang and colleagues identify a related failure mode: when a model only critiques and revises its own answer, with no outside view, it rarely departs from that answer once committed to it, and self-critique tends to just reconfirm a wrong answer rather than

correct it. The fix is the one debate already supplies, bringing in an opposing view from outside forces the model to genuinely weigh a different answer [13]. Role decomposition answers a different need: agents do not each answer the same question, they take on different pieces of the task. MetaGPT and ChatDev assign different deliverables, such as code or a design document, to different roles, turning coordination from a contest over one answer into an assembly problem [14, 15].

Adaptive graph generation covers four distinct methods for producing a task-specific connection structure. G-Designer learns the connections directly from data with a graph neural network, while ARG-Designer and GoAgent generate the graph autoregressively, selecting and connecting agents or agent groups step by step. AgentConductor instead adapts toward better-performing structures through execution feedback, and AFlow searches systematically through possible execution orders, that is, who runs before whom [5, 7, 8, 9, 16]. Budget routing keeps the agent set fixed instead: MasRouter, BAMAS, and ATOM route among existing agents under a cost or budget constraint, shaping the architecture around cues about the task itself, such as its difficulty or the expertise it needs, rather than producing a new structure [6, 17, 18]. The two families therefore differ in whether the result is a brand-new connection structure or a choice among ready-made configurations.

Structures mean:

- *same claim* — readers address one shared proposition.
- *role decomposition* — owners contribute different artifacts.

Boundaries mean:

- *verification* — an independent reading exists before merge.
- *ownership* — each mandated contribution retains an owner.

C.2 Selection, Measurement, and Oversight Evidence

Current selectors condition architecture on task or resource signals. G-Designer uses task features and token cost; MasRouter routes through a cascaded controller spanning collaboration mode, role allocation, and LLM selection; AgentConductor adapts graph density; BAMAS treats task specification and budget as policy inputs [5, 6, 8, 17]. These systems establish that dynamic choice is feasible. They do not use the evidence artifact required by an approved compliance case as the selection axis.

Output-profile classes draw on measurement theory rather than agent count. Nominal and categorical decisions support binary or qualitative forms, while measured quantities require a value, unit, and attribution [28, 29, 30]. The profile fixes presentation. Assurance separately fixes the evidence structure.

Kroll and colleagues ground auditability as procedural regularity rather than full transparency: showing that a decision followed a pre-committed, verifiable process, for example through cryptographic commitments or software verification, is a stronger and more practical guarantee than disclosing the underlying code or data outright [31]. The EU AI Act turns this into a concrete legal requirement in Article 12, which obliges high-risk systems to keep automatically generated

logs that make their lifecycle traceable [1].

Chen and colleagues show that adding more LLM calls to a compound system does not reliably improve outcomes; scaling behavior flattens or reverses depending on the task [32]. Cemri and colleagues (MAST) examine roughly 150 real multi-agent traces and group failures into three categories: specification errors, inter-agent misalignment, and missing verification [33]. Tran and Kiela show that, under an equal thinking-token budget, a single agent outperforms multi-agent systems on multi-hop reasoning [19]. More agents, on their own, do not amount to more assurance.

Human oversight is concentrated because attention is finite. Reviewers may accept recommendations they cannot meaningfully evaluate, while approval quality falls under repeated intermediate decisions [37, 38]. Scalable-oversight work likewise treats placement and information quality as central [41, 42]. Kenton and colleagues compare protocols, such as debate and consultancy, in which a weak model judges a strong one, and find that oversight accuracy improves when it concentrates where information asymmetry actually surfaces rather than at every step. Kale and colleagues propose a strategy for reliable weak-to-strong monitoring in which a weak monitor escalates only the small number of risky or uncertain agent actions to human or stronger review, instead of reviewing every action.

Legal-domain agent taxonomy studies report the same hallucination and verifiability failures in contract and compliance tasks that appear in general-purpose agents. A separate study of tool-invoking agents shows that an agent can complete a task's technical goal while entirely missing the regulation it was meant to honor – success and compliance are not the same thing [43, 44].

The two current pipeline families are a product of the everyday automations requested so far – not a ceiling on the architecture itself. EPOCH's actual design goal is to always keep building hybrid models and new pipelines possible: agents built for everyday tasks stay flexibly recombinable, along with the topology-selection mechanism and the ways they communicate. A document-family output, for example, might also need to interpret based on data. In that case, agents already built for earlier tabular cases are already available: say, one that connects to a data warehouse such as Snowflake, one that pulls the data deterministically, and one that interprets it. They come from the reusable agent library described in Section 3.2.1. The one short process left is integrating those existing agents into the new automation, and the case profiler and topology selector do most of that integration themselves.

C.3 EU AI Act Design Mapping

References mean:

- Articles refer to Regulation (EU) 2024/1689 [1].
- They serve as a process-design reference here.

Limits mean:

- Intended purpose determines high-risk classification.
- This thesis makes no deployment-specific legal determination.

Reference	Process requirement	Design connection
Article 9	Lifecycle risk management	Approved boundaries and retained decision history.
Article 11	Technical documentation	Structured artifacts and declared interfaces.
Article 12	Automatic event logging	Timestamped shelves and reconstructable lineage.
Article 14	Human oversight and intervention	Approval before binding decisions become active.
Article 15	Accuracy, robustness, and cybersecurity	Deterministic computation and visible limitations.

Table C.2: Design-oriented mapping of selected EU AI Act requirements. The table does not classify the evaluated system as high-risk.

Regulation (EU) 2026/1744 changed the relevant application timetable [50]. The timing does not remove the design value of documentation, logging, and intervention. It makes the articles a forward-looking reference unless a deployment falls within an applicable legal category.

Appendix D

System Architecture and Agent Contracts

This appendix records implementation-level contracts removed from the pipeline-level narrative in Chapter 3.

D.1 Agent Types and Shelf Records

Type	Permitted role	Validation boundary
LLM	Interpret evidence and produce structured judgments.	Pydantic checks shape and type; code does not verify prose against an answer key.
Deterministic	Parse, join, calculate, or summarize reproducibly.	The same approved input produces the same output without an LLM call.
Hybrid	Coordinate LLM judgment with deterministic work.	Each component follows the validation rule of its own type.

Table D.1: The three agent types and their validation boundaries.

Every agent's own folder under `src/epoch_switch/agents/` holds both halves of its contract side by side: the LLM prompt files (`.md`) an LLM or Hybrid agent actually sends, and the deterministic or Hybrid Python logic that runs alongside them — for example, `agents/data/d2_missing_value/` carries D2's prompt files next to `d2_data_quality_engine.py`, its deterministic component.

Each agent stores its active artifact under its own registry-keyed shelf. Table D.2 lists the identifiers one real record carries, drawn from the UC2.1 `D3` artifact (Appendix E.7).

Promotion moves an earlier active record to the shelf's archive rather than deleting it. Rejection archives the candidate together with its decision reason, so the audit trail shows the final state, the alternatives that existed, and how each was resolved. Only the orchestration layer, the code that runs the pipeline step by step, reads across shelves: it pulls each agent's output from that agent's own shelf and passes it forward explicitly, and no agent discovers undeclared context or opens another agent's store on its own. That restriction keeps the lineage visible and stops

Field	Example value	What it names
registry_id	uc2_1_ets1_scope_memo_annual	Which overall request the record belongs to.
run_id	2026-09-11_164841Z_regulation_cli	Which execution of that request produced it.
agent_id	D3	Which agent wrote the record.
artifact_id	d3-5051b05ef216	The record itself.
payload_sha256	9e382043b5b0a62d...	A SHA-256 digest of the exact payload content.
input_refs	d1-0652f34144b1, e25b17c608d1..., same run_id	Each upstream record consumed, by identifier, digest, and run.
status, review_round	active, 1	Whether this is the live record, and which review cycle produced it.
llm_provenance	null	Null for deterministic agents; model, reasoning, and token counts for LLM agents.

Table D.2: The identifiers and references one shelf record carries, shown on the UC2.1 D3 artifact.

a convenient cross-read from silently widening an agent's evidence boundary.

D.2 Role-to-Identifier Map

Chapter 3 uses functional role names so that the pipeline can be read without prior knowledge of its implementation codes. Table D.3 provides the corresponding identifiers used in shelf records, prompts, and the frozen figures.

Functional role	Implementation identifiers	Responsibility
Regulation analysis	R1, R2, RD1, RD2, RC1.1, RC1.2	Finds requirements, tests evidentiary support, and compares regulation versions.
Report mapping	RM1.1, RM1.2	Links earlier disclosure requirements to Siemens report sections.
Data mapping and execution	DA1, D1, D2, D3, RD3	Maps available data, executes approved operations, checks quality, and builds deterministic results.
Case profiling and selection	CP1, TS1	Describes the approved case and selects its topology.
Topology analysts	P1–P6, C1	Interpret one scope, provide independent readings, or own domain contributions.
Reconciliation and synthesis	S1, C2	Reconciles independent readings or combines domain-owned results.
Finalization	F1, F2	Produces the reviewed tabular or document deliverable.
External perspective	EX1, EX2	Adds advisory official-source commentary at the two stage-final gates.

Table D.3: Functional roles and their implementation identifiers.

D.3 External Perspective Source Allowlist

EX1 and EX2 (Section 3.2.4) may search only the fixed set of official institutions in Table D.4 – the same list that constrains the web-search tool's allowed_domains parameter, so the model cannot search outside it.

The two instances answer different questions rather than repeating one check. EX1 asks whether

an official outside source changes how the evidence boundary should be reviewed, and it runs immediately before the Stage 1 human gate. EX2 applies the same test to the completed deliverable, immediately before the Stage 2 human gate. This pairing is mandatory for every use case, regardless of pipeline family: a pipeline definition that adds a human gate without its corresponding external perspective agent violates this architecture.

Domain	Institution	Scope
eur-lex.europa.eu	EUR-Lex	EU Official Journal, consolidated legislation
efrag.org	EFRAG	ESRS standard setter
commission.europa.eu	European Commission	EU policy and legislative proposals
esma.europa.eu	ESMA	EU sustainability-reporting supervision
eea.europa.eu	European Environment Agency	EU environmental data and assessments
echa.europa.eu	European Chemicals Agency	EU chemicals and pollution (ESRS E2)
bafa.de	BAFA	EnEfG Sec. 8/16 administration (UC1)
dehst.de	DEHSt	German Emissions Trading Authority (UC2)
umweltbundesamt.de	Umweltbundesamt (UBA)	German Federal Environment Agency
bmuv.de	BMUV	German Federal Ministry for the Environment
ifrs.org	IFRS Foundation / ISSB	Global sustainability disclosure standards
globalreporting.org	GRI	Global Reporting Initiative standards
unfccc.int	UNFCCC	UN climate treaty body
unep.org	UNEP	UN Environment Programme
ipcc.ch	IPCC	Intergovernmental Panel on Climate Change
oecd.org	OECD	Environmental and economic policy analysis
iea.org	IEA	International Energy Agency
epa.gov	US EPA	US Environmental Protection Agency

Table D.4: The complete official-institution allowlist EX1 and EX2 may search: EU bodies, German authorities (Siemens' primary jurisdiction), and global standard setters and science bodies.

D.4 Stage 1 Pipeline Contracts

Tabular statuses mean:

- in_scope — approved evidence supports assessment.
- blocked — required evidence is unavailable.
- excluded — outside the approved boundary.

Blind passes mean:

- X.1 sees authoritative text but no candidates.
- X.2 compares both proposals with the source.
- Only X.2 assigns the pair's binding status.

The first pass's payload physically omits the candidate fields, a structural constraint rather than a prompt instruction, so nothing to anchor on reaches it. RC1.1/RC1.2 and RM1.1/RM1.2 are not themselves mandatory; what is mandatory is this blind-pass structure wherever anchoring risk

Family	Step	Receives	Produces
Tabular	Regulation discovery (R1)	Demand and regulation text	Candidate findings and operational fields.
Tabular	Data mapping (DA1)	Regulation fields and runtime catalog	Mapping and handoff for human approval.
Tabular	Scope review (R2)	Regulation findings and approved mapping	Supported, blocked, and excluded findings.
Tabular	Data execution and quality (D1/D2)	Approved handoff and scope boundary	Data product and claim-relevant quality verdict.
Document	Corpus preparation (RD1/RD2)	Public regulation and report corpus	Extracted text and non-binding candidates.
Document	Change classification (RC1.1/RC1.2)	Requirement text; candidates only in the second pass	Final old-to-new change status.
Document	Report mapping (RM1.1/RM1.2)	Requirement and report text; candidates only in the second pass	Final reported status.
Document	Action join (RD3)	Final change and report-mapping rows	Outer join, orphan side, and fixed action.
Shared	Profile and selection (CP1/TS1)	Approved family-specific evidence	Case profile, scores, and approved selection.

Table D.5: Pipeline-level Stage 1 input and output contracts.

exists, and later registries or evolving agent structures may apply it through different role pairs.

The two blind-pass pairs use the same information boundary. The first change reader (RC1.1) sees only the complete authoritative 2025 and 2026 requirement text. The second reader (RC1.2) receives that independent proposal beside the deterministic candidates and decides the binding change_status from the source paragraphs. Likewise, the first report reader (RM1.1) sees requirement and Siemens report text without the index proposal. The second reader (RM1.2) compares both routes and assigns the binding reported_status. The first-pass shelves remain available for audit but do not appear in the human gate payload.

The deterministic action join (RD3) performs a full outer join on the 2025 disclosure-requirement identifier. It retains old requirements without a 2026 counterpart, new requirements without a 2025 origin, merged mappings, and duplicate old identifiers. From change_status and reported_status alone, it assigns new_report_required, report_rewrite_required, report_update_required, status_review_required, or no_action. It does not call an LLM or revise either input classification.

The gate display for either family is intentionally smaller than the complete audit record built above: a reviewer sees the binding conclusions and the evidence needed to assess them, while exploratory proposals (blocked findings, first-pass blind readings, unmatched requirements) remain on their own shelves. This balance concentrates attention at the gate without discarding material a later reconstruction needs.

D.5 Selection and Stage 2 Bindings

The case profiler (CP1) separates routing, escalation, and descriptive context. The central routing field is assurance_artifact; supporting fields describe dual-method need and cross-functional ownership. The topology selector (TS1) receives the approved profile and returns three integer scores summing to 100, the selected topology, rationale, alternatives, sources, and limitations. The profile and selection remain separate shelf artifacts so that a profiling error can be distinguished from a routing error.

For tabular paths, the deterministic result builder (D3) preserves the grain required by the registered computation. Current outputs include site–rule, site–year, division, or portfolio records. The builder passes location and geographic attributes through approved joins; topology roles interpret these records but do not recompute them.

Path	Binding	Dependency	Final gate
Tabular Direct	D3 → P1 → F1	Dependent analytical chain.	EX2 beside F1.
Tabular Debate	D3 → (P2 ‖ P3) → S1 → F1	Independent reads before reconciliation.	EX2 beside F1.
Tabular Coalition	D3 → C1 → C2 → F1	Domain fan-out before synthesis.	EX2 beside F1.
Document UC4	(P4 ‖ P5 ‖ P6) → F2	Three fixed posture reads of one RD3 join.	EX2 beside F2.

Table D.6: Current Stage 2 bindings. UC4 runs all personas regardless of TS1.

Dependencies mean:

- → — the later step receives the earlier result.
- ‖ — independent readers converge later.

Gate rules mean:

- Only the final topology agent opens the Stage 2 gate.
- EX2 never enters the promoted payload.

The document assessors (P4, P5, and P6) all receive the same RD3 rows. They may assign required_urgent, required, or not_required as action_priority, but they cannot change action_needed. The document finalizer (F2) exposes priority disagreement in persona_divergence, reports one reconciled agreed_priority per row, and adds a deterministic standard-level summary. Both external perspective agents remain advisory: EX1 runs before the Stage 1 boundary gate, while EX2 runs beside the completed Stage 2 candidate.

The gate each path opens is visible in the published viewer. Each page below shows the exact candidate a reviewer approved, with the finalizer's own record beside the advisory external commentary.

Approved Stage 2 deliverables:

- UC1 (Direct, F1): https://emrecanulu.com/epoch/uc1/stage2/f1/.
- UC2 (F1): https://emrecanulu.com/epoch/uc2/stage2/f1/.
- UC2.1 (Debate, F1): https://emrecanulu.com/epoch/uc2-1/stage2/f1/.
- UC3 (Coalition, F1): https://emrecanulu.com/epoch/uc3/stage2/f1/.
- UC4 (document, F2): https://emrecanulu.com/epoch/uc4/stage2/f2/.

D.6 Post-Hoc Architecture Audit (/system-check)

Chapter 4's stale-shelf defect (Section 4.2) was found by reading run artifacts by hand. `/system-check` exists so that the next such defect does not require the same manual effort. It is a read-only, rule-by-rule audit run after a pipeline run or an agent/use-case change, checking a fixed catalog of 26 numbered rules (R01–R26) drawn from `AGENTS.md` and the code itself. Among them: agent-local shelf placement, `input_refs` provenance, the isolated-memory boundary, blind-pass pair wiring, the topology and gate shape, and LLM-provenance recording. It reports findings; it does not repair them, and any fix it recommends is applied only in a separate, user-approved pass.

Appendix E

UC2.1 Tabular Replication Artifacts

E.1 UC2.1 Registry Entry

Field	UC2.1 value	Purpose
ID	`uc2_1_ets1_scope_memo_annual`	Stable lookup key for the request and its agent-owned shelves.
Natural request	ETS 1 monitoring-assurance memo for large manufacturing sites, FY2022–FY2024, using two CO_2e methods and legal citations.	States the user's question without prescribing operational fields.
Regulation	`REG-1`; `CELEX_32003L0087_EN_TXT.pdf`	Names a known legal source that seeds R1's review.
Site filter	`size_tier = large`; `bu_rc` = two business units	States the organizational population initially known to the user.
Time window	2022-01-01 to 2024-12-31; annual grain.	States the requested observation period and reporting frequency.
Expected output	Per-site, per-year scope decision, two CO_2e figures, discrepancy flag, trend, provenance, legal citations, and limitations.	Defines the content the user expects from the final deliverable.
Output profile	`qualitative`; `structured_memo`.	Records the user's preferred output class and presentation form.
Assurance profile	`law_reference_required`; `dual_method_required = true`; `elevated`; no formal sign-off.	Records the user's initial evidence and verification expectation.
Computation spec	`dual_method`; emissions source; 5% tolerance; energy-derived versus reported Scope 2 CO_2e.	Records the deterministic comparison requested for later execution.

Table E.1: Nine-row summary of the UC2.1 registry entry. Complete registry record URL: https://emrecanulu.com/epoch/uc2-1/registry/.

Fields mean:

- ID: the stable key shared by registry-linked artifacts.
- Natural request: the question written in user-facing language.
- Scope and time fields: the sites, dates, and reporting grain initially requested.
- Expected output: the content the completed pipeline should deliver.

Profiles and boundaries mean:

- Output profile: the user's preferred answer class and delivery form.
- Assurance profile: the initial evidence and verification expectation.
- Computation spec: the requested deterministic calculation shape.
- Registry boundary: the entry contains no discovered field list or approved scope.

E.2 UC2.1 R1 Artifact

Item	ID	Class	Shortened artifact content	Grounding
Finding	RF-001	Scope	ETS 1 applies to Annex I activities and uses the installation as its compliance unit.	REG-1; CELEX_32003L0087_EN_TXT.pdf, p. 3
Finding	RF-006	Method	Annex IV permits calculation from activity data, emission factor, and oxidation factor, or a corroborated measurement.	REG-1; CELEX_32003L0087_EN_TXT.pdf, p. 13
Finding	RF-008	Verification	Annual reports require reliable data and a verifier's conclusion that emissions are not materially misstated.	REG-1; CELEX_32003L0087_EN_TXT.pdf, p. 14
Possible field	PF-001	Entity; core	site_identifier sets one installation as the primary reporting grain.	RF-001, RF-003, RF-004, RF-005, RF-010
Possible field	PF-007	Measure; core	amount_consumed_mwh provides annual activity data for the energy-derived calculation.	RF-006, RF-007; MWh per site and year
Missing context	—	Ungrounded	The supplied regulation does not define a numeric 5% materiality threshold for discrepancy flags.	Annex V states only "not materially misstated"

Table E.2: Six-row excerpt from the UC2.1 R1 artifact. Complete 48-entry artifact URL: https://emrecanulu.com/epoch/uc2-1/r1/.

Complete artifact contains:

- 9 selected source pages.
- 15 regulatory findings.
- 19 possible data fields.
- 5 missing-context records.
- 48 substantive entries in total.

Prefixes mean:

- RF: a regulatory finding grounded in the legal source.
- PF: a possible field that DA1 must map to the data catalog.
- REG-1: R1's internal reference to the English Directive PDF.

Data terms mean:

- Site: the regulated installation and reporting unit, not simply an office.
- MWh: megawatt-hour, the consumed-energy unit used as activity data.

Classes mean:

- Scope, Method, and Verification: applicability, procedure, and assurance finding types.
- Entity and Measure: identifier-like and numeric data roles.
- Core: a field required to answer the request.
- Ungrounded: a requested element absent from the supplied legal source.

Grounding means:

- CELEX_32003L0087_EN_TXT.pdf is the English PDF of EU Directive 2003/87/EC; its page number points to the supporting evidence.
- An Annex is a formal section of the Directive. Annex I defines scope, Annex IV monitoring methods, and Annex V verification rules.
- An emission factor converts the MWh activity data into tonnes of carbon dioxide equivalent.
- A materiality threshold is the numeric cutoff for a discrepancy large enough to matter. The Directive gives no 5% cutoff, so R1 records it as missing regulatory context.

E.3 UC2.1 DA1 Mapping Artifact

Possible field	Status	Primary catalog target	Source binding or rule	Mapping meaning
PF-001 site_ identifier	Mapped	Entity grain: site	locations. location_id; joins facts through location_id	Uses the regulated installation as the reporting entity.
PF-002 reporting_ year	Mapped	Time grain: year	dist_ie_energy_ emissions_raw. fiscal_year	Provides annual records; fiscal year remains a provenance caveat.
PF-003 annex_i_ activity_ type	Unmapped	—	No catalog target	The database has no Annex I activity classification, so DA1 records the gap.
PF-007 amount_ consumed_ mwh	Mapped	Measure: amount_ consumed_mwh	Emissions table; sum; scope = Scope 2; is_missing flag	Supplies the annual activity data for Method A.
PF-009 scope2_ market_ proxy_t	Mapped	Derived measure: scope2_market_ proxy_t	Sum of consumption times CO_2 factor for Scope 2 rows	Produces the energy-derived emissions value for Method A.
PF-018 business_ unit	Mapped	Filter: bu_rc_name	locations. bu_rc_name; alternatives: bu_rc, bu_rc_group	Exposes granular and grouped business-unit options for human review.

Table E.3: Six-row excerpt from the UC2.1 DA1 mapping artifact. Complete 19-field artifact URL: https://emrecanulu.com/epoch/uc2-1/da1/.

Complete approved package contains:

- The runtime catalog snapshot used for mapping.
- Every mapping row with its mapped or unmapped status.
- One human decision per mapping row and the approved mappings.
- The final approved handoff data request.

Statuses mean:

- Mapped: at least one defensible exact catalog target exists.
- Unmapped: no defensible target exists; the absence remains visible.

Targets and bindings mean:

- Primary target: the catalog field proposed for execution.
- Alternative target: another plausible option retained for review.
- Source binding: the exact table and column behind a target.
- Aggregation and filters: how the requested rows become one measure.
- Quality flag: a catalog signal that exposes a known data issue.

Suggested request means:

- The executable combination of domain, grains, dates, filters, measures, and quality policy proposed for human approval.

E.4 UC2.1 R2 Scope Artifact

State and ID	Requirement type	R1 references	Field references	Assessment meaning
In scope R2F-001	Method	RF-006, RF-007, RF-008	Approved: PF-001, PF-002, PF-007, PF-009-011	Supports an annual, per-site comparison of energy-derived and reported Scope 2 figures. It does not establish verified ETS 1 emissions.
In scope R2F-002	Verification	RF-008	Approved: PF-009, PF-010	Supports a discrepancy flag as an internal consistency check. The 5% rule is not a Directive threshold or formal verification.
In scope R2F-003	Obligation	RF-005, RF-011	Approved: PF-001, PF-002	Supports a per-site, annual reporting structure. The database uses fiscal year rather than the Directive's calendar year.
In scope R2F-004	Responsible party	RF-010, RF-001	Approved: PF-001, PF-018	Identifies sites and business units for reporting, but does not confirm the legal operator or permit holder.
Blocked RF-001	Scope	—	Related: PF-001, PF-003, PF-006, PF-018, PF-019	Annex I activity and ETS 1 scope status are unresolved, so substantive installation coverage cannot be concluded.
Excluded RF-012	Penalty	—	Related: PF-017	Excess-emissions penalty calculation is outside the registry request and its allowance field is unapproved.

Table E.4: Six-row excerpt from the UC2.1 R2 scope artifact. Complete 22-entry artifact URL: https://emrecanulu.com/epoch/uc2-1/r2/.

Complete approved artifact contains:

- 4 in-scope assessments with stable R2F identifiers.
- 8 blocked and 2 excluded R1 findings.
- 8 limitations carried into downstream interpretation.
- Human acknowledgement of all blocked and excluded findings.

Scope states mean:

- In scope: approved database data supports the stated assessment boundary.
- Blocked: the request remains relevant, but required database data is unavailable or unapproved.
- Excluded: the finding lies outside the approved assessment boundary.

Identifiers and fields mean:

- R2F: a scoped assessment created by R2.
- RF: the source regulatory finding produced by R1.
- PF: an R1 possible field reviewed through DA1.
- Approved fields support an in-scope statement; related fields explain a blocked or excluded classification.

Evidence and audit grounding mean:

- Operational evidence is database data available through the approved DA1 mapping. A blocked row lacks required operational data.
- Legal evidence is the R1 source file, page, and excerpt retained by each in-scope row.
- Assessment boundary states what the approved data can and cannot prove.
- Stored artifact IDs and hashes identify the exact R1 and DA1 inputs.
- After approval, only in-scope findings enter isolated memory.

E.5 UC2.1 D1–D2 Artifacts and UC1 Derived-Facts Example

Use case	Component	Artifact item	Stored value	Meaning
UC2.1	D1	Approved request	Site × year; 2022–2024; Scope 2, two-business-unit, and large-site filters; `energy_mwh`, `scope2_market_proxy_t`, and `scope2_location_t`	Executes the human-approved DA1 handoff without replanning it.
UC2.1	D1	Data-product summary	Emissions and energy tables with one row per site-year (counts withheld); time window not clamped	Records the observable result and execution boundary without storing raw rows in the shelf.
UC2.1	D2	Quality verdict	`warning`; `increase_assurance`; completeness `*.**`; maximum severity `high`	Data remains usable, but downstream interpretation must retain material quality caveats.
UC2.1	D2	Claim partition	Assessable: `R2F-001-004`; blocked: none	All approved R2 claims can proceed; unrelated quality findings do not affect the verdict.
UC1	Derived facts	Threshold configuration	`energy_mwh`; `enefg_8` = 7,500 MWh; `enefg_16` = 2,500 MWh; `max` aggregation	Shows the threshold-shaped input that activates the optional deterministic builder.
UC1	Derived facts	Runtime context	`deterministic_threshold_comparison`; assessable = true; completeness `*.**`; blocked summary empty	Produced at runtime for CP1 and not promoted to a separate agent shelf.

Table E.5: Six-row excerpt combining the UC2.1 D1–D2 execution package with a UC1 derived-case-facts example. Complete artifact URLs are listed below.

Complete records:

- UC2.1 D1 artifact URL: https://emrecanulu.com/epoch/uc2-1/d1/.
- UC2.1 D2 artifact URL: https://emrecanulu.com/epoch/uc2-1/d2/.
- UC1 threshold-computation URL: https://emrecanulu.com/epoch/uc1/d3/.

Components mean:

- D1: deterministic execution of the approved data request.
- D2: hybrid quality profiling and claim-level adjudication.
- Derived facts: optional deterministic threshold context for CP1.

Quality terms mean:

- Completeness measures usable operational database data, not legal citations.
- Assessable means D2 found enough usable data for the approved claim.
- `increase_assurance` continues the pipeline with stronger caveats.

Why two use cases are shown:

- UC2.1 shows D1–D2 execution for a dual-method request; it has no configured threshold-derived payload.
- UC1 shows the optional derived-facts path because it defines two energy thresholds.

E.6 UC2.1 CP1–TS1 Selection Artifacts

The approved D2 package becomes a case profile before D3 builds the deterministic records for Stage 2. CP1 records the routing signals and their sources, while TS1 maps the approved profile to one execution shape. Table E.6 reproduces the values that selected Debate for UC2.1.

Agent	Artifact item	Stored value	Meaning
CP1	Routing evidence	assurance_artifact = reconciliation_record; dual_method_required = true	One annual Scope 2 claim requires two method reads and one reconciled conclusion.
CP1	Ownership	cluster_count = 1; cross_functional_need = false	One team owns the result, so the case does not require Coalition.
CP1	Context and output	elevated; risk and uncertainty medium; verification; qualitative/structured_memo	Fixes the assurance, task, uncertainty, and deliverable context used by the selector.
TS1	Topology scores	Direct = 17; Debate = 78; Coalition = 5	Retains the selected shape and both alternatives instead of hiding them behind one label.
TS1	Selected topology	Debate	Binds D3's output to independent P2/P3 reads, S1 reconciliation, and F1.

Table E.6: Five-row excerpt from the UC2.1 CP1 and TS1 artifacts of the UC2.1 evaluation run. Complete artifact URLs are listed below.

Complete records:

- CP1 profile URL: https://emrecanulu.com/epoch/uc2-1/cp1/.
- TS1 selection URL: https://emrecanulu.com/epoch/uc2-1/ts1/.

Selection terms mean:

- Routing evidence describes the assurance object the final result must preserve.
- Ownership separates one reconciled claim from distributed organizational contributions.
- Scores preserve the selector's rejected alternatives beside its chosen topology.

The two records preserve separate decisions: CP1 describes the approved case, while TS1 selects its execution shape. Debate wins because one claim is read through two methods before reconciliation, not because several teams own separate results. The next artifact is therefore D3's method-level computation, which supplies the rows those independent readers inspect.

The same pair of records exists for every use case. Each selection page carries that case's own scores and rationale, and each overview page shows the full approved chain that reached it, from regulation discovery through the selector.

Topology decision and the approved chain behind it:

- UC1 — https://emrecanulu.com/epoch/uc1/ts1/, chain at https://emrecanulu.com/epoch/uc1/overview/.
- UC2 — https://emrecanulu.com/epoch/uc2/ts1/, chain at https://emrecanulu.com/epoch/uc2/overview/.
- UC2.1 — https://emrecanulu.com/epoch/uc2-1/ts1/, chain at https://emrecanulu.com/epoch/uc2-1/overview/.
- UC3 — https://emrecanulu.com/epoch/uc3/ts1/, chain at https://emrecanulu.com/epoch/uc3/overview/.
- UC4 — https://emrecanulu.com/epoch/uc4/ts1/, chain at https://emrecanulu.com/epoch/uc4/overview/.

E.7 UC2.1 D3 Computation Artifact

Site–year	Method A (t CO_2e)	Method B (t CO_2e)	Absolute delta	Difference	Above 5% tolerance
Site A–2023	***.****	***.****	***.****	**.**%	Yes
Site B–2023	***.****	***.****	***.****	**.**%	Yes
Site C–2022	***.****	***.****	***.****	**.**%	Yes
Site D–2022	***.****	***.****	***.****	**.**%	No
Site E–2023	***.****	***.****	***.****	**.**%	No
Site F–2024	***.****	***.****	***.****	**.**%	No

Table E.7: Six-row excerpt from the UC2.1 D3 artifact. Site identities, all geographic attributes, and all computed values are replaced by stable aliases and masks.

Complete record:

- D3 artifact URL: https://emrecanulu.com/epoch/uc2-1/d3/.
- Full result: all site-year rows in the approved scope.
- Rows above the configured five-percent tolerance are flagged.

Fields mean:

- Method A is the energy-derived Scope 2 calculation.
- Method B is the reported Scope 2 figure.
- Difference uses the larger method figure as its denominator.

E.8 UC2.1 P2–P3 Independent Reads

Site–year	D3 comparison	P2: energy-derived read	P3: reported-method read
Site A–2023	***.**** t CO_2e; **.**%; flagged	Plausible consumption-derived estimate, but factor internals do not support a reliable figure.	Reported figure diverges materially; reconcile the methods before relying on either result.
Site B–2023	***.**** t CO_2e; **.**%; flagged	Annex-IV-style calculation is traceable, but factor or activity inputs may explain the gap.	Reported value raises a consistency concern against the independently calculated figure.
Site C–2022	***.**** t CO_2e; **.**%; flagged	Difference may reflect an unresolved factor basis; calculation evidence remains incomplete.	Reported method also leaves its factor basis unclear, so the difference needs review.
Site D–2022	***.**** t CO_2e; **.**%; not flagged	Methods are effectively aligned, although the factor basis remains unverified.	Reported figure closely matches Method A; monitoring evidence still limits assurance.
Site E–2023	***.**** t CO_2e; **.**%; not flagged	Difference stays within tolerance, but the reason for the change remains unknown.	Reported result is close to Method A; retain the unexplained driver as a caveat.
Site F–2024	***.**** t CO_2e; **.**%; not flagged	Difference is negligible, while the wider trend cause remains unverified.	Reported and derived figures align; the available evidence does not explain the trend.

Table E.8: Aligned excerpt from the independent UC2.1 P2 and P3 artifacts. The same anonymized aliases used in Table E.7 replace site identity and geography; D3 values are masked.

Complete records:

- P2 artifact URL: https://emrecanulu.com/epoch/uc2-1/stage2/p2/.
- P3 artifact URL: https://emrecanulu.com/epoch/uc2-1/stage2/p3/.
- Each complete artifact contains the same D3 rows.

Independence means:

- P2 treats Method A as its primary analytical perspective.
- P3 treats Method B as its primary analytical perspective.
- Neither reader sees the other output before S1.

E.9 UC2.1 S1 Reconciliation Artifact

Site–year	Materiality verdict	Delta interpretation	Reconciled conclusion
Site A–2023	Material	A **.**% gap indicates a possible method or input inconsistency.	The methods do not reconcile; escalate the source difference before reliance. This is not formal verification.
Site B–2023	Material	The **.**% gap may reflect factor vintage, activity data, or a method change.	The methods do not reconcile; review both calculation paths and retain the limitation.
Site C–2022	Potentially material	The **.**% gap exceeds tolerance and likely reflects a different factor basis.	Review the factor basis before accepting one figure. This is not formal verification.
Site D–2022	Not material	The **.**% gap is negligible at the reported scale.	The methods reconcile closely; retain the unverified evidence-basis caveat.
Site E–2023	Not material	The **.**% gap remains within the configured tolerance.	The figures reconcile for this test, but the reason for change remains unresolved.
Site F–2024	Not material	The **.**% gap indicates practical agreement between methods.	The methods reconcile; retain the wider trend limitation separately.

Table E.9: Six-row excerpt from the UC2.1 S1 reconciliation artifact. Stable aliases replace all site identity and geography, while the reconciliation uses the masked D3 values shown in Table E.7.

Complete record:

- S1 artifact URL: https://emrecanulu.com/epoch/uc2-1/stage2/s1/.
- Full result: every reconciled site-year row.
- The record retains its chain-level caveats.

Reconciliation means:

- S1 sees P2 and P3 together for the first time.
- D3 remains authoritative for every numerical field.
- S1 combines interpretations; it does not verify reported data formally.

E.10 UC2.1 F1 Final Deliverable

Site–year	Material	Scope determination	Reader-facing entry
Site A–2023	Yes	The two Scope 2 methods do not reconcile.	Review the factor, activity data, and reported source before relying on the result.
Site B–2023	Yes	The method difference requires escalation.	Reconcile factor vintage, activity inputs, and method changes; preserve the uncertainty.
Site C–2022	Yes	The difference exceeds the configured tolerance.	Review the factor basis and calculation evidence before choosing a reporting figure.
Site D–2022	No	The two methods reconcile within tolerance.	No discrepancy action is required; retain the evidence-basis limitation for audit review.
Site E–2023	No	The two methods reconcile within tolerance.	Accept the comparison result while keeping the unexplained change as a limitation.
Site F–2024	No	The two methods reconcile within tolerance.	No discrepancy action is required; investigate the wider trend only if decision-relevant.

Table E.10: Six-entry excerpt from the UC2.1 F1 structured memo. The same stable aliases remove site identity and geography; numerical conclusions trace to the masked D3 artifact.

Complete record:

- F1 artifact URL: https://emrecanulu.com/epoch/uc2-1/stage2/f1/.
- Full memo: one entry per site-year.
- Summary: the D3 flags, the material entries, and the limitations.
- Citation list: nine approved evidence references.

Finalization means:

- F1 converts S1's reconciliation into the requested structured memo.
- Material entries can include cases beyond D3's tolerance flag.
- Human approval applies to the complete F1 deliverable.

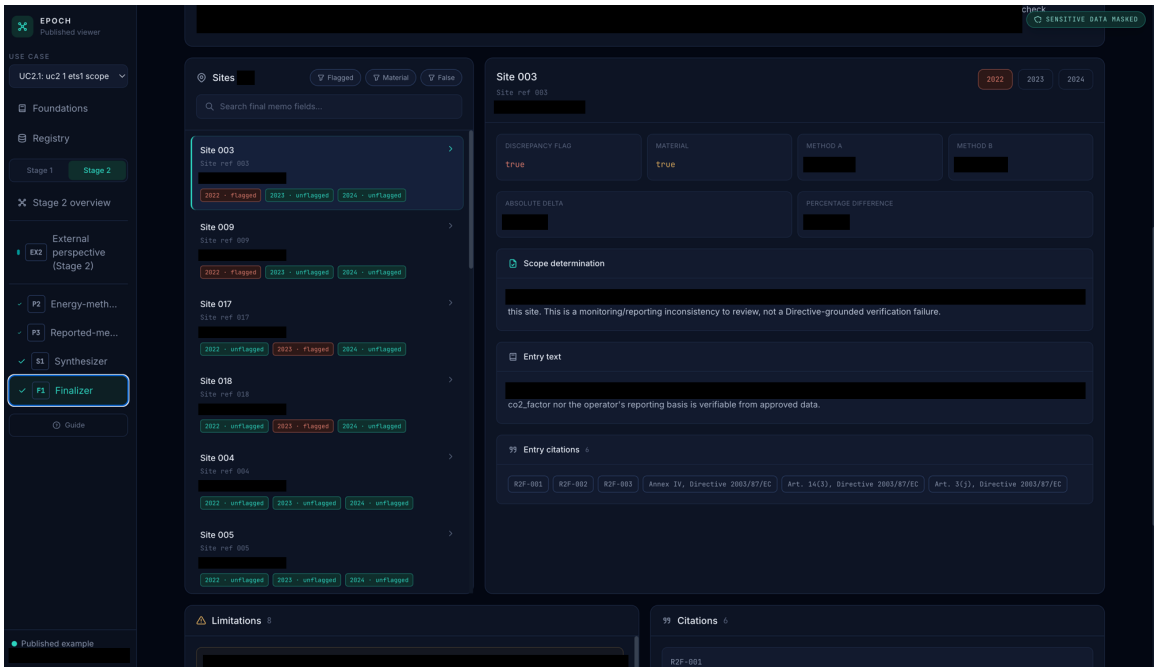


Figure E.1: The same F1 record as rendered in the published viewer at `https://emrecanulu.com/epoch/uc2-1/stage2/f1/`: site filtering, the Method A/Method B comparison, the scope determination, and the entry's citations and limitations together in one view. Site identity and numerical values are masked, matching the excerpt above.

E.11 UC2.1 External Commentary Artifacts

`EX1` runs immediately before the Stage 1 gate and `EX2` immediately before the Stage 2 gate, each reading the same allowlisted official sources (Appendix D.3) independently of the pipeline. Neither ever enters the promoted payload.

Gate	Stance	Relevance	Commentary	Grounding
EX1 (Stage 1)	challenges	high	EU ETS 1 prices direct on-site emissions, not the Scope 2 (purchased-electricity) figures both of the memo's methods actually use.	European Environment Agency; European Commission
EX1 (Stage 1)	flags_gap	high	The memo cites the original 2003/87/EC directive text, but the operative monitoring law for the reporting window is the Monitoring and Reporting Regulation, which supersedes its Annex IV formula.	EUR-Lex
EX1 (Stage 1)	supports	medium	The independent record confirms the memo's annual, per-installation grain and its treatment of the reported figure are consistent with how the scheme actually operates.	DEHSt
EX2 (Stage 2)	challenges	high	EU ETS 1 covers direct installation emissions, not the purchased-electricity (Scope 2) fields both of the memo's methods rest on.	European Environment Agency; EUR-Lex
EX2 (Stage 2)	flags_gap	high	The memo's many unexplained zero and "shared computation" entries would be a monitoring-completeness failure under the MRR, which it never cites.	EUR-Lex
EX2 (Stage 2)	supports	medium	DEHSt's risk-assessment framing supports the memo's "risk-identification follow-up" routing, though the obligation actually sits with the operator's monitoring plan.	DEHSt

Table E.11: Six-bullet excerpt from the UC2.1 `EX1`/`EX2` external-commentary artifacts (gold v002): three of `EX1`'s bullets from the Stage 1 gate and three of `EX2`'s bullets from the Stage 2 gate, one per stance shown.

Stance means:

- `challenges` — the source contradicts a claim the memo makes.
- `flags_gap` — the memo cites the wrong or an outdated instrument.
- `supports` — the source independently confirms the memo's structure.

Relevance and grounding mean:

- Relevance is the model's own read of how much the point matters.
- Grounding names the allowlisted institution the point traces to; every bullet in the complete 11/10-entry artifacts is web-verified this way.

Appendix F

UC4 Document Evaluation Materials

UC4 is the public document-family case used for the real-world expert evaluation. This appendix fixes the source hierarchy and the artifact chain used in that evaluation. Appendix G reproduces the completed review materials and returned responses.

F.1 Source and Version Hierarchy

Source	Role	Authority in UC4
2025-amended ESRS E1–E5	Baseline regulation text	Authoritative text for the requirements against which Siemens prepared FY2025 reporting.
2026 Revised ESRS E1–E5	Comparison regulation text	Authoritative target for the forward-looking requirement comparison.
Siemens FY2025 Sustainability Statement	Report evidence	Authoritative source for whether and how Siemens reported a 2025 disclosure requirement.
EFRAG comparison helpers	Matching lead	Non-binding guidance concerning a different version pair; never evidence for final status.

Table F.1: UC4 source hierarchy. Authoritative paragraphs may override every deterministic or LLM-generated matching proposal.

Regulation roles mean:

- Baseline describes the requirements used for FY2025 reporting.
- Target describes the prospective 2026 change.

Grounding means:

- Final statuses require authoritative paragraph text.
- Helper documents can propose a lead but cannot settle a row.

F.2 Frozen Artifact Chain

The frozen expert-review record is `2026-09-07_163102Z_regulation_cli/run_001` under the public experiment root. The run was not overwritten after review began. Its final `F2` output supplied the fixed deliverable assessed by both UC4 reviewers.

Stage	Artifact	Evaluation role
RD1/RD2	Extracted corpus and candidates	Reproducible text boundary and non-binding matching leads.
RC1.1/RC1.2	Blind and binding change rows	Independent proposal followed by authoritative change classification.
RM1.1/RM1.2	Blind and binding exposure rows	Independent proposal followed by authoritative reporting classification.
RD3	Full outer join	Fixed orphan and action classification reviewed at Stage 1.
CP1/TS1	Profile and topology decision	Recorded selection signal for the approved document case.
P4/P5/P6	Three priority assessments	Independent conservative, balanced, and maximum-assurance readings.
F2	Final comparison and summary	Expert evaluation object and Stage 2 human-gate candidate.

Table F.2: UC4 artifact chain used by the real-world evaluation design.

F.3 Expert Review Materials

Section 3.6.2 defines the completed expert-review protocol. Appendix G reproduces its common review form and the case-specific packet pages used in the evaluation. The resulting scores and written assessments remain in Section 4.3.

Appendix G

Peer Review Packets

Every packet begins with the same Page 1 instrument. Section G.1 reproduces its empty form once, Section G.2 transcribes what came back on those filled forms, and the next five sections group the remaining report pages by use case. Each report page retains its original A4 landscape geometry; UC4's shared body appears once because its two assessments differ only on Page 1.

G.1 Page 1: Common Review Form (All Use Cases)

Empty scoring and written-assessment form shared by all five use cases.

SIEMENS × EPOCH

Annual ETS 1 Scope 2 Assurance Memo

Peer Review

Siemens Internal – Confidential. Estimated review time: 10–15 minutes. * Required

Read the system overview and deliverable, then return here to complete the form.

Assessor: *

1. Correctness and grounding *
Are the claims accurate and actually supported by the sources cited? **Score:** *1 = worst / 5 = good*

2. Scope fit *
Does it answer the question that was asked? **Score:** *1 = worst / 5 = good*

3. Clarity and usability *
Could you act on this without asking the author? **Score:** *1 = worst / 5 = good*

4. Honesty about limitations *
Does it state plainly what it could not do or verify? **Score:** *1 = worst / 5 = good*

5. Overall confidence *
Would you rely on this in a real work decision? **Score:** *1 = worst / 5 = good*

6. Siemens implementation fit *
Could Siemens integrate this output into its existing systems and workflows? **Score:** *1 = worst / 5 = good*

7. Flexibility *
Could this output adapt to different scopes, data, or regulatory changes? **Score:** *1 = worst / 5 = good*

8. Evidence structure fit *
Is the evidence organized at the right depth for this output and its assurance needs? **Score:** *1 = worst / 5 = good*

9. Auditability and human control *
Can you trace key conclusions and see where human review affected the process? **Score:** *1 = worst / 5 = good*

10. Approach improvement *
Compared with normal manual or single-analyst work, does this approach improve traceability and confidence without unnecessary complexity? **Score:** *1 = worst / 5 = good*

Estimated manual production time * — As a domain expert, how long would an equivalent human-produced report take?

What this work did best *

What this work did worst *

Comments on your scores (optional)

Anything else — issues, ideas, questions (optional)

G.2 Returned Review Responses

Section 4.3 reports the mean score per criterion across all six reviews. This section transcribes every returned score and every written response in full, sourced verbatim from the transcription pipeline's own output (`evaluation/peer_review/reviews/`) rather than re-typed from the PDFs. Reviewer names are anonymized to single letters (`A`, `B`, `C`), matching the masking already applied at transcription time; site names, division names, and numerical values remain masked in the four tabular packets' own report pages (Sections G.3–G.6), per Section 3.6.2. Free-text answers are quoted exactly as returned, typing errors included, because they are evidence, not prose to be corrected.

Use case	Rev.	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	Avg
UC1	A	4	3	4	5	4	4	4	3	3	4	3.8
UC2	A	4	4	3	5	3	3	3	3	3	4	3.5
UC2.1	B	4	4	4	4	4	4	4	4	4	4	4.0
UC3	B	3	4	3	4	3	5	5	4	3	5	3.9
UC4	B	4	4	3	4	3	4	4	4	3	4	3.7
UC4	C	5	5	4	4	4	5	5	4	4	4	4.4
Mean		4.00	4.00	3.50	4.33	3.50	4.17	4.17	3.67	3.33	4.17	3.88

Table G.1: Every score from every returned review. C1–C10 are the ten criteria of Table 3.4, in the same order; `Avg` is that review's own mean, and the bottom row is the per-criterion mean across all six reviews.

Only two reviews carry a written note on individual criteria, both from Reviewer A, on UC1 and UC2. The remaining four reviews (UC2.1, UC3, and both UC4 reviews) left every per-criterion note field blank; those reviewers answered only the free-text fields transcribed further below. Table G.2 reproduces the fourteen notes that exist.

Case	Criterion	Sc.	Reviewer's note (verbatim)
UC1	Correctness and grounding	4	the report uses FYs instead of calendar years and does not sorts out whether that matters
UC1	Scope fit	3	It answers which sites are above the line, which are below, and by how much, but not whether they already have an ISO 50001 in place
UC1	Clarity and usability	4	there is no summary at the top, you have to scroll through the pages of sites to figure out which ones matter most. A "top 10 priority sites" view woud be helpful
UC1	Siemens implementation fit	4	it doesn't say who inside Siemens would be responsible for the approval steps
UC1	Flexibility	4	it only works for Germany and can't compare across countries or jurisdictions
UC1	Evidence structure fit	3	a worked example showing the actual math for one site would be helpful
UC1	Auditability and human control	3	the design includes human review at key checkpoints. But in this run, all checkpoints were auto-approved, so we don't actually know what a human reviewer would have caught
UC1	Approach improvement	4	the system can't access context an expert would know, like "this site just installed an energy management system last month"
UC2	Clarity and usability	3	the table is dense and technical and the column headers (Method A, Method B, Abs Delta, Pct Diff, Discrepancy Flag, Material) require domain knowledge to interpret. Also, thre is no plain-language summary at the top. Someone who isn't deep in this methodology would struggle to extract the key takeaways quickly
UC2	Overall confidence	3	without the year field you can't anchor any finding to a specific reporting period. The emission factor means Method A's numbers are unverifiable from this document alone
UC2	Siemens implementation fit	3	the output is called "assurance memo", but the labelling overpromises relative to what's actually produced, because there is no verifier, verificaiton conclusion, a monitoring plan, etc.
UC2	Flexibility	3	The "Debate" topology is a good architectural choice and could be reused for other cross-checks. But it only works for Scope 2, only for ETS 1, and only for sites where both Method A and Method B data exist in Siemens Database. Can't adapt to Scope 1, Scope 3, or non-ETS sites without rebuilding
UC2	Evidence structure fit	3	the emission factor used in Method A is not traceable, only the output (CO_2e) but not the input (which factor, which source). This is relevant for an assurance memo. Also, no full Method A calculation for one site is shown
UC2	Auditability and human control	3	human gates exist but were auto-approved. The external-view agents (EX1/EX2) came with challenges, like the 30 April surrender deadline, the 2003 Directive consolidation risk, the ESMA supervisory connection, but these are advisory only and don't change the output

Table G.2: The fourteen criterion-level notes returned, both from Reviewer A (UC1 and UC2). Typing errors are the reviewer's own and are kept verbatim.

Tables G.3–G.4 transcribe the five free-text fields every reviewer answered: the manual production-time estimate, what the work did best, what it did worst, any comments on the scores, and anything else the reviewer wanted to add.

Case (rev.)	Field	Reviewer's answer (verbatim)
UC1 (A)	Manual time	2–3 working days because the time need to pull the Siemens Database data, look up every site, check each one against both thresholds, write up findings, cross-reference the regulation text, and document assumptions
	Did best	The limitations section lists 14 specific and actual issues rather than generic. It also flags data quality issues, like misspelled cities, empty zip codes, and methodological gaps (FY versus calendar year). Also, the findings at site level go beyond provide context about each site's situation, which makes the output usable rather than just technically correct.
	Did worst	Not worst but the 6-page table of individual sites without an executive view at the top saying "here are the 10 sites that matter most and here's why." would be helful to save time in scrolling through everything to figure out where to focus or prioritise quickly.
	Comments on scores	(blank)
	Anything else	1) a one-page summary at the top with top 10 priority sites, key numbers, and the "so what" before the full table, and 2) one worked example showing the complete math for a single site: which raw data rows went in, how the 3-year average was calculated, and how it was compared to the threshold. Also whether the near-breach logic could be tested with a lower sensitivity band (e.g., 90% instead of 95%) to see if any sites fall into it
UC2 (A)	Manual time	1-2 full working days, considering pulling the Siemens Database data for both methods, computing Method A (MWh $\times$ co2_factor) for each site, comparing against Method B, writing up findings per site, and cross-referencing the Directive
	Did best	Having two independent routes to the same number and then comparing them, the site-level findings explain why they match (or don't), including plausible root causes like factor-vintage differences or data-completeness gaps.
	Did worst	the "year" field is empty for every singe row and, thus, you cannot anchor any finding to a specific calendar year or fiscal year, considering that the document is called "assurance memo"
	Comments on scores	(blank)
	Anything else	1) fix the empty year field and 2) make the emission factor visible and traceable per site and 3) consider renaming the output from "assurance memo" to something more like "internal reconciliation check"

Table G.3: Every free-text answer from the UC1 and UC2 reviews, quoted verbatim. Continued in Table G.4 for UC2.1, UC3, and UC4.

Case (rev.)	Field	Reviewer's answer (verbatim)
UC2.1 (B)	Manual time	1-2 working days
	Did best	Clear and concise findings, with relevant references and fast analysis of complex data.
	Did worst	Too much text; a short summary or overview would be very helpful.
	Comments on scores	(blank)
	Anything else	(blank)
UC3 (B)	Manual time	2–3 working days, including the evaluation of the supporting evidence.
	Did best	Summarizing the results and adding key findings with relevant references.
	Did worst	Ideally, the results should be kept concise, with less text and complexity.
	Comments on scores	(blank)
	Anything else	*Color Key and Data Status* — is there any existing reference or standard wording for the Color Key and Data Status columns in the report? Ideally, the same terminology should be used consistently if these two elements are linked to each other. *Provenance* — would it be helpful to define standardized evaluation categories instead of using long text descriptions? This could ensure that all relevant parameters are assessed consistently and that each assessment has a clear conclusion. Potential categories could include completeness, evidence, outliers, and other relevant quality criteria. *Human Expert Review / Assurance* — could we include a final column summarizing the points that require review by a human expert, for example in cases of incompleteness, outliers, or insufficient evidence? This could provide a clear overview of the items requiring expert validation and support an assurance-ready process.
UC4 (B)	Manual time	4-5 working days
	Did best	Providing an overview of changes, mapping requirements to company statements, and highlighting action items and agent conflicts that may require human review.
	Did worst	The report is extensive and not concise enough. A short summary highlighting the key changes, requirement alignment, action items, and conflicts requiring human review would be more helpful.
	Comments on scores	It is possible to see whether a human review took place and whether the final output was changed by a human.
	Anything else	(blank)
UC4 (C)	Manual time	8h
	Did best	Really great and fast overview with high accuracy
	Did worst	Not related to the model: in general, it requires much more company specific information in order to analyze deeper impacts on specific businesses
	Comments on scores	(blank)
	Anything else	(blank)

Table G.4: Every free-text answer from the UC2.1, UC3, and UC4 reviews, quoted verbatim, continued from Table G.3. UC3's "Anything else" answer is one sticky-note comment the transcription pipeline recorded twice (once matched to this field, once left unmatched by page position); it is reproduced once here and re-broken into its three intended parts, which ran together in the source PDF annotation.

G.3 UC1: German Energy Threshold Priority Screen

Pages 2–11: system design, final `F1` output, references, and assurance.

System Design — German Energy Threshold Priority Screen

Selection rule: $\pi(\text{case, output profile, assurance profile}) \rightarrow \text{topology}$

Selected: Direct — Dependent steps deepen one result through a single ownership chain.

Derived case and assurance profile (CP1)

Task: Compliance check **Output:** Binary / yes no alert **Assurance:** Routine **Evidence shape:** Single attestation **Uncertainty:** Medium **Independent methods:** No **Cross-functional:** No **Ownership clusters:** 1

One-flow view

Stage 1

R1 → DA1 → Human: data mapping → R2 → EX1 → Human: regulatory scope

Stage 2

D1 → D2 → D3 → CP1 → Human: case profile → TS1 → Human: topology

P1 → F1 → EX2 → Human: final output

What each agent does

R1 — Identifies regulation-driven information needs from the registered request.	**DA1** — Maps those needs to available data and prepares the approved handoff.
R2 — Fixes the regulatory findings that downstream work may use.	**D1** — Loads exactly the human-approved data request using deterministic code.
D2 — Checks data quality, coverage, and readiness before interpretation.	**D3** — Computes the deterministic case facts used by the output agents.
CP1 — Turns approved evidence into the case, output, and assurance profile.	**TS1** — Selects Direct, Debate, or Coalition from the approved case profile.
P1 — Interprets the result through one dependent analytical chain.	**F1** — Formats the completed tabular analysis as the requested deliverable.
EX1/EX2 — Add official-source outside views before the two principal review gates.	

Human gates — A reviewer approves the data boundary, regulatory scope, case profile, topology, and finished output before each becomes binding.

External View — EX1 and EX2 provide official-source second reads at the review boundary. They are advisory only: reviewers see them, but they never change the promoted result.

External View snapshots — run 2026-09-07_173946Z_regulation_cli_02

EX1 — Stage 1

1. High / Challenges The administering authority defines the addressee 'Unternehmen' for §§ 8 and * EnEfG as the smallest legally independent unit that keeps books and prepares a balance sheet, including its branches, subsidiaries' establishments and plant parts. [BAFA]

2. High / Flags gap BAFA's administrative guidance now requires that an EnMS or environmental management system cover at least ** % of a company's total final energy consumption, and the latest revision of the guidance (Stand **.**.2026) explicitly clarifies the site reference ('Standortbezug') within that ** % rule. [BAFA]

3. Medium / Flags gap Official guidance has added explicit rules for borderline consumption cases — a rule on short-term exceedance of *.* GWh per year, and a rule releasing companies that durably fall below *.* GWh per year from the related publication duty. [BAFA]

4. Medium / Adds context The Commission's guidance recommendation on Article ** states that the obligations exclude no sector on the basis of activity, naming Emissions Trading System installations and IPPC licence holders as examples that are not carved out, and it places responsibility for assessing consumption and reporting threshold breaches on the enterprise itself. [EUR-Lex]

EX2 — Stage 2

1. High / Flags gap The *,*** MWh line is not only a waste-heat cut-off. [BAFA]

2. High / Flags gap BAFA's February 2025 revision of the EnEfG guidance introduced a ** % coverage rule: an energy or environmental management system under § 8 EnEfG must cover at least ** % of the company's total final energy consumption, mirroring the energy-audit rule under § 8 EDL-G. [BAFA]

3. Medium / Challenges The deadline framing in current official guidance differs from the transitional date commonly quoted. [BAFA]

4. Medium / Flags gap Clearing the *.* GWh company cut-off does not make every heat stream at a site reportable. [BAFA]

5. Low / Adds context The macro picture argues for treating this as a real efficiency exercise rather than a paperwork screen. [Umweltbundesamt (UBA)]

Registered task and filters

Use case	German Energy Threshold Priority Screen
Requested output	For each German site: has it crossed the EnEfG priority threshold, by how much, and what does that trigger?
Threshold / rule	Priority thresholds (a site-level planning signal only – the legal duty sits on the company): 7,500 MWh/yr under Section 8 (energy-management system scope) and 2,500 MWh/yr under Section 16 (waste-heat avoidance and public register). A site within 95% of a threshold is flagged near-breach before it is actually crossed.
How it's calculated	The 3-year average is the mean of each site's annual energy use (the Siemens Database column amount_consumed_mwh, 2022-2024) – one number per site, then compared against both thresholds.
Scope	country=DE | 2022-**-** to 2024-**-** (year) | REG-2
Source	Live pipeline run; gate mode: auto-approved (–auto) (2026-09-07).

Color key: red = exception / urgent change amber = warning / material change green = acceptable / compliant

Table terms:

Status	obligated = already past a threshold; near-breach = within 95% of it; compliant = clear of both.
Gap	how far the site's 3-year average energy use sits from the threshold (negative means still under it).
Signals	alert = a threshold is crossed now; near-breach = approaching it, not yet crossed.

Site Alerts

Site	Rule	Status	Gap	Signals	Finding
Location Display: ***** *** ******** *. *. *****, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	*** ******** motor plant averages **,*** MWh, roughly **,*** MWh above the *,*** MWh §8 screen — high-priority EnMS scope candidate.
Location Display: ***** *** ******** *. *. *****, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	The same site clears the *,*** MWh waste-heat cut-off by about **,*** MWh, so avoidance screening and BfEE reporting apply.
Location Display: ***** *******, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******* site averages *,*** MWh, roughly *,*** MWh below the *,*** MWh §8 screen — no EnMS priority flag.
Location Display: ***** *******, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**.**	**Alert:** Yes **Near Breach:** No	The same ******* site clears the *,*** MWh waste-heat cut-off by only **.* MWh — flagged, but extremely margin-sensitive.
Location Display: ***** ********, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	******** plant averages **,*** MWh, some *,*** MWh above the §8 screen — moderate exceedance, priority EnMS scope candidate.
Location Display: ***** ********, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	******** also clears the *,*** MWh waste-heat cut-off by *,*** MWh; both EnEfG workstreams can be scoped together.
Location Display: ***** *****, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	***** electronics plant averages **,*** MWh, **,*** MWh above the §8 screen — firmly in the priority EnMS population.
Location Display: ***** *****, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	***** exceeds the waste-heat cut-off by **,*** MWh; the flag is largely evidentiary given publicly reported waste-heat use.

Site	Rule	Status	Gap	Signals	Finding
Location Display: ***** *********, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	********* campus averages **,*** MWh, about **,*** MWh above the §8 screen — a top-priority candidate for EnMS scope.
Location Display: ***** *********, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	********* exceeds the waste-heat cut-off by **,*** MWh, ranking among the highest ***** for screening and register reporting.
Location Display: ***** **********, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	********** component plant averages **,*** MWh, **,*** MWh above the §8 screen — high-priority candidate for ISO ***** scope.
Location Display: ***** **********, Germany (Europe) **Location Id:** **	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	********** clears the *,*** MWh waste-heat cut-off by **,*** MWh, making it a priority for BfEE register reporting.
Location Display: ***** ********* *. ****, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	********* switchgear plant averages **,*** MWh, **,*** MWh above the §8 screen — a clear priority EnMS candidate.
Location Display: ***** ********* *. ****, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	********* exceeds the waste-heat cut-off by **,*** MWh, ranking high for avoidance screening and the ** March submission.
Location Display: ***** ***********, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	********** rail-automation works averages **,*** MWh, **,*** MWh above the §8 screen — priority candidate for EnMS scope.
Location Display: ***** ***********, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	********** clears the waste-heat cut-off by **,*** MWh; its central-city position makes heat offtake plausible for recovered streams.
Location Display: ***** ******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******-******* site averages *,*** MWh, *,*** MWh below the §8 screen — no EnMS flag, but a narrow margin.
Location Display: ***** ******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	The same ****** site exceeds the *,*** MWh waste-heat cut-off by *,*** MWh — obligated on waste heat only.
Location Display: ***** *******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	*******-********** test centre averages **,*** MWh, *,*** MWh above the §8 screen — flagged for EnMS scope consideration.
Location Display: ***** *******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	The test centre clears the waste-heat cut-off by **,*** MWh, though dispersed traction losses limit practical recovery value.
Location Display: ***** *******-*********, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	******* rail works averages **,*** MWh, roughly ********** the §8 screen — among the strongest EnMS priorities here.
Location Display: ***** *******-*********, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	******* exceeds the waste-heat cut-off by **,*** MWh; nearby CHEMPARK heat infrastructure makes external offtake a realistic option.
Location Display: ***** *******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	******* site averages **,*** MWh, *,*** MWh above the §8 screen — flagged, though with a comparatively modest margin.
Location Display: ***** *******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	******* clears the waste-heat cut-off by *,*** MWh; burner and product test benches are plausible recoverable heat sources.
Location Display: ***** ******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	******-********** switchgear works averages **,*** MWh, *,*** MWh above the §8 screen — solidly in the EnMS priority group.
Location Display: ***** ******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	The site exceeds the waste-heat cut-off by **,*** MWh; dense ********** heat infrastructure makes network offtake physically conceivable.

Site	Rule	Status	Gap	Signals	Finding
Location Display: ***** ******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	***.**	**Alert:** Yes **Near Breach:** No	********** site averages *,*** MWh, only *** MWh above the §8 screen — the tightest §8 margin in the portfolio.
Location Display: ***** ******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	The same site clears the waste-heat cut-off by *,*** MWh, a far more robust margin than its §8 position.
Location Display: ***** *******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	******-****** locomotive works averages **,*** MWh, *,*** MWh above the §8 screen — a clear EnMS priority candidate.
Location Display: ***** *******, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	***** exceeds the waste-heat cut-off by **,*** MWh; co-located data-centre cooling is a continuous low-grade heat source.
Location Display: ***** ****, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******* service site averages *,*** MWh, some *,*** MWh below the §8 screen — no EnMS priority flag.
Location Display: ***** ****, Germany (Europe) **Location Id:** ***	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	The same ******* site still exceeds the *,*** MWh waste-heat cut-off by *,*** MWh, triggering register-reporting candidacy.
Location Display: ***** ****, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	***.**	**Alert:** Yes **Near Breach:** No	**** ********** site averages *,*** MWh, just *** MWh above the §8 screen — flagged on a thin margin.
Location Display: ***** ****, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	The same site clears the waste-heat cut-off by *,*** MWh, a robust flag for avoidance screening and reporting.
Location Display: ***** ******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	****** electronics complex averages **,*** MWh, roughly ********** the §8 screen — one of the largest EnMS signals.
Location Display: ***** ******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	****** exceeds the waste-heat cut-off by **,*** MWh, about ********** the exemption level — a top waste-heat priority.
Location Display: ***** ****, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	***.**	**Alert:** Yes **Near Breach:** No	**** **** plant averages *,*** MWh, only *** MWh above the §8 screen — flagged but volume-sensitive.
Location Display: ***** ****, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	The **** plant clears the waste-heat cut-off by *,*** MWh, so waste-heat candidacy is far more robust than §8.
Location Display: ***** ****-********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	****-******** site averages *,*** MWh, *,*** MWh below the §8 screen — no EnMS priority signal at all.
Location Display: ***** ****-********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-***.**	**Alert:** No **Near Breach:** No	****-******** also stays *** MWh under the *,*** MWh waste-heat cut-off, so neither EnEfG screen flags it.
Location Display: ******* (******) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	This record shows zero metered consumption across 2022-2024, so the below-threshold §8 result reflects missing data, not performance.
Location Display: ******* (******) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	The same zero-consumption record screens below the waste-heat cut-off; treat as a data-completeness question rather than an exemption.
Location Display: ***** **********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	********** site averages *** MWh, under ********** the §8 screen — no EnMS priority candidacy arises.
Location Display: ***** **********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	The same site sits *,*** MWh below the waste-heat cut-off, at roughly **% of the exemption level — unflagged.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	******** campus roll-up averages **,*** MWh, some **,*** MWh above the §8 screen — a clear EnMS scope priority.

Site	Rule	Status	Gap	Signals	Finding
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	The ******** campus exceeds the waste-heat cut-off by **,*** MWh — a stable, non-borderline waste-heat reporting candidate.
Location Display: ***** *******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	****** ****** roll-up averages **,*** MWh, **,*** MWh above the §8 screen — a first-order EnMS priority candidate.
Location Display: ***** *******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	The ****** roll-up exceeds the waste-heat cut-off by **,*** MWh; adjacent residential development offers a plausible heat sink.
Location Display: ***** ******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	******-******* roll-up averages **,*** MWh, **,*** MWh above the §8 screen, but rests on only two years' data.
Location Display: ***** ******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	The same roll-up clears the waste-heat cut-off by **,*** MWh; a district wastewater heat scheme offers a credible sink.
Location Display: ***** ******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	********** parcel averages about *,*** MWh, some *,*** MWh below the §8 screen — no EnMS flag.
Location Display: ***** ******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	The same parcel exceeds the waste-heat cut-off by *,*** MWh, so avoidance screening and register reporting are prioritised.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	********* ********** campus averages roughly *,*** MWh, about *,*** MWh below the §8 screen — no EnMS priority.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	The same campus exceeds the waste-heat cut-off by roughly *,*** MWh; converter test benches are plausible concentrated heat sources.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******** ****** averages *,*** MWh, some *,*** MWh below the §8 screen — clearly not an EnMS priority.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-***.**	**Alert:** No **Near Breach:** No	******** sits just *** MWh below the waste-heat cut-off — a tight compliant margin worth re-testing on fuller data.
Location Display: ***** ******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	The ****** record shows zero metered consumption, so this compliant §8 result is a data-completeness question, not a finding.
Location Display: ***** ******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******'s zero record also screens below the waste-heat cut-off; read it as not assessable rather than exempt.
Location Display: ***** *******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******* ****** averages roughly *,*** MWh, about *,*** MWh below the §8 screen — no EnMS priority signal.
Location Display: ***** *******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-***.**	**Alert:** No **Near Breach:** No	******* sits *** MWh below the waste-heat cut-off; the single-year basis means this reading needs refreshing.
Location Display: ***** *******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******* branch averages *,*** MWh, *,*** MWh below the §8 screen — ******-***** load, no EnMS flag.
Location Display: ***** *******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	The ******* branch stays *,*** MWh under the waste-heat cut-off, roughly half the exemption level — no flag.
Location Display: ***** **********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	********** branch averages *** MWh, *,*** MWh below the §8 screen — the widest margin among ****** *****.

Site	Rule	Status	Gap	Signals	Finding
Location Display: ***** **********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	********** sits *,*** MWh below the waste-heat cut-off, about **% of it — firmly on the exempt side.
Location Display: ***** *****, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	***** branch averages *,*** MWh, *,*** MWh below the §8 screen; its single year straddles an ********** .
Location Display: ***** *****, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	***** stays *,*** MWh under the waste-heat cut-off, about **% of it — no waste-heat priority flag.
Location Display: ******* (******) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	An organisational record with zero metered consumption; the compliant §8 result reflects absent data rather than low energy use.
Location Display: ******* (******) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	The same zero-consumption record screens below the waste-heat cut-off; no location context is available to interpret it.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	******** ********** averages **,*** MWh, roughly ******** the §8 screen — a clear priority for EnMS scope.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	The ******** plant exceeds the waste-heat cut-off by **,*** MWh; test and cooling loads are typical recoverable sources.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	German ***********-**** roll-up averages *,*** MWh, *,*** MWh below the §8 screen — no EnMS scope flag.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	The same roll-up exceeds the waste-heat cut-off by *,*** MWh, landing in the *.*–*.* GWh waste-heat-only band.
Location Display: ***** ******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	The ********* high-rise averages *,*** MWh, *,*** MWh below the §8 screen — ****** use since production ended.
Location Display: ***** ******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	The high-rise stays *,*** MWh under the waste-heat cut-off; ****** and training use yields little recoverable heat.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	********* traction factory averages **,*** MWh, *,*** MWh above the §8 screen — a clear EnMS scope candidate.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	**,***.**	**Alert:** Yes **Near Breach:** No	The traction plant clears the waste-heat cut-off by **,*** MWh; test beds and winding processes dissipate substantial heat.
Location Display: ***** **********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	********** site averages *,*** MWh, *,*** MWh below the §8 screen — no realistic path to EnMS candidacy.
Location Display: ***** **********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	********** sits *,*** MWh under the waste-heat cut-off, roughly **% of it — no waste-heat flag.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******* ********* averages *** MWh, *,*** MWh below the §8 screen — software-****** load, no EnMS flag.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******* stays *,*** MWh under the waste-heat cut-off, about ********** the exemption level — no flag.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******** service site averages *,*** MWh, *,*** MWh below the §8 screen — a stable compliant result.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-***.**	**Alert:** No **Near Breach:** No	******** sits just *** MWh below the waste-heat cut-off, about **% of it — a monitoring case.

Site	Rule	Status	Gap	Signals	Finding
Location Display: ***** *****, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	***** gearbox works averages *,*** MWh, *,*** MWh below the §8 screen — no EnMS priority candidacy.
Location Display: ***** *****, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	***** exceeds the waste-heat cut-off by *,*** MWh; gearbox machining gives it genuine recoverable waste-heat potential.
Location Display: ***** ************, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	************ site averages *** MWh, *,*** MWh below the §8 screen — small service-scale footprint, no flag.
Location Display: ***** ************, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	************ sits *,*** MWh under the waste-heat cut-off, below a fifth of it — a very wide margin.
Location Display: ***** *********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	********* ****** campus averages *,*** MWh, *,*** MWh below the §8 screen — building-services load only, no flag.
Location Display: ***** *********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-***.**	**Alert:** No **Near Breach:** No	********* sits only *** MWh below the waste-heat cut-off, about **% of it — a monitoring case.
Location Display: ***** *******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******* branch averages *,*** MWh, *,*** MWh below the §8 screen — sales and service profile, no flag.
Location Display: ***** *******, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-*,***.**	**Alert:** No **Near Breach:** No	******* stays *,*** MWh under the waste-heat cut-off, roughly **% of it — no waste-heat priority.
Location Display: ***** ****, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	*******-******* branch averages about *,*** MWh, *,*** MWh below the §8 screen — no EnMS candidacy here.
Location Display: ***** ****, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-***.**	**Alert:** No **Near Breach:** No	The ******* branch sits *** MWh below the waste-heat cut-off — closer to that line than to §8.
Location Display: ***** *********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	********* ····· ****** averages *,*** MWh, *,*** MWh below the §8 screen — a low-consumption certified asset.
Location Display: ***** *********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	compliant	-***.**	**Alert:** No **Near Breach:** No	The ********* ****** sits *** MWh under the waste-heat cut-off — the tighter of the two rules here.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	compliant	-*,***.**	**Alert:** No **Near Breach:** No	********* logistics centre averages *,*** MWh, *,*** MWh below the §8 screen — no EnMS scope flag.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	The same logistics centre exceeds the waste-heat cut-off by *,*** MWh, so waste-heat screening and reporting are flagged.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_8 **Rule Label:** §8 EnMS obligation (ISO ***** / EMAS)	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	**********-********** campus averages **,*** MWh, *,*** MWh above the §8 screen — flagged as an EnMS scope candidate.
Location Display: ***** ********, Germany (Europe) **Location Id:** ****	**Rule Id:** enefg_16 **Rule Label:** §16 waste-heat obligation	obligated	*,***.**	**Alert:** Yes **Near Breach:** No	The ····· campus clears the waste-heat cut-off by *,*** MWh, so both EnEfG screens flag this location.

Limitations

#	Limitations
*	Grain mismatch (mandatory): EnEfG §8 and §16/§17 place their duties on the company (Unternehmen), not on individual *****. This per-site screen is an internal prioritisation and planning instrument; it does not establish, confirm or refute statutory applicability for any site, and a 'compliant' site result does not create or evidence any company-level exemption.
*	In aggregate the German site portfolio plainly exceeds both the *,*** MWh and the *,*** MWh company-level cut-offs, so the ** compliant rows must not be read as evidence of exemption from the EnMS, waste-heat or reporting duties.
*	Scope boundary: the screen tests thresholds only. It does not establish whether an ISO ***** energy management system or EMAS registration is already in place, whether waste heat is actually avoided, reduced or reused, or whether waste-heat information has been transmitted to the Bundesstelle für Energieeffizienz.
*	Deadline boundary: for §8-flagged ***** the statutory §8(*) timeline (** July 2025, or at the latest ** months after attaining the status) can only be quoted as a reference. Which branch applies cannot be derived from the 2022-2024 window and no certification dates are available, so no conclusion on deadline compliance is possible.
*	Method caveat: all rows use fiscal years rather than the 'abgeschlossene Kalenderjahre' the statute requires, and completeness of the metered carrier data against the statutory Gesamtendenergieverbrauch is not verified. This deviation applies uniformly to all ** obligated and ** compliant classifications.
*	Uneven data basis by site type: roughly a third of ***** — largely the real-estate and ****** rows — rest on a single fiscal year (2024), and one roll-up on 2022-2023 only, even though the column is labelled a three-year average. The ******/real-estate segment is therefore screened on a materially weaker basis than the manufacturing segment.
*	Three records (**-** **, **-** **, *** ***) report exactly *.* MWh and are classified compliant on both rules; six of the ** compliant results are therefore artefacts of absent or unassigned metering rather than demonstrated low consumption.
*	The site list mixes grains: individual buildings sit alongside roll-ups ('*** DE (****** *********)', '*** DE - *******', '*** DE - ********', '*** ********** ***** *******', '**-** …'), so consumption may be double-counted or unevenly scoped and per-site rankings are not directly comparable.
*	The near-breach band is not operative in this run (n_near_breach = *), yet several rows sit within about **% of their threshold in either direction — for example ******* ** ** *** * at +*.*% above the §16 cut-off, ****** ********** at +**% and **** at +*% above the §8 screen, and ******** and ********* at **% and **% of the §16 cut-off. Threshold sensitivity is therefore invisible in the aggregate counts.
**	Framing tension: *,*** MWh is defined by the regulation as the statutory exemption cut-off in §16(*)/§17(*), whereas the applied rule is labelled a waste-heat obligation threshold; passing the cut-off signals candidacy, it does not by itself create a duty.
**	Forward-looking policy sensitivity: a June 2026 cabinet draft would raise the EnMS trigger to roughly **.* GWh and rework the waste-heat duties. It is not yet law, the current *,*** MWh value was correctly applied, and only the largest ***** would remain flagged were the draft enacted.
**	Recurring master-data defects: empty zip_code fields with the postcode embedded in the city string, missing coordinates, an apparent city misspelling ('**********'), and postcode/city inconsistencies (********** *****, ***** ***** vs *****). These point to a common upstream data issue rather than site-specific errors.
**	The whole portfolio is a single country and region (******* / ******), so no cross-jurisdictional comparison is possible.
**	Site-level narrative context (plant profiles, waste-heat sources, district-heating availability) drawn from public sources is contextual only and, where marked unverified in the upstream chain, is not a regulatory finding.

References

#	References
*	R2F-001 — EnEfG.pdf, p. *: "Unternehmen mit einem jährlichen durchschnittlichen Gesamtendenergieverbrauch innerhalb der letzten drei abgeschlossenen Kalenderjahre von mehr als *,* Gigawattstunden sind verpflichtet, ein Energie- oder Umweltmanagementsystem gemäß Absatz * Satz * oder Satz * einzurichten." (EnEfG §8(*) — basis for the *,*** MWh EnMS screen.)
*	R2F-002 — EnEfG.pdf, p. **: "Ausgenommen von der Pflicht zur Vermeidung von Abwärme nach Absatz * Satz * und der Pflicht zur Verwendung von Abwärme nach Absatz * Satz * sind Unternehmen, die einen jährlichen durchschnittlichen Gesamtendenergieverbrauch innerhalb der letzten drei abgeschlossenen Kalenderjahre Jahre von *,* Gigawattstunden oder weniger haben." (EnEfG §16(*) — basis for the *,*** MWh waste-heat cut-off.)
*	R2F-002 — EnEfG.pdf, p. **: "Unternehmen sind verpflichtet, unabhängig vom Vorliegen einer konkreten Anfrage die in Absatz * aufgeführten Informationen zu anfallender Abwärme an die Bundesstelle für Energieeffizienz bis zum **. März eines jeden Jahres zu übermitteln ... Ausgenommen ... sind Unternehmen, die einen jährlichen durchschnittlichen Gesamtendenergieverbrauch innerhalb der letzten drei abgeschlossenen Kalenderjahre von *,* Gigawattstunden oder weniger haben." (EnEfG §17 — annual waste-heat reporting and its exemption.)
*	R2F-003 — EnEfG.pdf, p. *: "**. Gesamtendenergieverbrauch: ist die Gesamtmenge an Endenergie, die über alle Sektoren in einem vorgegebenen Zeitraum verbraucht wurde" (definition of the threshold measure).
*	R2F-003 — EnEfG.pdf, p. *: "*. Endenergie: derjenige Teil der eingesetzten Primärenergie, der den Verbrauchern nach Abzug von Energiewandlungs- und Übertragungsverlusten zur Verfügung steht ... **. Energiemanagementsystem: ein System, das den Anforderungen der DIN EN ISO *****, Ausgabe Dezember 2018 entspricht" (carrier scoping and EnMS definition).
*	R2F-004 — EnEfG.pdf, p. *: "Unternehmen, die bis zum Ablauf des **. November 2023 den Status eines Unternehmens nach Absatz * erlangt haben, müssen ein Energie- oder Umweltmanagementsystem bis zum Ablauf des **. Juli 2025 eingerichtet haben. Unternehmen, die ab dem **. November 2023 den Status ... ********, müssen ... spätestens ** Monate nach dem Zeitpunkt, zu dem sie diesen Status erlangt haben, eingerichtet haben." (EnEfG §8(*) — reference timeline attached to §8-flagged *****.)
*	R2F-005 — EnEfG.pdf, p. *: "*. die Einrichtung von Energie- oder Umweltmanagementsystemen für Unternehmen, ... *. die Vermeidung, Verwendung sowie Auskunft über Abwärme für Unternehmen." (duties addressed to companies, not ***** — basis for the grain-mismatch limitation.)

References and Assurance

How to read the identifiers used in this report

Token	What it means
R*F-###	A regulatory finding: one thing the regulation requires that this data can actually be tested against. Numbered in the order the scope reviewer produced them.
location_id	Siemens' own internal site number, carried through unchanged so a figure can be traced back to the source system.
rule_id	The threshold rule applied to the row (for example enefg_8 = the *.* GWh energy-management threshold).

Source documents

Role	Ref	File	SHA-256 (truncated)
Regulation (baseline)	REG-2	EnEfG.pdf	c***d**dc**a**a*...

External View (EX1/EX2) allowed sources

Domain	Institution	Scope
eur-lex.europa.eu	EUR-Lex	EU Official Journal, consolidated legislation
efrag.org	EFRAG	ESRS standard setter
commission.europa.eu	European Commission	EU policy and legislative proposals
esma.europa.eu	ESMA	EU sustainability-reporting supervision
eea.europa.eu	European Environment Agency	EU environmental data and assessments
echa.europa.eu	European Chemicals Agency	EU chemicals and pollution (ESRS E*)
bafa.de	BAFA	EnEfG Sec. */** administration (UC*)
dehst.de	DEHSt	German Emissions Trading Authority (UC*)
umweltbundesamt.de	Umweltbundesamt (UBA)	German Federal Environment Agency
bmuv.de	BMUV	German Federal Ministry for the Environment
ifrs.org	IFRS Foundation / ISSB	Global sustainability disclosure standards
globalreporting.org	GRI	Global Reporting Initiative standards
unfccc.int	UNFCCC	UN climate treaty body
unep.org	UNEP	UN Environment Programme
ipcc.ch	IPCC	Intergovernmental Panel on Climate Change
oecd.org	OECD	Environmental and economic policy analysis
iea.org	IEA	International Energy Agency
epa.gov	US EPA	US Environmental Protection Agency

Data sources (Siemens Database)

Siemens Database table	File	Column(s) used for this deliverable's figures	Data as of (file recorded date)
**********	**********	amount_consumed_mwh (registry alias: energy_mwh), *-year mean per site	2026-**-**
**********	regions.csv	join/dimension key only, no measure	2026-**-**
**********	bu_rc_groups.csv	join/dimension key only, no measure	2026-**-**
**********	**********	join/dimension key only, no measure	2026-**-**

Assurance and traceability

Run identifier: 2026-09-07_173946Z_regulation_cli_02
Model / backend: claude-opus-5. Adaptive reasoning was enabled; no sampling parameters were sent.
Source documents: 1, listed above with recorded hashes where available.
Approval gates: both human gates were recorded but auto-approved – this run was invoked with `--auto`. EX1 and EX2 still ran before each gate, as the architecture requires; their advisory content never changed the promoted result.
Retention: every agent's full input and output record for this run is retained under ***
*** and can be shared on request. Each figure in this report traces to a stored shelf record through its artifact_id, payload_sha256, and run_id.

G.4 UC2: Quarterly ETS 1 Scope 2 Assurance Memo

Pages 2–13: system design, final `F1` output, references, and assurance.

System Design — Quarterly ETS 1 Scope 2 Assurance Memo

Selection rule: $\pi(\text{case},\ \text{output profile},\ \text{assurance profile}) \rightarrow \text{topology}$

Selected: Debate — Independent readings of the same claim are reconciled into one result.

Derived case and assurance profile (CP1)

Task: Verification **Output:** Qualitative / structured memo **Assurance:** Elevated **Evidence shape:** Reconciliation record **Uncertainty:** High **Independent methods:** Yes **Cross-functional:** No **Ownership clusters:** 1

One-flow view

Stage 1

R1 → DA1 → Human: data mapping → R2 → EX1 → Human: regulatory scope

Stage 2

D1 → D2 → D3 → CP1 → Human: case profile → TS1 → Human: topology

P2/P3 → S1 → F1 → EX2 → Human: final output

What each agent does

- **R1** — Identifies regulation-driven information needs from the registered request.
- **R2** — Fixes the regulatory findings that downstream work may use.
- **D2** — Checks data quality, coverage, and readiness before interpretation.
- **CP1** — Turns approved evidence into the case, output, and assurance profile.
- **P2/P3** — Read the same claim independently through its two calculation methods.
- **F1** — Formats the completed tabular analysis as the requested deliverable.
- **DA1** — Maps those needs to available data and prepares the approved handoff.
- **D1** — Loads exactly the human-approved data request using deterministic code.
- **D3** — Computes the deterministic case facts used by the output agents.
- **TS1** — Selects Direct, Debate, or Coalition from the approved case profile.
- **S1** — Reconciles the two independent method readings into one conclusion.
- **EX1/EX2** — Add official-source outside views before the two principal review gates.

Human gates — A reviewer approves the data boundary, regulatory scope, case profile, topology, and finished output before each becomes binding.

External View — EX1 and EX2 provide official-source second reads at the review boundary. They are advisory only: reviewers see them, but they never change the promoted result.

External View snapshots — run 2026-09-07_190505Z_regulation_cli

EX1 — Stage 1

1. High / Challenges The EU ETS * annual surrender deadline is no longer ** April. [DEHSt]

2. High / Challenges The claim that * * % threshold has no basis in EU emissions-trading law is only half true, and the coincidence could mislead * reader. [EUR-Lex]

3. Medium / Adds context Although no emission-factor field was approved, official grid-intensity data give * reviewer an external plausibility check on any implied factor. [Umweltbundesamt (UBA), European Environment Agency]

4. Medium / Adds context The scope boundary being locked here sits against an actively moving legislative background. [European Commission]

EX2 — Stage 2

1. High / Flags gap The Commission tabled * full EU ETS review proposal in July 2026 (COM(2026)***) that responds to the 2023 review clauses on stationary installations under Articles *gg, **b and ** of the ETS Directive, and it ****** that several of the measures covering stationary installations expand the scope of the EU ETS rather than simplify it. [EUR-Lex]

2. High / Flags gap The deliverable treats 'ETS * scope' as the only emissions-trading frame for * large-manufacturing portfolio, but ETS* now also reaches fuel combustion in manufacturing, energy and construction activities that are not covered by ETS *, with monitoring and reporting already running and surrender starting later this decade. [DEHSt, EUR-Lex]

3. Medium / Flags gap * further downstream currency issue: GRI has superseded Disclosures ***-* to ***-* of GRI ***: Emissions 2016 with GRI ***: Climate Change 2025, effective * January 2027, alongside GRI ***: Energy 2025. [GRI]

4. Medium / Adds context If these site-level Scope * figures are ever reused for sustainability reporting, the standards basis is about to shift: the Commission adopted the Revised ESRS on * July 2026, they become legally effective only on Official Journal publication after the scrutiny period, and EFRAG has flagged that the GHG emissions boundary has moved to the financial consolidation approach with * separate operational-control disclosure in defined circumstances. [EFRAG]

5. Low / Adds context German administrative practice shows that even fixed statutory reporting deadlines get operationally adjusted, which is worth noting alongside the memo's caution that no Art. **(*) or ** April conclusion follows from * fiscal-quarter window: DEHSt agreed with the responsible federal ministry that 2025 ETS* emission reports, statutorily due ** April 2026, be accepted up to ** May 2026. [DEHSt]

6. Low / Adds context For * large-manufacturing portfolio, an adjacent carbon-pricing regime entered its definitive phase during the very window this memo covers: CBAM's transitional period ran to end-2025 and the definitive phase, with gradual phasing in of carbon pricing on imported embedded emissions, started in January 2026, with * Commission proposal to extend scope to downstream goods from 2028. [EUR-Lex]

Registered task and filters

Use case	Quarterly ETS 1 Scope 2 Assurance Memo
Emissions concept	Scope 2 = the emissions caused by the electricity, heat, or steam a site buys and consumes. Siemens does not generate this power itself, but accounts for it because the site is the one drawing it from the grid. Method A estimates it from metered energy use (MWh) times the local grid's carbon factor; Method B is the CO2e figure Siemens reports directly for the same rows. Comparing the two is a cross-check on the same underlying claim, not two different things.
Requested output	For each site and quarter: do the two independently computed CO*e figures (energy-based vs. reported) agree, and if not, is the gap big enough to matter?
Threshold / rule	Discrepancy check: a gap over 5% of the higher figure is mechanically flagged for review. Whether that gap is actually material is a separate judgement call made by the reconciling agent, not an automatic rule.
How it's calculated	Both methods read the same emissions-table rows for that site and period (Scope 2 only). Method A = amount_consumed_mwh × co2_factor (Siemens Database columns, computed); Method B = co2e_t, taken as reported, with no calculation applied.
Scope	size_tier=large; bu_rc=**, *** | 2025-**-** to 2026-**-** (quarter) | REG-1
Source	Live pipeline run; gate mode: auto-approved (–auto) (2026-09-07).

Color key: red = exception / urgent change amber = warning / material change green = acceptable / compliant

Table terms:

Discrepancy flag	yes if the two methods differ by more than 5% of the higher figure (mechanical check only).
Material	the reconciling agent's own judgement on whether that gap is big enough to matter for compliance – the 5% flag is only a starting signal, not the final word.

Entries

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ******* / ****** / ***** ******** **Location Id:** **	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods agree at *.* t CO_2e, but under R2F-004 that agreement is arithmetically trivial because both draw on the same (empty) Scope * rows. The site-quarter is data-absent rather than zero-emitting and should be presented as an open completeness item; no Art. **(*) or Annex IV determination is supported.	**Entry Text:** Neither method had activity data or reported tonnage; the step to ***.** t next quarter suggests reporting-population onboarding. Nil consumption cannot be distinguished from missing data, and the emission factor is unverifiable. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ******** **Location Id:** **	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The activity-data-derived and reported Scope * figures reconcile exactly at ***.**** t CO_2e, satisfying the internal consistency indicator described in R2F-004. This evidences internal arithmetic consistency only, not independent corroboration or * verified emissions determination.	**Entry Text:** No gap exists: both figures are alternative framings of the same Scope * rows. The emission factor is not verifiable, and the discontinuity against the nil prior quarter remains an open completeness question. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ******** **Location Id:** **	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two methods reconcile at approximately ***.** t CO_2e with only * rounding-level residual between them. The result is internal arithmetic consistency under R2F-004 — not independent corroboration, since both figures come from the same Scope * row population.	**Entry Text:** Only * fourth-decimal rounding residual separates the methods, * precision artefact rather than divergence. The emission factor is unverifiable, and the quarter-on-quarter rise is attributed to consumption volume rather than factor change. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ******** **Location Id:** **	2025 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B reconcile exactly at ***.**** t CO_2e, satisfying the R2F-004 consistency indicator. The quarter should nevertheless not be relied on as * complete period total without confirmation of activity-data coverage.	**Entry Text:** No methodological divergence — both measures use the same rows. The figure sits far below the next two quarters, * step neither seasonality nor grid-factor drift explains; possible partial-period capture. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ******** **Location Id:** **	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two figures reconcile exactly at ***.**** t CO_2e, meeting the internal consistency check under R2F-004. This supports an internal monitoring-assurance statement only, not * verified Annex IV emission determination.	**Entry Text:** No gap: both are views of the same Scope * rows, with the level attributed to winter-quarter purchased electricity. The emission factor and any verification status are not inspectable from the approved data. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ******** **Location Id:** **	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The energy-derived and reported Scope * figures reconcile exactly at ***.**** t CO_2e, the site's highest quarter in the window. Under R2F-004 this evidences internal arithmetic consistency only and supports no Annex V verification opinion.	**Entry Text:** Zero delta by construction, since both measures share source rows. The continued rise reflects metered consumption or wider coverage rather than * factor effect; the absence of * flag is not assurance. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V (verification)
Location Display: ******* / ****** / ***** ********** **Location Id:** ***	2025 Q*	**Method A:** **.** **Method B:** **.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B agree exactly at **.**** t CO_2e, so the R2F-004 consistency indicator is satisfied for this site-quarter. The figure is nevertheless provisional and should carry an explicit completeness caveat rather than serve as * trend baseline.	**Entry Text:** The reconciliation question is temporal, not methodological: this quarter is an order of magnitude below the next, too large * break for seasonality or factor drift. Probable partial-period activity-data capture. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ********** **Location Id:** ***	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The figures reconcile exactly at ***.**** t CO_2e, meeting the internal consistency check under R2F-004. The quarter is usable as * Scope * proxy for internal monitoring, though trend inference across the FY2025 Q* boundary remains unreliable.	**Entry Text:** Methods agree exactly because both are computed over identical Scope * rows. Both analysts locate the completeness defect in the prior quarter rather than here; the emission factor remains uninspectable. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*)

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ******* / ****** / ***** ************ **Location Id:** ***	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two figures reconcile exactly at ***.**** t CO_2e, the highest value in the batch. Under R2F-004 this is an internal consistency result only — not independent corroboration and not an Annex V verification opinion.	**Entry Text:** Zero delta reflects shared source rows. The roughly **% rise over Q* is attributed to higher metered consumption or wider coverage, since the German grid factor has been falling; implied consumption is inferred, not observed. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V (verification)
Location Display: ******* / ****** / ***** ****** **Location Id:** ***	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods return *.* t CO_2e, an informationally empty agreement rather than * reconciled emissions figure. Location *** should be presented as data-absent for this quarter and escalated for * Scope * row-coverage review, not counted as * zero-emitting site.	**Entry Text:** No delta arises because neither method had data. * persistent nil at an occupied German site on * carbon-bearing grid is more consistent with missing Scope * rows than * genuine zero. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ****** **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B both return *.* t CO_2e, and the nil cannot be affirmed as * real zero on the approved data. The conclusion is * completeness escalation for location *** rather than any numeric statement.	**Entry Text:** Absence of * gap reflects absence of data on both sides. * nil in the seasonally highest-load winter quarter is * strong indicator of missing Scope * records; the non-flag is uninformative. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V (freedom from inconsistencies)
Location Display: ******* / ****** / ***** ****** **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two methods agree at *.* t CO_2e for * third consecutive quarter, giving an entirely null series across the window. The site must be reported as data-absent and excluded from any quantitative portfolio conclusion.	**Entry Text:** Nothing sits on either side of the comparison; most plausibly no Scope * rows exist for this location. Absent rows cannot be distinguished from * zero-factor treatment within the approved scope. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ******* **Location Id:** ***	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both figures stand at *.* t CO_2e and reconcile only in the trivial sense that the same absent data flowed into each. The quarter is unresolved rather than compliant and should be carried as * data-completeness item.	**Entry Text:** Both methods operate on the same empty Scope * row set. * nil at * physical German site drawing grid supply points to * coverage gap rather than * documented zero. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V (freedom from inconsistencies)
Location Display: ******* / ****** / ***** ******* **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods return *.* t CO_2e, agreeing only because the same absent data underlies both. No affirmative Art. **(*) total can be stated; the row is flagged alongside the other nil quarters for location ***.	**Entry Text:** The zero delta reflects two derivations of the same empty rows. Comparable German sites report several hundred tonnes this quarter, making * nil at an operating site * significant completeness outlier. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ******* **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B agree at *.* t CO_2e for * third consecutive quarter, giving an entirely null series across FY2025 Q* – FY2026 Q*. The site should be reported as data-absent, excluded from quantitative portfolio totals and escalated for Scope * row-coverage review.	**Entry Text:** * persistent nil across the full window at * physical operating German location is more consistent with missing Scope * rows than * verified zero; monitoring-plan and supply data were out of scope. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC, Annex IV (calculation methodology and Part B 'Total emissions'), Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V (verification)

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ******* / ****** / ***** *******-********* **Location Id:** ***	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both figures agree to within *.**** t and are reconciled at approximately *.*** t CO_2e for the quarter. This is * documented Scope * proxy and * recorded reported total — neither an Annex IV emission determination nor evidence that the Art. **(*) duty has been discharged.	**Entry Text:** The *.**** t gap is rounding or stored-factor precision, not methodological divergence. * roughly one-tonne quarterly figure is small for an operating industrial site and may indicate partial metering coverage. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC Annex IV (Activity data × Emission factor × Oxidation factor), Directive 2003/**/EC Annex IV part B 'Total emissions', Directive 2003/**/EC Art. **(*)
Location Display: ******* / ****** / ***** *******-********* **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two methods reconcile at approximately *.*** t CO_2e, the site's peak quarter in this window. The figure is * Scope * purchased-electricity tonnage in Annex IV form and as recorded, not * verified installation emission.	**Entry Text:** The *.**** t gap is precision noise from * shared derivation. The roughly *.*× step from the prior quarter is driven by activity data, not any documented grid-factor change, and needs meter-coverage review. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B 'Total emissions', Directive 2003/**/EC Art. **(*)
Location Display: ******* / ****** / ***** *******-********* **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	Both methods return *.**** t CO_2e and are fully reconciled. The exact agreement evidences arithmetic consistency only and does not constitute Art. ** verification or an Annex V verification opinion.	**Entry Text:** No delta, since both measures are computed from the same Scope * rows. The seasonal fall from the Q* peak is plausible, but attribution to seasonality rather than metering change is unverified. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B, Directive 2003/**/EC Art. **(*)
Location Display: ******* / ****** / *-**** **** **Location Id:** ***	2025 Q*	**Method A:** **.** **Method B:** **.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two methods agree exactly at **.**** t CO_2e and are reconciled. The combined figure is * purchased-electricity Scope * tonnage at * named ******** site, not an ETS * installation emission determination.	**Entry Text:** Zero delta reflects shared derivation, not independent agreement. The value is roughly fifteen times below the next quarter at the same site, suggesting partial-period or partial-scope metering; review recommended. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B 'Total emissions', Directive 2003/**/EC Art. **(*)
Location Display: ******* / ****** / *-**** **** **Location Id:** ***	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods return ***.**** t CO_2e and are fully reconciled; this site dominates the batch's Scope * total. The figure is * recorded and calculation-consistent purchased-electricity tonnage, not * verified Art. **(*) return or an Annex IV determination.	**Entry Text:** The zero delta shows only one shared computation over the same rows. The roughly fifteenfold step change from the prior quarter is an open reconciliation item for the underlying MWh series. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B 'Total emissions', Directive 2003/**/EC Art. **(*)
Location Display: ******* / ****** / *-**** **** **Location Id:** ***	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two methods are identical at ***.**** t CO_2e and reconciled, with the site showing * level shift followed by continued growth. Under R2F-003 this is the recorded reported Scope * total; it supports no Art. ** verification or Art. **(*) surrender conclusion.	**Entry Text:** No gap to explain — both resolve to the same recorded quantity. The Q* rise runs counter to falling seasonal ******** grid intensity, implying volume growth that should be corroborated against meter records. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B, Directive 2003/**/EC Art. **(*)
Location Display: ******* / ****** / *-**** **** **Location Id:** ***	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two methods agree at *.**** t CO_2e, so the figures are reconciled arithmetically but carry weak evidential value. The reading is * probable data-completeness gap rather than * substantiated quarterly Scope * level.	**Entry Text:** At roughly ** kg for * full quarter, rounding and factor precision could dominate and no uncertainty statement bounds the value. Both analysts read it as partial-period or partially metered supply. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B 'Total emissions', Directive 2003/**/EC Art. **(*)

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ******* / ****** / *-**** **** **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B reconcile at approximately *.*** t CO_2e, with no inconsistency under the R2F-004 indicator. The combined picture is * small-to-mid-sized ******** site load recorded as purchased-electricity Scope * tonnes.	**Entry Text:** The *.**** t gap is precision noise from * single shared computation. The roughly **× discontinuity against the prior quarter most likely reflects that quarter's completeness and should be reconciled first. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B, Directive 2003/**/EC Art. **(*)
Location Display: ******* / ****** / *-**** **** **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two methods reconcile at approximately *.*** t CO_2e, completing * monotonically rising series across the window. The figure is reproducible in Annex IV calculation form but is not * monitored emission determination and is not verified under Art. **.	**Entry Text:** * *.**** t residual is precision noise. Using * period-average rather than consumption-weighted factor introduces unquantified error, and no Annex IV uncertainty element is available to bound it. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B, Directive 2003/**/EC Art. **(*)
Location Display: ****** / ******** / ********** ** *** *** **Location Id:** ***	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods return *.* t CO_2e; the figures are formally reconciled but substantively empty. The row is an unresolved data-completeness item, and as * ******** location no ETS * obligation is asserted or implied.	**Entry Text:** * null Annex IV product and an empty reported record cannot be distinguished from * data gap. *******'s grid is low-carbon but not zero-carbon, so any metered draw would yield * positive figure. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B 'Total emissions', Directive 2003/**/EC Art. **(*)
Location Display: ****** / ******** / ********** ** *** *** **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods report *.* t CO_2e, so the two readings agree but neither has testable content. The conclusion is * systematic activity-data or emission-factor population gap, to be resolved before portfolio aggregation.	**Entry Text:** * zero in the peak winter quarter is the least plausible point for * genuine nil. The value is identical across all three quarters, indicating * structural rather than quarter-specific cause. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B, Directive 2003/**/EC Art. **(*)
Location Display: ****** / ******** / ********** ** *** *** **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B both return *.* t CO_2e, closing an unbroken nil series across the window. The row evidences an unresolved data-completeness issue and should be excluded from inference about the site's actual Scope * profile.	**Entry Text:** * formally well-formed but empty calculation; two nil measures agree trivially and carry no assurance value. The absence of * discrepancy flag must not be presented as verification. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B, Directive 2003/**/EC Art. **(*)
Location Display: ****** ****** / ******** / **********/** *****-**** **Location Id:** ***	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods return *.* t CO_2e; the figures are trivially reconciled but evidence incomplete Scope * activity data rather than absence of emissions. As * US location it falls outside the Member State reach of the Directive.	**Entry Text:** The local supply retains * substantial fossil component, so any metered consumption would produce * non-zero figure. Neither analyst could distinguish * genuine nil from * data gap within scope. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B 'Total emissions', Directive 2003/**/EC Art. **(*)
Location Display: ****** ****** / ******** / **********/** *****-**** **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods report *.* t CO_2e, contributing nothing to the portfolio Scope * total for the quarter. The reading is * data-capture gap of structural character, not * decarbonisation outcome.	**Entry Text:** US practice relies predominantly on location-based grid-average factors, making * zero location-based figure for an active site hard to account for; three consecutive nils indicate * structural cause. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B, Directive 2003/**/EC Art. **(*)

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ****** ****** / ******** / **********/** *****-**** **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B both return *.* t CO_2e, completing * flat nil series across the batch. The row is untestable and should be carried as an open completeness item, excluded from any assurance narrative.	**Entry Text:** * residual fossil component persists in local supply throughout the window, so * genuine zero would require zero metered consumption. The unflagged consistency result carries no assurance content. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC Annex IV (calculation formula), Directive 2003/**/EC Annex IV part B, Directive 2003/**/EC Art. **(*)
Location Display: ******* / ****** / ***** ******* **Location Id:** ***	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B are arithmetically reconciled at *.* t CO_2e with no discrepancy flag. The reading is most plausibly * monitoring or data-loading gap; source-data tracing of the Scope * row population for location *** is recommended rather than reliance on the zero.	**Entry Text:** The energy leg had no MWh to multiply and the reported record was equally empty — either an unpopulated row set or * contractual zero. Neither explanation is documented in the approved data. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part B (Activity data × Emission factor × Oxidation factor; Total emissions), Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V
Location Display: ******* / ****** / ***** ******* **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two methods are reconciled at *.* t CO_2e with no discrepancy flag, showing that no Scope * consumption was captured for location *** in the quarter. This supports neither an emissions determination nor * statement about reporting compliance.	**Entry Text:** The Annex IV formula shape survives with * nil activity-data input. * German location-based reading in * winter quarter would ordinarily be positive, so the zero points to an empty record. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ******* **Location Id:** ***	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two figures agree at *.* t CO_2e and are reconciled arithmetically, closing * flat three-quarter zero series for location ***. Read together they show an absence of captured purchased-electricity data rather than an emissions performance outcome.	**Entry Text:** The consistency indicator passes trivially and evidences nothing. * verifier could test neither the reported activity data nor the calculations because both are nil; treat as * systematic monitoring gap. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Annex V, Directive 2003/**/EC, Art. **(*)
Location Display: ***** / **** ********* / ****** ****** ******** ******** **Location Id:** ****	2025 Q*	**Method A:** *.** **Method B:** -*.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two methods agree numerically to within *.**** t and D* records no discrepancy flag, but neither figure represents FY2025 Q* emissions at the ****** site. The ******** reported value should be traced to source and the quarter treated as incomplete or reversing.	**Entry Text:** Both measures collapse to about zero in * quarter that adjacent periods of *** t and *** t suggest should be substantial. The ******** reported value is * probable reversal or rounding residual. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V
Location Display: ***** / **** ********* / ****** ****** ******** ******** **Location Id:** ****	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	Both methods report ***.**** t CO_2e and are fully reconciled, with the figure sitting in * plausible range for the location. It remains * purchased-electricity Scope * proxy and internal reported record — not an Annex IV emission determination and not verified.	**Entry Text:** Exact agreement reflects computation over the same Scope * rows rather than corroboration. The applied emission factor cannot be confirmed as an official national, provincial or supplier value. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Art. **(*)
Location Display: ***** / **** ********* / ****** ****** ******** ******** **Location Id:** ****	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	Both methods report ***.**** t CO_2e and are fully reconciled with no discrepancy. Together they show * genuine and rising consumption signal at the ****** site, though the figure remains * Scope * proxy rather than an Annex IV determination.	**Entry Text:** Zero delta arises from shared source rows. Applying an annual-average grid factor to * fiscal quarter is an unaddressed simplification whose error is unquantified; the ~**% rise is unattributed. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ******* ***** **Location Id:** ****	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B are reconciled at *.* t CO_2e with no discrepancy flag, but the combined picture is an absence of monitored purchased-electricity data rather than an emissions result. The row should be escalated for source-data confirmation.	**Entry Text:** * zero on * ******** grid with positive published carbon intensity implies missing activity data or an unposted row set. No certificate evidence supports * market-based zero. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part * and Part B, Directive 2003/**/EC, Art. **(*)

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ******* / ****** / ******* ***** **Location Id:** ****	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two figures are reconciled at *.* t CO_2e with no discrepancy flag, showing that the ***** site contributed no Scope * tonnage to the record. This is best read as * data-population question rather than an operational outcome.	**Entry Text:** The Annex IV calculation is reproduced in form only, with no activity data, factor or oxidation factor available for inspection; the reported total is equally unsubstantiated without monitoring-plan evidence. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ******* ***** **Location Id:** ****	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods stand at *.* t CO_2e and are arithmetically reconciled, completing * flat zero series for location **** across the window. The site has no usable Scope * emissions record and should be treated as * systematic monitoring gap.	**Entry Text:** Zero-versus-zero produces * trivial pass on the consistency indicator. Annex V-style testing of activity data and calculations is impossible where both inputs are nil; the persistence is the finding. **Citations:** R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Annex V, Directive 2003/**/EC, Art. **(*)
Location Display: ****** / ****** / ***** ********** **Location Id:** ****	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two methods are reconciled at *.* t CO_2e with no discrepancy flag, and the combined reading is * probable activity-data gap at * site on * high-carbon grid. ****** is not an EU Member State, so no ETS * obligation attaches directly and the record should be handled as internal Scope * data.	**Entry Text:** On one of ******'s most carbon-intensive grids any purchased kWh would carry * high factor, so * nil is very unlikely to be * genuine result. Source confirmation is required. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part * and Part B, Directive 2003/**/EC, Art. **(*)
Location Display: ****** / ****** / ***** ********** **Location Id:** ****	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two figures are reconciled at *.* t CO_2e with no discrepancy flag, showing no captured Scope * consumption for * winter quarter on * lignite-dominated grid. Both analysts treat this as * monitoring gap rather than an emissions outcome; no ETS * conclusion follows.	**Entry Text:** The formula shape survives with no activity data, factor or oxidation factor for * verifier to test, and it cannot be confirmed whether * ******-******** grid factor was applied at all. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Art. **(*)
Location Display: ****** / ****** / ***** ********** **Location Id:** ****	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods report *.* t CO_2e and are arithmetically reconciled, closing * flat three-quarter series for location ****. This is an unexplained absence of Scope * records at an operating site, to be escalated as * data-completeness matter; no ETS * obligation is asserted for this non-EU location.	**Entry Text:** The zero-versus-zero comparison yields no assurance information. No emission factor, oxidation factor or uncertainty estimate exists against which the calculation could be assessed. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Annex V, Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ***** **Location Id:** ****	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two methods are reconciled at *.* t CO_2e for the quarter with no discrepancy flag, but the reconciled substance is an absence of monitored purchased-electricity data. No conclusion about installation status, emissions performance or reporting discharge is available; escalate for source-data confirmation.	**Entry Text:** The activity-data input was empty and the reported record correspondingly zero — either an unpopulated row set or an undocumented market-based treatment. Neither is evidenced in the approved data. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part * and Part B, Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ***** **Location Id:** ****	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B agree at *.* t CO_2e with no discrepancy flag, showing no Scope * tonnage captured where * positive location-based figure would be expected. The accounting basis and row completeness must be established before the value is relied upon.	**Entry Text:** The calculation follows Annex IV in shape only, with neither activity data nor * visible factor. Whether * location-based or market-based basis was applied is the unresolved determinant. **Citations:** R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ***** **Location Id:** ****	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods report *.* t CO_2e and are arithmetically reconciled, completing * flat three-quarter zero series for location ****. This is * systematic gap in Scope * data capture, to be escalated as * monitoring and completeness matter; no Annex V verification opinion is offered.	**Entry Text:** The zero delta is * trivial pass evidencing nothing. The concentration of exact-zero rows across this portfolio is * data-completeness question * tolerance-based check can never register. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-006, Directive 2003/**/EC, Annex IV Part B, Directive 2003/**/EC, Annex V, Directive 2003/**/EC, Art. **(*)

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ******* / ****** / *******, **. *****, ******** **Location Id:** ****	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B agree at *.* t CO_2e and are reconciled for the quarter. The agreed figure supports only that both approved Scope * measures are nil; it is not an Annex IV emission determination and does not evidence discharge of the Art. **(*) duty.	**Entry Text:** * zero cannot be distinguished from an absent or unposted Scope * record, and *******'s grid factor is materially above zero, so grid intensity does not explain the result. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV part *, Directive 2003/**/EC, Annex IV part B, Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / *******, **. *****, ******** **Location Id:** ****	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two methods reconcile exactly at *.* t CO_2e. This supports only that both approved Scope * measures are nil; the site-quarter should be confirmed as * genuine zero or corrected as * data gap before inclusion in any assurance statement.	**Entry Text:** Neither an operating site drawing grid power nor * peak ***** heating quarter would normally calculate to zero; both analysts attribute the result to an absent or suppressed consumption record. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV part B, Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / *******, **. *****, ******** **Location Id:** ****	2026 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two methods reconcile at *.* t CO_2e, completing * flat nil series across the FY2025 Q* – FY2026 Q* window. Per R2F-004 this agreement is internal arithmetic consistency only; location **** has no usable Scope * signal and should be treated as * data-completeness item.	**Entry Text:** * flat zero series more plausibly indicates absent activity data or an unsupported zero-factor assumption than genuine nil consumption; * defensible market-based zero would need certificate evidence not present here. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV part B, Directive 2003/**/EC, Annex V, Directive 2003/**/EC, Art. **(*)
Location Display: ***** / ********* / ****** ******* **Location Id:** ****	2025 Q*	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Method * and Method B agree at *.* t CO_2e so the consistency indicator passes, but the FY2025 Q* series for this site is most likely incomplete rather than genuinely nil. No Annex IV determination or Art. **(*) discharge can be inferred, and R2F-001 confines the row to identification grain only.	**Entry Text:** The same site records ***.** t and ***.** t in the following two quarters, so consumption resumes immediately; * heating-season nil on * coal-weighted grid is implausible. Treat as an open data gap. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV part *, Directive 2003/**/EC, Annex IV part B, Directive 2003/**/EC, Art. **(*)
Location Display: ***** / ********* / ****** ******* **Location Id:** ****	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two figures reconcile exactly at ***.**** t CO_2e and are unit-consistent under R2F-005. Per R2F-004 the agreement is an internal arithmetic consistency check, not independent corroboration, and the figure remains * purchased-electricity proxy rather than * verified Art. **(*) report.	**Entry Text:** Both methods are computed over the same Scope * rows. The magnitude is plausible for * ******* industrial-park site, but whether * provincial, regional or national factor was applied cannot be determined. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV part B, Directive 2003/**/EC, Annex V, Directive 2003/**/EC, Art. **(*)
Location Display: ***** / ********* / ****** ******* **Location Id:** ****	2026 Q*	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two figures reconcile at approximately ***.** t CO_2e with only * rounding-scale residual, the site's highest tonnage in the window. Under R2F-004 the near-exact agreement is internal consistency rather than independent corroboration, and it supports no Annex V verification opinion.	**Entry Text:** The *.**** t difference is decimal rounding, not divergence. The roughly **% quarter-on-quarter rise cannot be decomposed into consumption change versus factor change with the approved fields. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV part B, Directive 2003/**/EC, Annex V, Directive 2003/**/EC, Art. **(*)

Limitations

#	Limitations
*	Independence limitation (R2F-004): Method * (scope*_market_proxy_t) and Method B (scope*_location_t) are both derived from the same Scope * rows of the same emissions fact table. Portfolio-wide agreement therefore evidences internal arithmetic consistency, not independent corroboration of the reported tonnages, and must not be presented to readers as assurance.
*	The *% discrepancy threshold applied by the deterministic engine is * user-defined assurance convention with no basis in Directive 2003/**/EC, which sets no numeric materiality limit (R2F-004). The zero flag count is not * regulatory materiality clearance; all materiality verdicts in this memo are analyst judgements in context.
*	Where both methods return zero the consistency test passes trivially and carries no information whatsoever. * large share of the unflagged rows in this portfolio are nil-versus-nil comparisons of exactly this kind.
*	Company-vs-site grain: the approved data supports identification by site name and country only (R2F-001). No row determines that * site is * 'stationary technical unit' within Art. *(e), that it carries out an Annex I activity, or that it is an ETS * installation; address, postcode and Annex I activity type are not available, and zip_code is empty across the portfolio.

#	Limitations
*	Scope boundary (R2F-002, R2F-003): all figures are purchased-electricity Scope * tonnes, not emissions released from sources in an installation. Nothing here evidences that any operator has discharged the Art. **(*) reporting duty, and no figure is * verified emissions figure.
*	The co*_factor underlying Method * was not independently verifiable at any site, no oxidation factor is visible, no uncertainty estimate is available, and no carrier (media_type) decomposition is exposed. Annex IV part B requires all of these, so the assessment is confined to the structure of the calculation rather than its substance.
*	Monitoring-plan status, verifier status, emission-factor vintage and Art. ** verification evidence were all outside the approved data scope; no Annex V verification opinion and no statement that any report is satisfactory can be made.
*	Systematic zero-value coverage concern: several sites report *.* t on both methods for all three quarters — locations *** (****** K **), *** (**********), *** (**********), *** (**********), *** (******* KM), **** (*****), **** (**********), **** (***** T) and **** (*******). Both analysts independently read these as probable systematic Scope * row or activity-data gaps rather than genuine nil consumption; they should be presented as data-absent and excluded from quantitative conclusions.
*	Single-quarter nils beside substantial adjacent quarters: location **** (*******) records *.* t for FY2025 Q* immediately before quarters of ***.** t and ***.** t, and location **** (******) records * nil/******** Y2025 Q* before ***.** t and ***.** t. Both are most plausibly missing periods rather than idle quarters.
**	Data-hygiene exception: location ****, FY2025 Q* carries * ******** reported Scope * value (−*.**** t CO_2e) against * Method * value of *.*. * ******** total-emissions figure is not physically meaningful; D*'s pct_diff of *.* masks the sign anomaly and the entry should be traced to source.
**	Series-discontinuity concern: FY2025 Q* figures at locations ** (nil), ** (***.** t) and *** (**.** t) sit far below the same sites' FY2026 quarters; large unexplained step changes also appear at location *** (~**× from FY2025 Q* to FY2026 Q*), location *** (~**× over the same step) and location *** (~*.*×). The dual-method check cannot detect these because both measures move together; the underlying MWh series must be reconciled before any trend is presented as an emissions trend.
**	De minimis figures: location *** FY2025 Q* (*.**** t) and location *** FY2025 Q* (~*.*** t) are small enough for * full quarter that rounding and factor precision could dominate, and both are out of line with the same sites' later quarters — flagged for activity-data completeness review.
**	Periodicity (R2F-006) applies portfolio-wide: all figures are Siemens Database fiscal quarters (FY2025 Q* – FY2026 Q*), * monitoring-assurance artefact that does not align with the Directive's calendar-year basis under Art. **(*) and Art. **(*). The window does not cover * complete calendar year, so no surrender-compliance, ** April deadline or Art. ** verification conclusion is supported.
**	Unit caveat (R2F-005): figures are stated in t CO_2e, the Directive's unit for Annex II gases, but the tonnage cannot be attributed to any specific Annex II gas and the GWP conversion applied is not confirmable from the approved fields.
**	Jurisdictional caveat: sites in ****** (***), the ****** ****** (***), ***** (****, ****) and ****** (****) lie outside the Member State reach of Directive 2003/**/EC. For these locations the Art. **(*) framing is analogical only; no ETS * obligation is asserted or implied, and ******'s domestic MRV and carbon-pricing regime, not the Directive, is the operative local obligation.
**	Numerous site-level explanatory statements in the upstream analysis were expressly marked 'unverified / model-knowledge' — including operational status, occupancy, metering arrangements, supply contracts, certificate retirement, and inferred MWh magnitudes. None of these should be relied on as findings.

References

#	References
*	R2F-001
*	R2F-002
*	R2F-003
*	R2F-004
*	R2F-005
*	R2F-006
*	Directive 2003/**/EC, Art. *(e) (definition of 'installation')
*	Directive 2003/**/EC, Art. **(*) (surrender obligation, ** April, preceding calendar year)
*	Directive 2003/**/EC, Art. **(*)–(*) (monitoring and operator reporting duty)
**	Directive 2003/**/EC, Art. ** (verification)
**	Directive 2003/**/EC, Annex I (activities)
**	Directive 2003/**/EC, Annex II (greenhouse gases)
**	Directive 2003/**/EC, Annex IV part * (data identifying the installation)
**	Directive 2003/**/EC, Annex IV part B (Activity data × Emission factor × Oxidation factor; Total emissions; Uncertainty)
**	Directive 2003/**/EC, Annex V (verification criteria; freedom from inconsistencies)

References and Assurance

How to read the identifiers used in this report

Token	What it means
R*F-###	* regulatory finding: one thing the regulation requires that this data can actually be tested against. Numbered in the order the scope reviewer produced them.
location_id	Siemens' own internal site number, carried through unchanged so * figure can be traced back to the source system.

Cited regulatory findings in full

ID	Finding and source
R2F-001	Emissions records can be attributed to * named site and Member State, so the memo can be produced at the per-installation grain and can report the installation-identifying elements required by Annex IV part * that are available (name of the installation and its country); this is an identification and presentation grain only, not * determination that the site is an 'installation' within the meaning of Art. *(e). – REG-1, CELEX_*****L****_EN_TXT.pdf, p. *
R2F-002	Method (*) — the calculation-based figure in the form Annex IV prescribes (activity data × emission factor) — can be produced per site per fiscal quarter from the approved derived measure scope*_market_proxy_t over Scope * rows of the emissions fact table, and can be decomposed by energy carrier (media_type), reflecting Annex IV's requirement of * separate calculation per fuel. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. **
R2F-003	Method (B) — the directly reported total-emissions figure that Annex IV requires per activity and that Art. **(*) makes the object of the operator's reporting duty — can be produced per site per fiscal quarter from the approved derived measure scope*_location_t (co*e_t over Scope * rows of the same emissions fact table), expressed in tonnes of CO_2 equivalent. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. *
R2F-004	Annex V requires that reported data be free of inconsistencies and that verification address the reported activity data and the calculations leading to overall emissions; the two approved measures allow * per-site, per-quarter consistency indicator — the absolute delta between the activity-data-derived figure (scope*_market_proxy_t) and the reported figure (scope*_location_t), with * flag where the delta exceeds * % of the higher value. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. **
R2F-005	Both approved figures are expressed in tonnes of CO_2 equivalent, the unit the Directive uses for the greenhouse gases listed in Annex II, so the two methods are reported in * directive-consistent unit and are directly comparable to one another. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. **
R2F-006	The Directive anchors both the reporting duty (Art. **(*)) and the surrender obligation (Art. **(*), by ** April for the preceding calendar year) to verified calendar-year emissions; the approved time grain is the fiscal quarter (FY2025 Q* – FY2026 Q*), so the memo can state expressly that its quarterly figures are * monitoring-assurance artefact that does not align with the Directive's annual calendar-year basis. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. *

Source documents

Role	Ref	File	SHA-256 (truncated)
Regulation (baseline)	REG-1	CELEX_*****L****_EN_TXT.pdf	b*******e*fe*af*...

External View (EX1/EX2) allowed sources

Domain	Institution	Scope
eur-lex.europa.eu	EUR-Lex	EU Official Journal, consolidated legislation
efrag.org	EFRAG	ESRS standard setter
commission.europa.eu	European Commission	EU policy and legislative proposals
esma.europa.eu	ESMA	EU sustainability-reporting supervision
eea.europa.eu	European Environment Agency	EU environmental data and assessments
echa.europa.eu	European Chemicals Agency	EU chemicals and pollution (ESRS E*)
bafa.de	BAFA	EnEfG Sec. */** administration (UC*)

Domain	Institution	Scope
dehst.de	DEHSt	German Emissions Trading Authority (UC*)
umweltbundesamt.de	Umweltbundesamt (UBA)	German Federal Environment Agency
bmuv.de	BMUV	German Federal Ministry for the Environment
ifrs.org	IFRS Foundation / ISSB	Global sustainability disclosure standards
globalreporting.org	GRI	Global Reporting Initiative standards
unfccc.int	UNFCCC	UN climate treaty body
unep.org	UNEP	UN Environment Programme
ipcc.ch	IPCC	Intergovernmental Panel on Climate Change
oecd.org	OECD	Environmental and economic policy analysis
iea.org	IEA	International Energy Agency
epa.gov	US EPA	US Environmental Protection Agency

Data sources (Siemens Database)

Siemens Database table	File	Column(s) used for this deliverable's figures	Data as of (file recorded date)
**********	**********	amount_consumed_mwh × co*_factor → scope*_market_proxy_t; co*e_t → scope*_location_t (Scope * rows only)	2026-**-**
**********	regions.csv	join/dimension key only, no measure	2026-**-**
**********	bu_rc_groups.csv	join/dimension key only, no measure	2026-**-**
**********	**********	join/dimension key only, no measure	2026-**-**

Assurance and traceability

Run identifier: 2026-09-07_190505Z_regulation_cli
Model / backend: claude-opus-5. Adaptive reasoning was enabled; no sampling parameters were sent.
Source documents: 1, listed above with recorded hashes where available.
Approval gates: both human gates were recorded but auto-approved – this run was invoked with `--auto`. EX1 and EX2 still ran before each gate, as the architecture requires; their advisory content never changed the promoted result.
Retention: every agent's full input and output record for this run is retained under *** *** and can be shared on request. Each figure in this report traces to a stored shelf record through its artifact_id, payload_sha256, and run_id.

G.5 UC2.1: Annual ETS 1 Scope 2 Assurance Memo

Pages 2–9: system design, final `F1` output, references, and assurance.

System Design — Annual ETS 1 Scope 2 Assurance Memo

Selection rule: $\pi(\text{case, output profile, assurance profile}) \rightarrow \text{topology}$

Selected: Debate — Independent readings of the same claim are reconciled into one result.

Derived case and assurance profile (CP1)

Task: Verification **Output:** Qualitative / structured memo **Assurance:** Elevated **Evidence shape:** Reconciliation record **Uncertainty:** High **Independent methods:** Yes **Cross-functional:** No **Ownership clusters:** 1

One-flow view

Stage 1

R1 → DA1 → Human: data mapping → R2 → EX1 → Human: regulatory scope

Stage 2

D1 → D2 → D3 → CP1 → Human: case profile → TS1 → Human: topology

P2/P3 → S1 → F1 → EX2 → Human: final output

What each agent does

- **R1** — Identifies regulation-driven information needs from the registered request.
- **R2** — Fixes the regulatory findings that downstream work may use.
- **D2** — Checks data quality, coverage, and readiness before interpretation.
- **CP1** — Turns approved evidence into the case, output, and assurance profile.
- **P2/P3** — Read the same claim independently through its two calculation methods.
- **F1** — Formats the completed tabular analysis as the requested deliverable.
- **DA1** — Maps those needs to available data and prepares the approved handoff.
- **D1** — Loads exactly the human-approved data request using deterministic code.
- **D3** — Computes the deterministic case facts used by the output agents.
- **TS1** — Selects Direct, Debate, or Coalition from the approved case profile.
- **S1** — Reconciles the two independent method readings into one conclusion.
- **EX1/EX2** — Add official-source outside views before the two principal review gates.

Human gates — A reviewer approves the data boundary, regulatory scope, case profile, topology, and finished output before each becomes binding.

External View — EX1 and EX2 provide official-source second reads at the review boundary. They are advisory only: reviewers see them, but they never change the promoted result.

External View snapshots — run 2026-09-07_190508Z_regulation_cli

EX1 — Stage 1

1. High / Challenges The ** April surrender deadline that the gate treats as the operative Art. **(*) date has been superseded for the very years in the memo's window. [DEHSt]

2. High / Challenges Grounding the entire memo in * 2003 consolidation of Directive 2003/**/EC is * live legal-currency risk, not merely * terminological gap. [EUR-Lex]

3. Medium / Flags gap An independent, non-Siemens dataset exists that could partially answer the scope question this gate is about to record as indeterminate. [European Environment Agency]

4. Medium / Adds context The fiscal-versus-calendar-year limitation is more consequential than * rounding note, because the EU compliance cycle is built on fixed calendar dates. [EUR-Lex]

5. Low / Adds context Although this memo is scoped for internal compliance-team review, the same Scope * figures typically flow into * CSRD sustainability statement where supervisors are actively looking. [ESMA]

EX2 — Stage 2

1. High / Challenges The memo ****** that the Directive fixes no numeric materiality figure and treats the *% tolerance as * purely internal registry convention. [EUR-Lex]

2. High / Challenges The limitations section refers to 'the ** April deadline' for surrender under Art. **(*). [DEHSt]

3. Medium / Flags gap Factor vintage is * larger risk than the memo's per-entry commentary implies, because European grid intensity has been moving fast: the EEA reports that generating one kilowatt hour in the EU emitted **% less in 2024 than in 2023 and **% less than * decade ago. [European Environment Agency]

4. Medium / Flags gap The absent reporting-period label is more consequential against the real compliance cycle than * 'grain mismatch' framing suggests. [EUR-Lex]

Registered task and filters

Use case	Annual ETS 1 Scope 2 Assurance Memo
Emissions concept	Scope 2 = the emissions caused by the electricity, heat, or steam a site buys and consumes. Siemens does not generate this power itself, but accounts for it because the site is the one drawing it from the grid. Method A estimates it from metered energy use (MWh) times the local grid's carbon factor; Method B is the CO2e figure Siemens reports directly for the same rows. Comparing the two is a cross-check on the same underlying claim, not two different things.
Requested output	For each site and year (FY2022-FY2024): do the two independently computed CO*e figures (energy-based vs. reported) agree, and if not, is the gap big enough to matter?
Threshold / rule	Discrepancy check: a gap over 5% of the higher figure is mechanically flagged for review. Whether that gap is actually material is a separate judgement call made by the reconciling agent, not an automatic rule.
How it's calculated	Both methods read the same emissions-table rows for that site and period (Scope 2 only). Method A = amount_consumed_mwh × co2_factor (Siemens Database columns, computed); Method B = co2e_t, taken as reported, with no calculation applied.
Scope	size_tier=large; bu_rc=**, *** | 2022-**-** to 2024-**-** (year) | REG-1
Source	Live pipeline run; gate mode: auto-approved (–auto) (2026-09-07).

Color key: red = exception / urgent change amber = warning / material change green = acceptable / compliant

Table terms:

Discrepancy flag	yes if the two methods differ by more than 5% of the higher figure (mechanical check only).
Material	the reconciling agent's own judgement on whether that gap is big enough to matter for compliance – the 5% flag is only a starting signal, not the final word.

Entries

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ******* / ****** / ***** ********* **Location Id:** **	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** **.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two figures bracket * Scope * purchased-energy quantity of roughly *,***–*,*** t CO_2e and are reconciled for memo purposes, agreeing within the registry tolerance and unit-consistent under R2F-002. Under R2F-001 neither figure may be presented as an ETS * reportable installation emission.	**Entry Text:** The ~** t gap most plausibly reflects * factor-vintage or factor-basis difference between columns, not differing activity data; the embedded emission factor is not inspectable and no period label exists. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC, Annex IV (calculation and reported/measurement routes), Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V
Location Display: ******* / ****** / ***** ********* **Location Id:** **	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** **.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The energy-derived and reported figures converge on approximately *,***–*,*** t CO_2e of Scope * purchased-energy emissions and are reconciled. Under R2F-001 the figure must be presented as indirect purchased-energy emissions, not as an ETS * installation total.	**Entry Text:** The residual ~** t is explainable by rounding and minor factor differences; no data-quality defect appears, though the Annex IV formula remains incomplete and the reporting period is unlabelled. **Citations:** R2F-001, R2F-002, R2F-003, R2F-004, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC, Annex IV, Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V
Location Display: ******* / ****** / ***** ********** **Location Id:** ***	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	Both methods report approximately *,***.*–*,***.* t CO_2e and are reconciled arithmetically, but the reconciliation does not deliver the Annex IV corroboration R2F-005 contemplates because the two figures are not independent. The figure remains outside the installation-emission perimeter.	**Entry Text:** The *.***% gap suggests the columns reproduce rather than test one another; both analysts flagged this as * possible common-source or back-calculation artefact rather than corroboration. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV, Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ******* / ****** / ***** ****** **Location Id:** ***	–	**Method A:** **.** **Method B:** **.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	Both methods report approximately **.** t CO_2e of Scope * purchased-energy emissions and the figures are reconciled. The site contributes negligibly to the portfolio total, though the low absolute value should be confirmed as complete coverage before issue.	**Entry Text:** The delta carries no diagnostic information at this magnitude. Both analysts asked that the small figure be confirmed as genuine low consumption rather than partial-period or partial-meter coverage. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV, Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ******* **Location Id:** ***	–	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods return *.* t CO_2e, so the figures are arithmetically reconciled but jointly uninformative. The row should be treated as an open completeness item and resolved with the operator; under R2F-007 the approved contract cannot substantiate * nil return.	**Entry Text:** * zero product carries no information about activity data or factor, and * reported zero is ambiguous between nil consumption, an unpopulated row, and * contractually-zeroed figure. **Citations:** R2F-001, R2F-003, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC, Annex IV, Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V
Location Display: ******* / ****** / ***** *******-********* **Location Id:** ***	–	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The figures are numerically reconciled at *.* t CO_2e, but the reconciliation carries no assurance. The compliance team should treat this site-year as * data-completeness item for follow-up rather than as * confirmed nil emission.	**Entry Text:** The zero delta is trivial because both columns are zero; German grid intensity remains materially non-zero, so the value is unlikely to reflect genuinely decarbonised supply. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC Art. * (Scope), Directive 2003/**/EC Art. **(*), Directive 2003/**/EC Annex IV
Location Display: ******* / ****** / *-**** **** **Location Id:** ***	–	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** **.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	The two figures reconcile to within *.**% and are unit-consistent, so there is no internal-inconsistency finding under R2F-006. However, the near-threshold position means this is * screen pass, not * verification statement under Art. **/Annex V.	**Entry Text:** Factor choice is the most likely driver: production-mix versus consumption or residual-mix basis materially affects the ******** location-based figure, and the embedded factor cannot be inspected. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC Art. **(*), Directive 2003/**/EC Annex IV, Directive 2003/**/EC Annex V
Location Display: ******* / ****** / *-**** **** **Location Id:** ***	–	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two figures agree at *.* t CO_2e, but the agreement is vacuous and supports no positive finding of nil emissions. The site is recommended for data-completeness follow-up before the value is relied on.	**Entry Text:** Both analysts read the zero as absent determinable data rather than nil purchased energy; even *******'s low grid intensity cannot mathematically produce an exact zero from real consumption. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC Art. **(*), Directive 2003/**/EC Annex IV
Location Display: ****** / ******** / ********** ** ***** *** **Location Id:** ***	–	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The figures reconcile at *.* t CO_2e, but the agreement carries no evidential weight. Because the site lies outside the EU ETS * territorial scope, the Art. **(*) and Annex IV framing is structural correspondence only, not * compliance obligation.	**Entry Text:** An exact zero is inconsistent with any positive ******* grid factor applied to real consumption, so the row should be carried as * group-reporting data gap. **Citations:** R2F-001, R2F-003, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC Art. * (Scope), Directive 2003/**/EC Art. **(*), Directive 2003/**/EC Annex IV
Location Display: ********** **Location Id:** ***	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The figures are trivially reconciled at *,***.**** t CO_2e and raise no material-misstatement signal on the R2F-006 screen, but the reconciliation is non-informative. The site sits outside the Directive's installation-emission perimeter and outside the EU ETS * territorial scope.	**Entry Text:** Exact identity confirms only common source, not corroboration. All geographic attributes are empty, so location was inferred from the city and address strings and no country is asserted. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC Art. **(*), Directive 2003/**/EC Annex IV, Directive 2003/**/EC Annex V
Location Display: ****** ****** / ******** / **********/** *****-**** **Location Id:** ***	–	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods return *.* t CO_2e and the row passes the tolerance mechanically, but the zero is not * defensible reported quantity without evidence of nil consumption or * retired attribute certificate. Obtain the underlying MWh and factor before presenting it as * genuine zero.	**Entry Text:** The columns agreeing at zero is an artefact of * common empty source record; * nil figure on * partly carbon-free grid suggests an unpopulated record. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation route), Directive 2003/**/EC, Art. **(*)

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ******* / ****** / ***** ******* **Location Id:** ***	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The energy-derived and reported figures are reconciled, agreeing to within *.**%, and the site-year is internally consistent for the R2F-006 cross-check. Both figures remain indirect Scope * purchased-energy emissions outside the installation-emission perimeter.	**Entry Text:** * rounding-order divergence consistent with two columns of one consumption record; no follow-up is required on the delta, but factor inspectability and gas composition remain unresolved. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC, Annex IV (calculation route; Part B activity data and total emissions), Directive 2003/**/EC, Art. **(*)
Location Display: ***** / **** ********* / ****** ****** ******** ******** **Location Id:** ****	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two figures are reconciled and the site-year is internally consistent under the R2F-006 cross-check. The site is outside the EU ETS * territorial scope and the figures are indirect purchased-energy emissions presented for comparability only.	**Entry Text:** The rounding-order difference is unremarkable; the substantive uncertainty is whether * national or ******** provincial factor underlies both columns, which would shift them together. **Citations:** R2F-001, R2F-003, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC, Annex IV (calculation route), Directive 2003/**/EC, Art. **(*)
Location Display: ****** **Location Id:** ****	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The figures are reconciled and the site-year is internally consistent under R2F-006. However, under R2F-007 the row falls short of the Annex IV Part * identification element because neither country nor postcode is available, so figures are presented without an asserted country.	**Entry Text:** The smallest gap in the batch is rounding-order. Empty country and region fields mean grid-factor appropriateness is untestable and the ***** attribution is inferred, not approved. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC, Annex IV (calculation route; Part * identification data), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ******* ***** **Location Id:** ****	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** ***.** **Pct Diff:** **.*% **Discrepancy Flag:** Yes **Material:** Yes	The two figures do not reconcile: * **.**% divergence means this site-year must be characterised as internally inconsistent under R2F-006. This is an internal-consistency observation only, not * verification statement under Art. **/Annex V; * factor-basis and consumption-data review is required.	**Entry Text:** The reported column appears to carry * heavier or older grid factor than the derived column, but no emission-factor, oxidation-factor or method-provenance attribute is approved to confirm this. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation route), Directive 2003/**/EC, Annex V (internal-consistency / material misstatement), Directive 2003/**/EC, Art. **(*)
Location Display: ****** / ****** / ***** ********** **Location Id:** ****	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** ***.** **Pct Diff:** *.*% **Discrepancy Flag:** Yes **Material:** Yes	The two figures bracket Scope * emissions between roughly *,*** t and *,*** t CO_2e but do not reconcile, and the internal-consistency cross-check fails. Treat this site-period as internally inconsistent and seek the underlying activity data and emission factor before either figure is used.	**Entry Text:** * gap of this size between two columns of one fact table points to * stale or differing emission factor; the level of each figure is individually credible. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation route), Directive 2003/**/EC, Art. **(*), Directive 2003/**/EC, Annex V (material misstatement)
Location Display: ****** ***** ********** **Location Id:** ****	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	The two methods return the same annual figure of approximately *,***.** t CO_2e and are reconciled arithmetically and unit-consistent. The agreement provides no independent corroboration, so the site should be treated as unverified rather than confirmed.	**Entry Text:** Difference only in the fourth decimal place, reflecting shared derivation. Country and region fields are empty, so no jurisdictional grid or regulatory context is asserted for this row. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC, Annex IV (calculation route), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / ***** ***** **Location Id:** ****	–	**Method A:** *.** **Method B:** *.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** Yes	Both methods report *.* t CO_2e, so the figures are arithmetically reconciled but the reconciliation is uninformative. The site should be escalated for source-data review before the zero is presented as * Scope * result.	**Entry Text:** The cause of the zero is not inspectable: nil consumption, missing submission, * null factor and * market-based zero mis-mapped into the location-based column cannot be distinguished. **Citations:** R2F-001, R2F-003, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC, Annex IV (calculation route and report content), Directive 2003/**/EC, Art. **(*)
Location Display: ******* / ****** / *******, **. *****, *** **** **Location Id:** ****	–	**Method A:** *,***.** **Method B:** *,***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	Both methods return *,***.**** t CO_2e; the figures are fully reconciled, unit-consistent and presentable at per-site annual granularity, subject to the standing limitation that the agreement is not independent corroboration. It is not an ETS * reportable installation emission.	**Entry Text:** Exact identity reflects the shared-source limitation. Neither analyst could establish which ***** factor basis — consumption or gross-supply, national or supplier-specific — was applied. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, Directive 2003/**/EC, Annex IV (calculation route), Directive 2003/**/EC, Art. **(*)

Site	Period	Methods	Difference / signals	Scope determination	Finding / references
Location Display: ***** / **** ********* / ****** ******* **Location Id:** ****	–	**Method A:** ***.** **Method B:** ***.**	**Abs Delta:** *.** **Pct Diff:** *.*% **Discrepancy Flag:** No **Material:** No	Both methods return ***.**** t CO_2e, so the figures are reconciled and unit-consistent. The arithmetic is consistent but the factor basis is unverified; * factor-selection error would move both figures together without triggering the discrepancy test.	**Entry Text:** National ******* and ******* provincial factors differ substantially, so the absolute level carries * factor-selection risk this cross-check is structurally incapable of detecting. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006, R2F-007, Directive 2003/**/EC, Annex IV (calculation route and report content), Directive 2003/**/EC, Art. **(*)

Limitations

#	Limitations
*	Perimeter: every figure derives from Scope * purchased-energy rows and lies outside the Directive's installation-emission perimeter (R2F-001). No figure may be presented as an ETS * reportable installation emission, and no positive determination that any site carries out an Annex I activity is supported.
*	Grain mismatch: the demand asks for * per-site, per-year series across FY2022–FY2024, but the 'year' field is empty on every row supplied. No entry can be anchored to * reporting period, so the year-over-year trend analysis requested cannot be delivered and no quarter-on-quarter or year-on-year movement can be stated anywhere in the portfolio. R2F-003 separately records that the approved time grain is the Siemens Database fiscal year, not the calendar year that Art. **(*) and Art. **(*) attach to.
*	Method independence: the two series are two columns of the same fact table drawn from the same Scope * rows (R2F-005). Close or exact agreement evidences column consistency within one source, not convergence of two independent determination systems, and must not be presented as Annex IV corroboration.
*	Factor inspectability: the co*_factor is embedded inside the derived measure and is not an approved, inspectable field, and no oxidation factor exists in the approved mapping (R2F-005, R2F-007). Method * therefore does not reproduce the full Annex IV formula at any site, factor choice, vintage and regional basis cannot be assessed, and * common factor error would not be caught by the cross-check.
*	Threshold basis: the *% discrepancy threshold is * registry convention; the Directive ****** no numeric materiality figure (R2F-006). The flag is * screening signal, not * legal determination — this matters most at the near-threshold ****** entry, where * marginally different tolerance would have reversed the result.
*	No verification: no verifier, verification status, reported uncertainty, monitoring plan or reporting-status attribute is approved. Nothing here constitutes * verification statement under Art. ** or Annex V, it cannot be concluded that emissions were monitored in accordance with the Commission guidelines or that any Art. **(*) report was made, and no conclusion on surrender compliance or the ** April deadline follows (R2F-003, R2F-004, R2F-006).
*	Zero-value entries: where both methods return zero the tolerance test is vacuous and discrepancy_flag=false carries no assurance. Nil consumption, missing activity data, * null or absent emission factor, an unreported period and * market-based zero mis-mapped into the location-based column cannot be distinguished within the approved field set. These entries should be resolved with the operator as completeness items.
*	Small-value entries: at least one site reports * very small absolute figure (≈**.** t CO_2e). This was flagged independently by both interpretation stages for confirmation that it reflects genuinely low consumption rather than partial-period or partial-meter coverage.
*	Geographic completeness: several entries have empty country_name and cdp_region fields (location IDs ***, ****, ****). For these the Annex IV Part * identification element cannot be reconstructed (R2F-007), no country is asserted, and grid-factor appropriateness is untestable. One further entry records * city string ('******* *****') inconsistent with the national postcode format, and one site's name ('**********') differs from its recorded city ('***** *******') without reconciliation.
**	Gas composition: per R2F-002 the two figures are unit-consistent in metric tonnes CO_2e and directly comparable, but it cannot be confirmed which greenhouse gases are contained in the CO_2e aggregate, nor that only Annex II gases are included, at any site.
**	Territorial scope: several sites are outside the EU entirely (******, *****, ******, the ****** ******, and on inference ******). For these the Directive's territorial scope is not engaged and the Art. **(*)/Annex IV framing is structural correspondence only.
**	External context: grid-intensity values cited by the interpretation stages are public national or regional benchmarks used for plausibility commentary only. They are not evidence of the factor actually embedded in either column of the reported data.
**	Site characterisation: no activity-type, capacity, load-profile or supply-arrangement attribute is approved for any site. Characterisations of sites as manufacturing, office or depot premises, and any assumption about grid, district-heat or contracted supply, are unverified model knowledge and are not asserted as findings.

References

#	References
*	R2F-001
*	R2F-002
*	R2F-003
*	R2F-004

#	References
*	R2F-005
*	R2F-006
*	R2F-007
*	Directive 2003/**/EC, Art. * (Scope)
*	Directive 2003/**/EC, Art. **(*) (surrender obligation)
**	Directive 2003/**/EC, Art. **(*) (monitoring and reporting)
**	Directive 2003/**/EC, Art. ** (verification)
**	Directive 2003/**/EC, Annex I (activities)
**	Directive 2003/**/EC, Annex II (greenhouse gases)
**	Directive 2003/**/EC, Annex IV (calculation and measurement/reported routes; Part * identification, Part B activity data and total emissions)
**	Directive 2003/**/EC, Annex V (verification criteria; material misstatement)

References and Assurance

How to read the identifiers used in this report

Token	What it means
R*F-###	* regulatory finding: one thing the regulation requires that this data can actually be tested against. Numbered in the order the scope reviewer produced them.
location_id	Siemens' own internal site number, carried through unchanged so * figure can be traced back to the source system.

Cited regulatory findings in full

ID	Finding and source
R2F-001	The approved emissions rows are restricted by the GHG-Protocol scope qualifier to 'Scope *' (purchased energy). Because the Directive applies to emissions from Annex I activities and Annex II gases at the installation, the Scope * market-proxy and location-based figures produced by this contract are indirect purchased-energy emissions and must be reported as such, not as ETS * reportable installation emissions. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. *
R2F-002	Both annual figures are expressed in metric tonnes of CO_2 equivalent, the unit the Directive defines for the annual emission figure; the approved reported measure (scope*_location_t, co*e_t) and the energy-derived measure (scope*_market_proxy_t) are therefore unit-consistent and directly comparable. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. *
R2F-003	The approved contract delivers emission figures at one row per site per year across FY2022–FY2024, with the site identified by identifier, name and country, which reproduces the per-installation, per-year reporting granularity that Art. **(*) requires of an operator's emissions report. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. *
R2F-004	Art. **(*) fixes the surrender quantity as the total emissions of each installation during the preceding calendar year, surrendered by ** April; the approved contract can present, per site and per year, the annual aggregate figure that corresponds in structure to that per-installation annual quantity. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. *
R2F-005	Annex IV recognises two determination routes for the same annual emission figure — * calculation route (activity data × emission factor × oxidation factor) and * measurement/reported route corroborated by * supporting calculation. The approved contract supports both routes for the Scope * perimeter: Method * as the energy-derived measure scope*_market_proxy_t (amount_consumed_mwh × co*_factor on Scope * rows) and Method B as the reported measure scope*_location_t (co*e_t on Scope * rows), per site and per year, with an optional breakdown by energy carrier (media_type). – REG-1, CELEX_*****L****_EN_TXT.pdf, p. **
R2F-006	The absolute delta between the energy-derived figure (scope*_market_proxy_t) and the reported figure (scope*_location_t) can be computed per site per year and flagged where |delta| exceeds * % of the higher value. This cross-check is an internal-consistency test of the reported activity data and the calculations leading to overall emissions, and it operationalises — for the memo's purposes only — the Annex V condition that * report may be called satisfactory only where total emissions are not materially misstated. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. **
R2F-007	Annex IV prescribes the content of the per-installation emissions report. The approved contract supports * partial reconstruction of that content: installation name and country (identification), energy activity data in MWh, total emissions in tonnes CO_2e, and the energy carrier to which each row relates. – REG-1, CELEX_*****L****_EN_TXT.pdf, p. **

Source documents

Role	Ref	File	SHA-256 (truncated)
Regulation (baseline)	REG-1	CELEX_*****L****_EN_TXT.pdf	b*******e*fe*af*...

External View (EX1/EX2) allowed sources

Domain	Institution	Scope
eur-lex.europa.eu	EUR-Lex	EU Official Journal, consolidated legislation
efrag.org	EFRAG	ESRS standard setter
commission.europa.eu	European Commission	EU policy and legislative proposals
esma.europa.eu	ESMA	EU sustainability-reporting supervision

Domain	Institution	Scope
eea.europa.eu	European Environment Agency	EU environmental data and assessments
echa.europa.eu	European Chemicals Agency	EU chemicals and pollution (ESRS E*)
bafa.de	BAFA	EnEfG Sec. */** administration (UC*)
dehst.de	DEHSt	German Emissions Trading Authority (UC*)
umweltbundesamt.de	Umweltbundesamt (UBA)	German Federal Environment Agency
bmuv.de	BMUV	German Federal Ministry for the Environment
ifrs.org	IFRS Foundation / ISSB	Global sustainability disclosure standards
globalreporting.org	GRI	Global Reporting Initiative standards
unfccc.int	UNFCCC	UN climate treaty body
unep.org	UNEP	UN Environment Programme
ipcc.ch	IPCC	Intergovernmental Panel on Climate Change
oecd.org	OECD	Environmental and economic policy analysis
iea.org	IEA	International Energy Agency
epa.gov	US EPA	US Environmental Protection Agency

Data sources (Siemens Database)

Siemens Database table	File	Column(s) used for this deliverable's figures	Data as of (file recorded date)
**********	**********	amount_consumed_mwh × co*_factor → scope*_market_proxy_t; co*e_t → scope*_location_t (Scope * rows only)	2026-**-**
**********	regions.csv	join/dimension key only, no measure	2026-**-**
**********	bu_rc_groups.csv	join/dimension key only, no measure	2026-**-**
**********	**********	join/dimension key only, no measure	2026-**-**

Assurance and traceability

Run identifier: 2026-09-07_190508Z_regulation_cli
Model / backend: claude-opus-5. Adaptive reasoning was enabled; no sampling parameters were sent.
Source documents: 1, listed above with recorded hashes where available.
Approval gates: both human gates were recorded but auto-approved – this run was invoked with `--auto`. EX1 and EX2 still ran before each gate, as the architecture requires; their advisory content never changed the promoted result.
Retention: every agent's full input and output record for this run is retained under *** *** and can be shared on request. Each figure in this report traces to a stored shelf record through its artifact_id, payload_sha256, and run_id.

G.6 UC3: EMEA Divisional Scope 2 Consolidation

Pages 2–6: system design, final `F1` output, references, and assurance.

System Design — EMEA Divisional Scope 2 Consolidation

Selection rule: $\pi(\text{case},\ \text{output profile},\ \text{assurance profile}) \rightarrow \text{topology}$ **Selected:** Coalition — Separately owned sub-results are assembled into one result.

Derived case and assurance profile (CP1)

Task: Multi owner consolidation **Output:** Quantitative / numeric measurement **Assurance:** Regulatory **Evidence shape:** Distributed signoff **Uncertainty:** High **Independent methods:** Yes **Cross-functional:** Yes **Ownership clusters:** 6

One-flow view

Stage 1

R1 → DA1 → Human: data mapping → R2 → EX1 → Human: regulatory scope

Stage 2

D1 → D2 → D3 → CP1 → Human: case profile → TS1 → Human: topology

C1 → C2 → F1 → EX2 → Human: final output

What each agent does

R1 — Identifies regulation-driven information needs from the registered request.

R2 — Fixes the regulatory findings that downstream work may use.

D2 — Checks data quality, coverage, and readiness before interpretation.

CP1 — Turns approved evidence into the case, output, and assurance profile.

C1 — Produces the division-level data-volume and evidence assessment.

F1 — Formats the completed tabular analysis as the requested deliverable.

DA1 — Maps those needs to available data and prepares the approved handoff.

D1 — Loads exactly the human-approved data request using deterministic code.

D3 — Computes the deterministic case facts used by the output agents.

TS1 — Selects Direct, Debate, or Coalition from the approved case profile.

C2 — Assembles the separately owned divisional results into one consolidation.

EX1/EX2 — Add official-source outside views before the two principal review gates.

Human gates — A reviewer approves the data boundary, regulatory scope, case profile, topology, and finished output before each becomes binding.

External View — EX1 and EX2 provide official-source second reads at the review boundary. They are advisory only: reviewers see them, but they never change the promoted result.

External View snapshots — run 2026-09-07_160902Z_regulation_cli

EX1 — Stage 1

1. High / Challenges ESRS E* requires gross Scope * to be disclosed on both a location-based and a market-based basis, not one or the other. [EFRAG]

2. High / Challenges The CSRD personal scope has been rewritten since the Directive text this assessment appears to rest on. [EUR-Lex, European Commission]

3. Medium / Flags gap GHG metrics are carved out of the partial-coverage relief that the revised standards otherwise grant. [EFRAG]

4. Medium / Adds context District heating is the weakest link in any purchased-energy inventory that carries no factor provenance. [BAFA]

5. Low / Supports The direction of travel at the standard setter favours a narrower, well-evidenced disclosure over a broad thinly-supported one. [EFRAG]

EX2 — Stage 2

1. High / Challenges Every legal citation supporting this report points to Directive (EU) 2022/**** in its original form, but that Directive has since been amended. [EUR-Lex, European Commission]

2. High / Challenges Reporting a location-based-only Scope * figure is treated by the standard setters as a substantive shortfall rather than a presentational caveat. [EFRAG]

3. Medium / Flags gap The phrase 'full mandatory assurance' should be pinned to a specific assurance level before sign-off, because the level is currently limited assurance and the framework has moved. [ESMA, EUR-Lex]

4. Medium / Flags gap Nothing in the divisional provenance chain records which emission factor produced each tCO_2e figure, and factor values are themselves restated between official editions. [Umweltbundesamt (UBA)]

5. Low / Adds context Any provenance mapping built on 2023 ESRS paragraph and datapoint references will need re-tagging for FY2027 machine-readable reporting. [EFRAG]

Registered task and filters

Use case	EMEA Divisional Scope 2 Consolidation
Emissions concept	The GHG Protocol splits a company's emissions into three scopes: Scope 1 is fuel a site burns directly; Scope 2 is the electricity, heat, or steam a site buys and consumes – Siemens does not generate it, but is accountable for it as the buyer; Scope 3 is everything else across the value chain. This use case covers Scope 2 only, consolidated per division and then across the EMEA group.
Requested output	For each division: what is its own Scope * total, is the evidence behind it solid, and did the division actually sign off on its own number?
How it's calculated	Each division's sub-total is the sum of co2e_t (Siemens Database column, Scope 2 rows only) across that division's own sites for the year – a single reported figure, not a dual-method cross-check.
Scope	region=EMEA; bu_rc=**, ***, **, ***, *******, *** | 2024-**-** to 2024-**-** (year) | REG-4
Source	Live pipeline run; gate mode: auto-approved (–auto) (2026-09-07).

Color key: red = exception / urgent change amber = warning / material change green = acceptable / compliant

Table terms:

Data status adequate or thin – the owning division's own judgement on whether its row coverage is sufficient to trust the figure; there is no fixed row-count cutoff.

Division Results

Division	Subtotal tCO2e	Years	Rows	Data status	Provenance	Finding / references
*******	***.**	2024	*	thin	The entire sub-total rests on a single contributing data row (n_rows = *) for the one year in scope, so there is no internal corroboration or cross-site redundancy behind the ***.**** tCO_2e figure. On that basis the evidence volume is read as thin.	**Entry Text:** *******'s FY2024 Scope * figure of ***.**** tCO_2e is the smallest divisional contribution to the group, but it rests on one record only, so the level should be read as provisional pending broader site coverage. **Citations:** R2F-001, R2F-002, R2F-004, R2F-005
**	*,***.**	2024	*	thin	The sub-total rests on only * contributing data rows for the single year 2024, which is a small evidence base for a whole division's EMEA Scope * figure and is the basis for the 'thin' read. Coverage is complete for the one requested year (no zero-row or missing year), so the weakness is row depth rather than period gaps.	**Entry Text:** ** reports *,***.*** tCO_2e for FY2024 across ********** — a substantial share of the group quantum, yet thin row depth leaves the figure sensitive to any single mis-stated or missing site. **Citations:** R2F-001, R2F-002, R2F-003, R2F-005, R2F-006
**	***.**	2024	*	thin	The sub-total rests on just * contributing site-level rows for the single year 2024, which is a very low evidence base for a divisional aggregate and leaves the figure highly sensitive to any single erroneous or missing site record. Coverage is complete for the one year present but there is no second year against which to sanity-check the level.	**Entry Text:** **'s ***.**** tCO_2e for FY2024 draws on ********** ; the figure is arithmetically well-defined but its low row depth limits confidence that the division's EMEA footprint is complete. **Citations:** R2F-001, R2F-002, R2F-004, R2F-005, R2F-006
***	*,***.**	2024	**	adequate	The sub-total rests on ** contributing site-level rows for the single in-scope year FY2024, with no zero-row or missing year within the requested window, which is sufficient evidentiary volume for a divisional aggregate. On that basis the row coverage is read as adequate rather than thin.	**Entry Text:** *** contributes *,***.**** tCO_2e for FY2024, the largest divisional figure, and is the only division with adequate row coverage — ********** and no missing period in the requested window. **Citations:** R2F-001, R2F-003, R2F-004, R2F-005, R2F-006
***	*,***.**	2024	*	thin	The entire *,***.**** tCO_2e sub-total is carried by a single contributing data row for FY2024, giving no internal corroboration, no site-level decomposition to inspect, and no second year for comparison. On that row count alone the evidence behind this sub-total is judged thin rather than adequate.	**Entry Text:** ***'s *,***.**** tCO_2e is roughly **% of the group total yet derives entirely from one record, making it the portfolio's single most concentrated evidence exposure and provisional pending wider coverage. **Citations:** R2F-001, R2F-003, R2F-004, R2F-005, R2F-006

Limitations

#	Limitations
*	R2F-001 names ********** (**, ***, **, ***, *******, ***), but no consolidated row exists for *** in this run — *** does not appear in division_results. The reported total of **,***.**** tCO_2e therefore covers five divisions, not the full approved six-division perimeter; no cause for the absence is stated in the inputs and none is asserted here.
*	******* is flagged thin: its FY2024 sub-total of ***.**** tCO_2e is carried by a single contributing row (n_rows = *), giving no internal corroboration or site-level decomposition.
*	*** is flagged thin: its FY2024 sub-total of *,***.**** tCO_2e is carried by a single contributing row (n_rows = *), and this one record determines approximately **% of the group total.
*	** is flagged thin on row depth: * contributing rows for a whole divisional EMEA aggregate of *,***.*** tCO_2e, so a single mis-stated or missing site row would move the figure materially.
*	** is flagged thin on row depth: * contributing site-level rows behind ***.**** tCO_2e, leaving the divisional figure highly sensitive to any single erroneous or missing record.
*	Group-level concentration: divisions flagged thin account for approximately **% of the consolidated total, so the portfolio figure is materially exposed to a small number of unverified records; only *** (** rows) has adequate coverage.
*	Portfolio-wide period limitation: all five reported divisions cover FY2024 only, so no comparative period, year-on-year trend or group-level plausibility check exists.
*	Portfolio-wide energy-input gap: no division supplied the consumed Electricity or District Heating MWh contemplated by R2F-002, so the purchased-energy evidencing behind the group Scope * figure cannot be substantiated at any level of the roll-up.
*	Portfolio-wide measurement-basis caveat: every division reports the mapped location-based measure only (R2F-001 assessment boundary); the group total carries no declared location-versus-market accounting choice and no disclosure of offsets or contractual energy attributes.
**	Consolidation-perimeter caveat (R2F-003, R2F-005): the roll-up is an organisational bu_rc aggregation over a catalog-derived EMEA boundary; it cannot be confirmed to correspond to any statutory consolidation perimeter, and 'EMEA' is not a Directive-defined concept.
**	R2F-004 comparability caveat: divisional shares of the group total can be computed arithmetically, but for the four thin divisions any apparent deviation from the group profile may reflect incomplete site coverage rather than a genuine difference in Scope * impact or intensity.
**	Scope limitation: the RC country-companies, and any entity or division excluded from this use case by the approved handoff, are outside this portfolio's measurement perimeter and are not represented in any figure reported here.
**	This deliverable is prepared for compliance-team review and is not a formal regulatory filing; human sign-off is required per the assurance profile, and no divisional attestation is created or upgraded by this document.

References

#	References
*	R2F-001 — Directive (EU) 2022/**** (CELEX *****L****), p. ** and p. **: sustainability reporting standards must specify disclosure of scope *, scope * and, where relevant, scope * greenhouse gas emissions.
*	R2F-002 — Directive (EU) 2022/****, p. **: energy aspects should be duly considered in sustainability reporting standards, in particular in relation to climate-related matters.
*	R2F-003 — Directive (EU) 2022/****, p. **: parent undertakings of a large group shall include sustainability information in the consolidated management report through a clearly identifiable dedicated section.
*	R2F-004 — Directive (EU) 2022/****, p. **: where significant differences exist between the impacts of the group and of one or more subsidiary undertakings, an adequate understanding of those units must be provided.
*	R2F-005 — Directive (EU) 2022/****, p. **: geographical or other contextual information should be disclosed to provide an understanding of principal sustainability impacts.
*	R2F-006 — Directive (EU) 2022/****, p. **, Art. *(*) phase-in: first wave applies to financial years starting on or after * January 2024, with which the reported FY2024 data set is period-aligned.

References and Assurance

How to read the identifiers used in this report

Token	What it means
R*F-###	A regulatory finding: one thing the regulation requires that this data can actually be tested against. Numbered in the order the scope reviewer produced them.

Cited regulatory findings in full

ID	Finding and source
R2F-001	For FY2024, Scope * greenhouse gas emissions in tCO_2e can be compiled for the approved EMEA sites, aggregated into a sub-total for each of the ▪▪▪▪▪▪▪▪▪▪▪▪▪▪▪▪ (**, ***, **, ***, *******, ***) and consolidated into a single group portfolio total, satisfying the substantive disclosure item 'scope * greenhouse gas emissions' that the sustainability reporting standards must require. – REG-4, CELEX_*****L****_EN_TXT.pdf, p. **
R2F-002	The purchased-energy inputs underlying the FY2024 Scope * figure can be reported as consumed MWh of electricity and of district heating per site and per division, evidencing the energy aspects that must be duly considered in climate-related sustainability disclosure. – REG-4, CELEX_*****L****_EN_TXT.pdf, p. **
R2F-004	Because the FY2024 Scope * quantum is available at divisional (bu_rc) grain, divisions whose Scope * impact differs significantly in size or intensity from the group as a whole can be identified and disclosed separately, addressing the requirement to give an adequate understanding of units within the consolidation whose impacts differ significantly from the group's. – REG-4, CELEX_*****L****_EN_TXT.pdf, p. **
R2F-005	The geographical and organisational context of the reported Scope * figures can be disclosed by identifying, for every contributing record, the site and its EMEA region membership, meeting the requirement that geographical or other contextual information be provided so that principal impacts can be understood. – REG-4, CELEX_*****L****_EN_TXT.pdf, p. **
R2F-003	A group-level consolidated FY2024 Scope * quantity can be assembled as an explicit site → division → group roll-up over the approved EMEA perimeter, providing the quantitative climate content that a parent undertaking of a large group must include in the dedicated sustainability section of its consolidated management report. – REG-4, CELEX_*****L****_EN_TXT.pdf, p. **
R2F-006	The approved reporting period (fiscal year 2024) can be stated as falling within the first CSRD phase-in wave, which applies to financial years starting on or after * January 2024, so the reported Scope * data set is period-aligned with the earliest mandatory sustainability-reporting financial year. – REG-4, CELEX_*****L****_EN_TXT.pdf, p. **

Source documents

Role	Ref	File	SHA-256 (truncated)
Regulation (baseline)	REG-4	CELEX_*****L****_EN_TXT.pdf	**e***eb***db*fb...

External View (EX1/EX2) allowed sources

Domain	Institution	Scope
eur-lex.europa.eu	EUR-Lex	EU Official Journal, consolidated legislation
efrag.org	EFRAG	ESRS standard setter
commission.europa.eu	European Commission	EU policy and legislative proposals
esma.europa.eu	ESMA	EU sustainability-reporting supervision
eea.europa.eu	European Environment Agency	EU environmental data and assessments
echa.europa.eu	European Chemicals Agency	EU chemicals and pollution (ESRS E*)
bafa.de	BAFA	EnEfG Sec. */** administration (UC*)
dehst.de	DEHSt	German Emissions Trading Authority (UC*)
umweltbundesamt.de	Umweltbundesamt (UBA)	German Federal Environment Agency

Domain	Institution	Scope
bmuv.de	BMUV	German Federal Ministry for the Environment
ifrs.org	IFRS Foundation / ISSB	Global sustainability disclosure standards
globalreporting.org	GRI	Global Reporting Initiative standards
unfccc.int	UNFCCC	UN climate treaty body
unep.org	UNEP	UN Environment Programme
ipcc.ch	IPCC	Intergovernmental Panel on Climate Change
oecd.org	OECD	Environmental and economic policy analysis
iea.org	IEA	International Energy Agency
epa.gov	US EPA	US Environmental Protection Agency

Data sources (Siemens Database)

Siemens Database table	File	Column(s) used for this deliverable's figures	Data as of (file recorded date)
**********	**********	co*e_t → scope*_location_t (Scope * rows), summed per division	2026-**-**
**********	regions.csv	join/dimension key only, no measure	2026-**-**
**********	bu_rc_groups.csv	join/dimension key only, no measure	2026-**-**
**********	**********	join/dimension key only, no measure	2026-**-**
**********	**********	join/dimension key only, no measure	2026-**-**

Assurance and traceability

Run identifier: 2026-09-07_160902Z_regulation_cli
Model / backend: claude-opus-5. Adaptive reasoning was enabled; no sampling parameters were sent.
Source documents: 1, listed above with recorded hashes where available.
Approval gates: both human gates were recorded but auto-approved – this run was invoked with `--auto`. EX1 and EX2 still ran before each gate, as the architecture requires; their advisory content never changed the promoted result.
Retention: every agent's full input and output record for this run is retained under *** *** and can be shared on request. Each figure in this report traces to a stored shelf record through its artifact_id, payload_sha256, and run_id.

G.7 UC4: Siemens 2026 ESRS Reporting Impact

Pages 2–6: system design, one example page of the final `F2` output, references, and assurance.

The full `F2` requirement table spans thirteen further pages. It reproduces text from the Siemens Sustainability Statement and the EFRAG standards, so only one page of it is shown here as a representative example.

System Design — Siemens 2026 ESRS Reporting Impact

Selection rule: $\pi(\text{case, output profile, assurance profile}) \rightarrow \text{topology}$

Selected: Debate — Independent readings of the same claim are reconciled into one result.

Derived case and assurance profile (CP1)

Task: Risk fusion **Output:** Qualitative / structured memo **Assurance:** Audit ready **Evidence shape:** Reconciliation record **Uncertainty:** Low **Independent methods:** No
Cross-functional: No **Ownership clusters:** 1

One-flow view

Stage 1

RD1/RD2 → RC1.1/RC1.2 + RM1.1/RM1.2 → RD3 → EX1 → Human: mapping

Stage 2

CP1 → Human: case profile → TS1 → Human: topology

P4/P5/P6 → F2 → EX2 → Human: final output

What each agent does

- **RD1/RD2** — Extract authoritative text and propose non-binding candidate mappings.
- **RM1.1/RM1.2** — Perform blind report matching, then decide Siemens reporting exposure.
- **CP1** — Turns approved evidence into the case, output, and assurance profile.
- **P4/P5/P6** — Assess the same evidence independently from three risk postures.
- **EX1/EX2** — Add official-source outside views before the two principal review gates.

- **RC1.1/RC1.2** — Perform blind regulatory matching, then decide the final change classification.
- **RD3** — Joins regulatory change and reporting status and derives the required action.
- **TS1** — Selects Direct, Debate, or Coalition from the approved case profile.
- **F2** — Reconciles the three risk-posture readings into the final impact report.

Human gates — A reviewer approves the joined document mapping, case profile, topology, and finished output before each becomes binding.

External View — EX1 and EX2 provide official-source second reads at the review boundary. They are advisory only: reviewers see them, but they never change the promoted result.

External View snapshots — run 2026-09-07_163102Z_regulation_cli

EX1 — Stage 1

1. **High / Challenges** The 2026 Revised ESRS were adopted by the Commission as a delegated act on 3 July 2026 but are not yet legally effective: both that act and the voluntary-standard act were transmitted to the European Parliament and Council for scrutiny, and EFRAG states the standards become legally effective only after publication in the Official Journal. [EFRAG]
2. **High / Challenges** ESMA's February 2026 opinion questions whether the very largest and most resourceful companies — those above 1,000 employees and EUR 450 million turnover — can reasonably claim they lack the skills, capabilities or resources to report quantitative information on anticipated financial effects, and notes that the 'undue cost or effort' relief applies on top of the other reliefs. [ESMA]
3. **Medium / Flags gap** EFRAG describes a 'revised GHG boundary' as one of the interoperability enhancements in the simplified standards, and the question of retaining the financial-control boundary despite the GHG Protocol's alternatives was raised publicly at EFRAG's December 2025 conference. [EFRAG]
4. **Medium / Flags gap** EFRAG has publicly identified 'substances of concern' as one of the most difficult areas in the revision, which is why an extended phase-in to 2030 was introduced. [EFRAG]
5. **Low / Adds context** The scope framework around these standards has itself changed: the Accounting Directive as amended by the Omnibus I Directive (EU) 2026/470 limits mandatory sustainability reporting to undertakings exceeding EUR 450 million net turnover and 1,000 employees, with a value-chain cap and a separate voluntary standard for those below. [EUR-Lex]

EX2 — Stage 2

1. **High / Challenges** External sources point strongly towards the anticipated-financial-effects content being consolidated into ESRS 2 rather than abolished, which cuts against reading the four 'Removed' rows (E2-6, E3-5, E4-6, E5-6) as a genuine release from obligation. [EFRAG]
2. **High / Flags gap** The authoritative status of the '2026' text used as the comparison target needs to be pinned down before sign-off. [ESMA, EFRAG]
3. **Medium / Adds context** An independent, datapoint-level cross-check of the DR mapping now exists and is still open for comment. [EFRAG]
4. **Medium / Adds context** Where the assessment anticipates restating previously published figures — notably recomputing recyclability percentages and re-basing waste metrics as proportions — ESRS 1 supplies the applicable rule and it should be cited in the eventual disclosure. [EFRAG]

Registered task and filters

Use case	Siemens 2026 ESRS Reporting Impact
Requested output	For every 2025-to-2026 ESRS requirement: did it change, did Siemens already report on it, and if the picture changed, what should Siemens do about it?
Threshold / rule	Only the two authoritative regulation texts (2025-amended and 2026-revised) decide whether a requirement changed; the EFRAG comparison files are leads only, never evidence (see Source documents on the last page).
Scope	2024-10-01 to 2025-09-30 (year) | REG-ESRS-E1, REG-ESRS-E2, REG-ESRS-E3, REG-ESRS-E4, REG-ESRS-E5
Source	Live pipeline run; gate mode: auto-approved (–auto) (2026-09-07).

Color key: red = exception / urgent change amber = warning / material change green = acceptable / compliant

Table terms:

Action needed	new report required / rewrite required / update required / status review required / no action – a fixed rule based only on whether the requirement changed and whether Siemens already reported it, not a judgement call.
Agreed priority	required urgent / required / not required – how urgently the three independent risk-posture reads agree Siemens should act, reconciled into one verdict.
Confidence	how sure the reconciling step is in its own reading of the evidence for that row.

Standard Summary

Standard	Change required	Action rows	Total rows	Highest priority
E1	Yes	11	11	required_urgent
E2	Yes	6	6	required_urgent
E3	Yes	5	5	required_urgent
E4	Yes	6	6	required_urgent
E5	Yes	6	6	required_urgent

Rows

Std.	2025 to 2026 requirements / sources	Change	Siemens reporting	Siemens impact	Decision / confidence
E1	**Dr Id 2025:** E1-1 **Requirement 2025:** Transition plan for climate change mitigation **Source 2025:** ESRS_E1_2025_amended.md E1-1 paras. 14, 16, 17, 14., 16. (a), 16. (b), 16. (c), 16. (e), 16. (f), 16. (g), 16. (h), 16. (i), 16. (j), 17. **Dr Id 2026:** E1-1 **Requirement 2026:** Transition plan for climate change mitigation **Source 2026:** ESRS_E1_2026.md E1-1 paras. 11-13, AR 2, 12. (a), 12. (b), 12. (c), 12. (d), 12. (e), 13., AR 2	**Change Status:** Modified **Change Description:** The transition plan DR is retained at the same DR number but materially restructured: the ten itemised sub-points of 2025 paragraph 16 are consolidated into five items, with targets, decarbonisation levers, investments/funding, board approval and embedding in strategy folded into a single 'key features' narrative (12(a)). The Taxonomy-alignment objective/plans disclosure (16(e)) and the EU Paris-aligned Benchmarks exclusion statement (16(g)) no longer appear in the 2026 text, while a new requirement on key assumptions and dependencies (12(c)) is added; the locked-in emissions assessment is retained but narrowed to key physical assets and products, and the 'no plan' statement now keys off whether the plan includes the key features of 12(a).	**Siemens Reported:** reported **Siemens Source Ref:** 2.2.2, 2.2.1, 2.2.5	All three postures agree the existing transition-plan narrative in sections 2.2.1/2.2.2/2.2.5 is largely reusable but must be re-mapped onto the five consolidated 'key features' of 2026 para. 12(a), with a genuinely new passage on key assumptions and dependencies (12(c)) and the locked-in emissions statement refocused on key physical assets and products. The Taxonomy-alignment (old 16(e)) and Paris-aligned Benchmark exclusion (old 16(g)) statements are no longer required; P6 alone recommends retaining them as clearly flagged voluntary content so the FY2025 level of detail is not lost. Priority split 2-1: P4 and P5 rate the row required, P6 required_urgent — treated as majority required.	**Action Needed:** report_update_required **Agreed Priority:** required **Agreed Effort:** medium **Effort Note:** All three personas rate effort medium — the content largely exists but the 12(c) assumptions-and-dependencies block is new drafting; agreed reading is medium. **Confidence:** 0.79 **Confidence Note:** Mapping is certain and Siemens reports the topic fully; only the urgency of the new 12(c) item is disputed. **Divergence:** E1-1 Transition plan: P4 and P5 rate priority required, P6 required_urgent — treated as majority required; effort agreed medium. P6-only supplemental: retain the Taxonomy-alignment (16(e)) and EU Paris-aligned Benchmarks (16(g)) statements as flagged voluntary content instead of deleting them.
E1	**Dr Id 2025:** E1-2 **Requirement 2025:** Policies related to climate change mitigation and adaptation **Source 2025:** ESRS_E1_2025_amended.md E1-2 paras. 22-25, 22., 24., 25., 25. (c), 25. (d) **Dr Id 2026:** E1-4 **Requirement 2026:** Policies related to climate change mitigation and adaptation **Source 2026:** ESRS_E1_2026.md E1-4 para. 20, 20.	**Change Status:** Modified **Change Description:** The policies DR keeps the same title and requirement to disclose climate change mitigation and adaptation policies under the generic policies module (MDR-P becomes GDR-P), but is reduced to a single sentence: the objective paragraph and the explicit list of areas the policies must address (mitigation, adaptation, energy efficiency, renewable energy deployment, other) are no longer present in the 2026 text. The DR number shifts from E1-2 to E1-4, which alone is not a substantive change.	**Siemens Reported:** reported **Siemens Source Ref:** 2.2.3	This is a renumbering (E1-2 to E1-4) and a module re-tag from MDR-P to GDR-P; the 2026 text is strictly narrower, so Siemens' existing policy descriptions in section 2.2.3 already satisfy it. P5 and P6 recommend keeping the explicit mitigation/adaptation/energy-efficiency/renewables coverage structure because it costs nothing and preserves auditability, while P4 sees no obligation at all. Priority split 2-1: P4 rates it not_required, P5 and P6 rate it required — treated as majority required.	**Action Needed:** report_update_required **Agreed Priority:** required **Agreed Effort:** low **Effort Note:** All three personas rate effort low — label and cross-reference changes only; agreed reading is low. **Confidence:** 0.80 **Confidence Note:** Change and reporting status are clear; the only judgment call is whether a renumbering counts as required action. **Divergence:** E1-2 to E1-4 Policies: P4 rates priority not_required (renumbering only), P5 and P6 rate it required — treated as majority required; effort agreed low., E1-2 (New): P4 and P6 rate priority required_urgent, P5 required — majority required_urgent. Effort split 2-1: P4 and P5 medium (consolidating existing internal risk material), P6 high (assume a from-scratch build) — majority medium.

References and Assurance

How to read the identifiers used in this report

Token	What it means
E1-1-style ids	ESRS Disclosure Requirement numbers, as published in the standard.

Source documents

Role	Ref	Document	Source URL	Retrieved
Baseline (2025 amended)	REG-ESRS-E1	ESRS E1 (10 November 2025 amended)	https://knowledgehub.efrag.org/eng/interactive/esrs-set-1/esrs-e1-climate-change/2025-11-10-quick-fix	2026-08-27
Target (2026 revised)	REG-ESRS-E1-2026	ESRS E1 (2026)	https://knowledgehub.efrag.org/eng/interactive/simplified-esrs/esrs-e1/delegated-act	2026-08-27
Siemens FY2025 report	E1 report section	Siemens Sustainability Statement for fiscal 2025, 2.2 Climate change and energy (pp. 21-38)	https://static.sw.cdn.siemens.com/siemens-disw-assets/public/2atz6iRXH3AVv3VgksCvmX/en-US/sustainability-statement.pdf	2026-08-27
Comparison helper (lead only, non-authoritative)	E1 helper	ESRS E1 — Comparative Table (December 2025)	https://api.knowledgehub.efrag.org/dev-doc-attachment/comparative-table-2025-december-e1.pdf	2026-08-27
Comparison helper (lead only, non-authoritative)	E1 helper	ESRS E1 — Log of Amendments (December 2025)	https://api.knowledgehub.efrag.org/dev-doc-attachment/log-of-amendments-2025-december-esrs-e1.pdf	2026-08-27
Baseline (2025 amended)	REG-ESRS-E2	ESRS E2 (10 November 2025 amended)	https://knowledgehub.efrag.org/eng/interactive/esrs-set-1/esrs-e2-pollution/2025-11-10-quick-fix	2026-08-27
Target (2026 revised)	REG-ESRS-E2-2026	ESRS E2 (2026)	https://knowledgehub.efrag.org/eng/interactive/simplified-esrs/esrs-e2/delegated-act	2026-08-27
Siemens FY2025 report	E2 report section	Siemens Sustainability Statement for fiscal 2025, 2.3 Pollution (pp. 39-42)	https://static.sw.cdn.siemens.com/siemens-disw-assets/public/2atz6iRXH3AVv3VgksCvmX/en-US/sustainability-statement.pdf	2026-08-27
Comparison helper (lead only, non-authoritative)	E2 helper	ESRS E2 — Comparative Table (December 2025)	https://api.knowledgehub.efrag.org/dev-doc-attachment/comparative-table-2025-december-e2.pdf	2026-08-27
Comparison helper (lead only, non-authoritative)	E2 helper	ESRS E2 — Log of Amendments (December 2025)	https://api.knowledgehub.efrag.org/dev-doc-attachment/log-of-amendments-2025-december-esrs-e2.pdf	2026-08-27
Baseline (2025 amended)	REG-ESRS-E3	ESRS E3 (10 November 2025 amended)	https://knowledgehub.efrag.org/eng/interactive/esrs-set-1/esrs-e3-water-and-marine-resources/2025-11-10-quick-fix	2026-08-27
Target (2026 revised)	REG-ESRS-E3-2026	ESRS E3 (2026)	https://knowledgehub.efrag.org/eng/interactive/simplified-esrs/esrs-e3/delegated-act	2026-08-27
Siemens FY2025 report	E3 report section	Siemens Sustainability Statement for fiscal 2025, 2.4 Water (pp. 43-45)	https://static.sw.cdn.siemens.com/siemens-disw-assets/public/2atz6iRXH3AVv3VgksCvmX/en-US/sustainability-statement.pdf	2026-08-27
Comparison helper (lead only, non-authoritative)	E3 helper	ESRS E3 — Comparative Table (December 2025)	https://api.knowledgehub.efrag.org/dev-doc-attachment/comparative-table-2025-december-e3.pdf	2026-08-27
Comparison helper (lead only, non-authoritative)	E3 helper	ESRS E3 — Log of Amendments (December 2025)	https://api.knowledgehub.efrag.org/dev-doc-attachment/log-of-amendments-2025-december-esrs-e3.pdf	2026-08-27
Baseline (2025 amended)	REG-ESRS-E4	ESRS E4 (10 November 2025 amended)	https://knowledgehub.efrag.org/eng/interactive/esrs-set-1/esrs-e4-biodiversity-and-ecosystems/2025-11-10-quick-fix	2026-08-27
Target (2026 revised)	REG-ESRS-E4-2026	ESRS E4 (2026)	https://knowledgehub.efrag.org/eng/interactive/simplified-esrs/esrs-e4/delegated-act	2026-08-27
Siemens FY2025 report	E4 report section	Siemens Sustainability Statement for fiscal 2025, 2.5 Biodiversity and ecosystems (pp. 46-48)	https://static.sw.cdn.siemens.com/siemens-disw-assets/public/2atz6iRXH3AVv3VgksCvmX/en-US/sustainability-statement.pdf	2026-08-27
Comparison helper (lead only, non-authoritative)	E4 helper	ESRS E4 — Comparative Table (December 2025)	https://api.knowledgehub.efrag.org/dev-doc-attachment/comparative-table-2025-december-e4.pdf	2026-08-27

Role	Ref	Document	Source URL	Retrieved
Comparison helper (lead only, non-authoritative)	E4 helper	ESRS E4 — Log of Amendments (December 2025)	https://api.knowledgehub.efrag.org/dev-doc-attachment/log-of-amendments-2025-december-esrs-e4.pdf	2026-08-27
Baseline (2025 amended)	REG-ESRS-E5	ESRS E5 (10 November 2025 amended)	https://knowledgehub.efrag.org/eng/interactive/esrs-set-1/esrs-e5-resource-use-and-circular-economy/2025-11-10-quick-fix	2026-08-27
Target (2026 revised)	REG-ESRS-E5-2026	ESRS E5 (2026)	https://knowledgehub.efrag.org/eng/interactive/simplified-esrs/esrs-e5/delegated-act	2026-08-27
Siemens FY2025 report	E5 report section	Siemens Sustainability Statement for fiscal 2025, 2.6 Resource use and circular economy (pp. 49-57)	https://static.sw.cdn.siemens.com/siemens-disw-assets/public/2atz6iRXH3AVv3VgksCvmX/en-US/sustainability-statement.pdf	2026-08-27
Comparison helper (lead only, non-authoritative)	E5 helper	ESRS E5 — Comparative Table (December 2025)	https://api.knowledgehub.efrag.org/dev-doc-attachment/comparative-table-2025-december-e5.pdf	2026-08-27
Comparison helper (lead only, non-authoritative)	E5 helper	ESRS E5 — Log of Amendments (December 2025)	https://api.knowledgehub.efrag.org/dev-doc-attachment/log-of-amendments-2025-december-esrs-e5.pdf	2026-08-27

External View (EX1/EX2) allowed sources

Domain	Institution	Scope
eur-lex.europa.eu	EUR-Lex	EU Official Journal, consolidated legislation
efrag.org	EFRAG	ESRS standard setter
commission.europa.eu	European Commission	EU policy and legislative proposals
esma.europa.eu	ESMA	EU sustainability-reporting supervision
eea.europa.eu	European Environment Agency	EU environmental data and assessments
echa.europa.eu	European Chemicals Agency	EU chemicals and pollution (ESRS E2)
bafa.de	BAFA	EnEfG Sec. 8/16 administration (UC1)
dehst.de	DEHSt	German Emissions Trading Authority (UC2)
umweltbundesamt.de	Umweltbundesamt (UBA)	German Federal Environment Agency
bmuv.de	BMUV	German Federal Ministry for the Environment
ifrs.org	IFRS Foundation / ISSB	Global sustainability disclosure standards
globalreporting.org	GRI	Global Reporting Initiative standards
unfccc.int	UNFCCC	UN climate treaty body
unep.org	UNEP	UN Environment Programme
ipcc.ch	IPCC	Intergovernmental Panel on Climate Change
oecd.org	OECD	Environmental and economic policy analysis
iea.org	IEA	International Energy Agency
epa.gov	US EPA	US Environmental Protection Agency

Assurance and traceability

Run identifier: 2026-09-07_163102Z_regulation_cli
Model / backend: not recorded by this agent chain (see AGENTS.md, LLM Provenance). Adaptive reasoning was enabled; no sampling parameters were sent.
Source documents: 25, listed above with recorded hashes where available.
Approval gates: both human gates were recorded but auto-approved – this run was invoked with `--auto`. EX1 and EX2 still ran before each gate, as the architecture requires; their advisory content never changed the promoted result.
Retention: every agent's full input and output record for this run is retained under `evaluation/opus-5/public/uc4_esrs_2026_impact/2026-09-07_163102Z_regulation_cli/run_001` and can be shared on request. Each figure in this report traces to a stored shelf record through its artifact_id, payload_sha256, and run_id.